



**VIDYA VIKAS INSTITUTE OF
ENGINEERING AND TECHNOLOGY,
MYSORE**

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VTU Previous Question Papers

BE – Basic Science

I to II Semester

Jun/Jul-2024

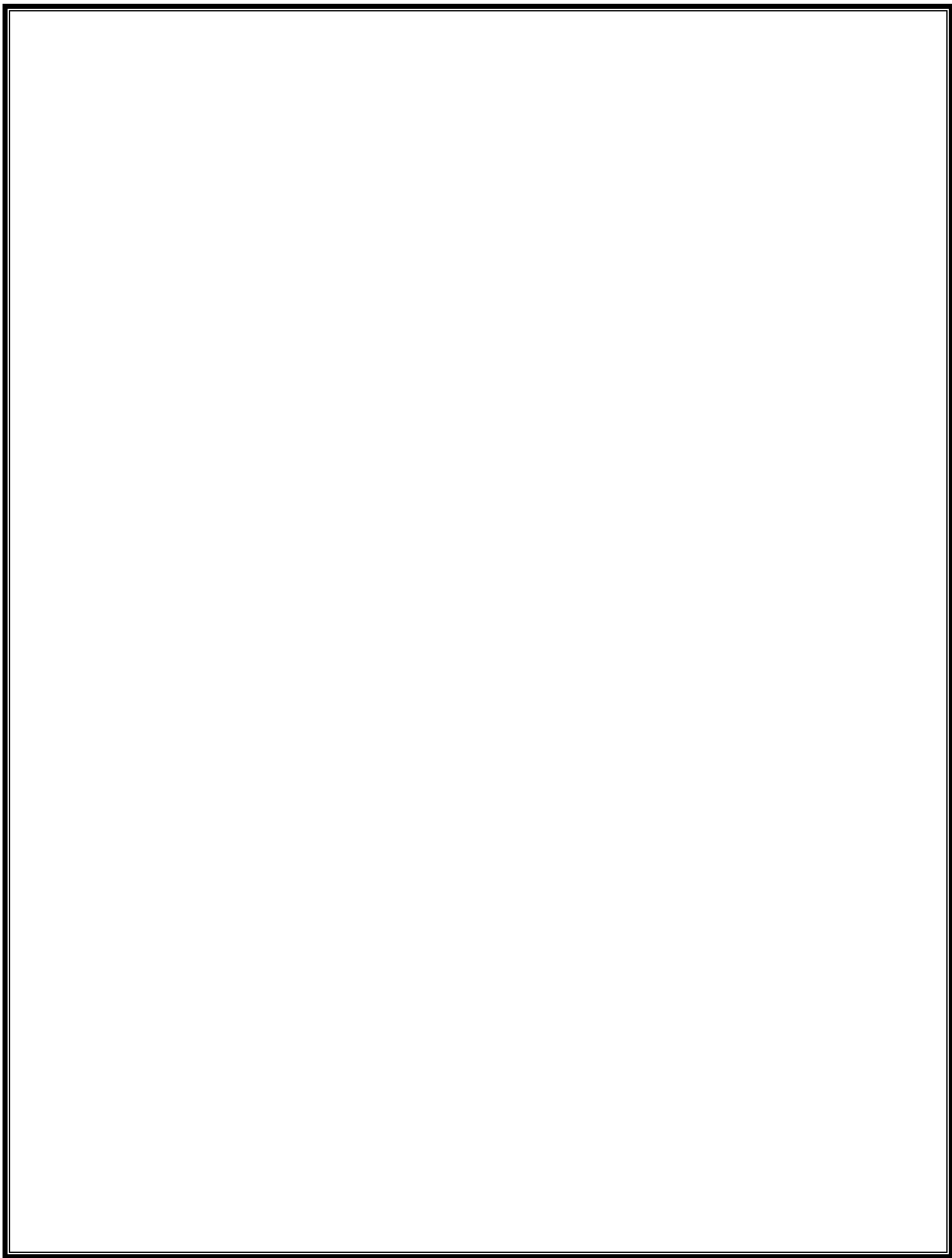
(2022 Scheme)

VVIET

Bannur road, Alanahalli, Mysore-570028

Sl.No.	Sub-Code	Subject Title	Exam Date
1	BCHEC102/ 202	Applied Chemistry for Civil Engineering Stream	June/July 2024
2	BCHE E102/202	Chemistry for EEE Stream	June/July 2024
3	BCHEM102 /202	Applied Chemistry for M E Stream	June/July 2024
4	BCHES102/202	Applied Chemistry for CSE Stream	June/July 2024
5	BEMEM103/ 203	Elements of Mechanical Engineering	June/July 2024
6	BENGK106/206/22BD17	Communicative English	June/July 2024
7	BESCK204D/BESCKD204	Introduction to Mechanical Engineering	June/July 2024
8	BETCK205B/BETCKB205	Green Buildings	June/July 2024
9	BETCK205F/BETCKF205	Waste Management	June/July 2024
10	BICOK107/207	Indian Constitution	June/July 2024
11	BIDTK158/258/22BD16	Innovation and Design Thinking	June/July 2024
12	BKBKK107/207	Ba lake Kannada	June/July 2024
13	BKSKK107/207	Samskruthika kannada	June/July 2024
14	BMATC101	Mathematics – I for Civil Engineering Stream	June/July 2024
15	BMATC201	Mathematics – II for Civil Engineering Stream	June/July 2024
16	BMATC201	Mathematics – II for Civil Engineering Stream	June/July 2024
17	BMATE101	Mathematics - I for EEE Stream	June/July 2024
18	BMATE201	Mathematics – II for E EE Stream	June/July 2024
19	BMATE201	Mathematics – II for EEE Stream	June/July 2024

20	BMATM101	Mathematics – I for Mechanical Engineering Stream	June/July 2024
21	BMATM201	Mathematics – II for Mechanical Engineering Stream	June/July 2024
22	BMATM201	Mathematics – II for Mechanical Engineering Stream	June/July 2024
23	BMATS101	Mathematics for CSE Stream – I	June/July 2024
24	BMATS201	Mathematics – II for CSE Stream	June/July 2024
25	BMATS201	Mathematics – II for CSE Stream	June/July 2024
26	BPHYC102/202	Applied Physics for Civil Engineering Stream	June/July 2024
27	BPHYC102/202	Applied Physics for Civil Engineering Stream	June/July 2024
28	BPHYE102 /202	Applied Physics for EEE Stream	June/July 2024
29	BPHY M102/202	Applied Physics for ME Stream	June/July 2024
30	BPHY S102/202	Applied Physics for CSE Stream	June/July 2024
31	BPLCK105B/ BPLCKB105	Introduction to Python Programming	June/July 2024
32	BPLCK205B / BPLCKB205	Introduction to Python Programming	June/July 2024
33	BPLCK205C/BPLCKC205	Basic s of Java Programming	June/July 2024
34	BPWSK106/206/22BD26	Professional Writing Skills in English	June/July 2024
35	BSFHK158/BSFHK258/22BD27	Scientific Foundations of Health	June/July 2024
36	BUHK408	Universal Human Values Course	June/July 2024
37	BCEDK103/203	COMPUTER AIDED ENGINEERING DRAWING	June/July 2024
38	BCEDK103/203	COMPUTER AIDED ENGINEERING DRAWING	June/July 2024
39	BCEDK103/203	COMPUTER AIDED ENGINEERING DRAWING	June/July 2024
40	BCEDK103/203	COMPUTER AIDED ENGINEERING DRAWING	June/July 2024



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BCHEC102/202

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024
Applied Chemistry for Civil Engineering Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Describe the manufacturing of cement by wet method.	7	L2	CO1
	b.	What are refractories? Mention the properties and applications of refractory materials.	7	L2	CO1
	c.	Mention the properties and applications of Aluminium and its alloys.	6	L2	CO1
OR					
Q.2	a.	Describe the preparation of Soda-lime glass.	7	L2	CO1
	b.	Explain the testing of cement by EDTA method.	7	L2	CO1
	c.	Mention the properties and applications of Iron and its alloys.	6	L2	CO1
Module – 2					
Q.3	a.	Explain construction, working and applications of methanol-oxygen fuel cell.	7	L2	CO2
	b.	Explain electrochemical corrosion of steel in concrete.	7	L2	CO2
	c.	Explain construction and working of Li-ion battery.	6	L2	CO2
OR					
Q.4	a.	Discuss the following types of corrosion: (i) Differential metal corrosion. (ii) Differential aeration corrosion.	7	L2	CO2
	b.	Describe the following corrosion control methods : (i) Galvanization (ii) Sacrificial anode method.	7	L2	CO2
	c.	Explain construction, working and applications of photovoltaic cells.	6	L2	CO2
Module – 3					
Q.5	a.	Explain softening of water by Ion exchange method.	7	L2	CO3
	b.	Define nanomaterials. Explain the synthesis of nanomaterials by Sol-gel method.	7	L2	CO3
	c.	In a COD test, 28.1 cm ³ and 14 cm ³ of 0.05 N FAS solutions were required for blank and sample titrations respectively. The volume of sample used is 25 cm ³ . Find the COD of the sample solution.	6	L3	CO3
OR					
Q.6	a.	50 ml of hard water sample is titrated with 0.015 M EDTA solution consumes 12 ml EDTA during titration. Now 250 ml of same hard water is boiled to 50 ml, filtered and diluted to 250 ml with distilled water. When 50 ml of boiled water titrated with 0.015 m EDTA, it consumes 8 ml EDTA during titration. Calculate temporary, permanent and total hardness of given water sample.	7	L3	CO3

	b.	Explain desalination of water by Electrodialysis method.	7	L2	CO3
	c.	What are carbon nano tubes? Mention the properties and applications of carbon nanotubes.	6	L2	CO3
Module – 4					
Q.7	a.	What is Geo polymer concrete? Mention the properties and applications of Geo polymer concrete.	7	L2	CO4
	b.	A polymer sample contain 5 molecules having molecular weight 2000 g/mol, 4 molecules having molecular weight 3000 g/mol and 3 molecules having molecular weight 4000 g/mol. Calculate the number average and weight average molecular mass of the polymer.	7	L3	CO4
	c.	Explain synthesis, properties and applications of nylon fibers.	6	L2	CO4
OR					
Q.8	a.	Define biodegradable polymers. Explain synthesis and applications of polylactic acid.	7	L2	CO4
	b.	Explain the properties and applications of fiber reinforced polymer composites.	7	L2	CO4
	c.	Describe synthesis properties and applications of epoxy resin.	6	L2	CO4
Module – 5					
Q.9	a.	What is phase rule? Explain the terms involved in it with example.	7	L2	CO5
	b.	Explain the estimation of acid mixture using conductometric sensor.	7	L2	CO5
	c.	Explain the principle of pH sensor and describe the determination of pH of soil sample using pH sensor.	6	L2	CO5
OR					
Q.10	a.	With the help of neat phase diagram, describe the lead-silver system.	7	L2	CO5
	b.	Describe the construction and working of pH sensor.	7	L2	CO4
	c.	Explain the estimation of Iron in FAS using potentiometric sensors.	6	L3	CO4

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BCHEE102/202

First/Second Semester B.E/B.Tech. Degree Examination, June/July 2024 Chemistry for EEE Stream

Time: 3 hrs.

Max. Marks:100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

3. VTU databook is permitted

Module – 1			M	L	C
1	a.	Explain classification of materials as conductors, insulators and semiconductors with the help of band theory.	7	L2	CO1
	b.	Explain the preparation, properties and commercial applications for grapheme oxide.	7	L2	CO1
	c.	Describe the purification of electronic grade silicon from quartz by float zone method.	6	L2	CO1
OR					
2	a.	What are conducting polymers? Explain the mechanism of conduction in polyethylene.	7	L2	CO1
	b.	What is electroless plating? Describe the electroless plating of copper in the manufacture of double-sided PCB.	7	L2	CO1
	c.	A polymer has the following composition 100 molecules of molecular mass 1000 g/mol, 200 molecules of molecular mass 2000g/mol, and 500 molecules of molecular mass 5000g/mol. Calculate the number and weight average molecular weight.	6	L3	CO1
Module – 2					
3	a.	What are Batteries? Explain the classification of batteries with suitable examples.	6	L2	CO2
	b.	Explain the construction and working of sodium-ion battery. Mention its applications.	7	L2	CO2
	c.	Explain the construction and working of vanadium flow battery. Mention its applications.	7	L2	CO2
OR					
4	a.	What are photovoltaic cells? Describe the construction and working of a PV cell. Mention its advantages and disadvantages.	7	L2	CO2
	b.	What are fuel cells? Explain the construction and working of methanol – oxygen fuel cell.	6	L2	CO2
	c.	Explain the construction and working of lithium – polymer battery. Mention its application.	7	L2	CO2
Module – 3					
5	a.	Define corrosion? Explain the electro chemical theory of corrosion taking iron as an example.	7	L2	CO3
	b.	Explain the differentiate metal differential aeration corrosion with an example.	7	L2	CO3
	c.	Calculate the CPR in both MPY and MMPY for a thick steel sheet of area 100 inch ² which experience a weight loss of 485g after one year. (Density of steel = 7.9g/cm ³).	6	L3	CO3

OR

6	a.	What is anodizing? Explain anodizing of aluminium. Mention its application.	7	L2	C03
	b.	Write a note on : i) Galvanizing ii) Sacrificial anode method.	7	L2	C03
	c.	What is e-waste? Describe the effects of e-waste on environment and human health.	6	L2	C03

Module – 4

7	a.	Describe the synthesis of nano-materials by sol-gel method with example.	7	L2	C04
	b.	Write a note on nanofibers and nanosensors.	7	L2	C04
	c.	What are QLED? Mention its properties along with their applications.	6	L2	C04

OR

8	a.	Describe the synthesis of nano-materials by co-precipitation method with an example.	7	L2	C04
	b.	What are nano-materials? Explain any two size dependent properties of nano-materials.	7	L2	C04
	c.	What are OLED's? Mention its properties and applications.	6	L2	C04

Module – 5

9	a.	What are reference electrodes? Explain the construction and working of calomel electrode.	7	L2	C03
	b.	Explain the working principle and applications of conductometric sensor.	7	L3	C03
	c.	What are concentration cells? A concentration cell is constructed by immersing two iron electrodes in 0.01m and 0.1m Fe SO ₄ solution represent the cell and calculate EMF of the cell at 298K.	6	L3	C03

OR

10	a.	What are ion-selective electrodes? Explain the construction and working principle of glass electrode.	7	L2	C05
	b.	Explain the working principle and applications of colorimetric sensor.	7	L3	C05
	c.	Explain how the P ^H of the given solution is determined using glass electrode.	6	L2	C05

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BCHEM102/202

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024
Applied Chemistry for ME Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

3. VTU Formula Hand Book is permitted.

Module – 1			M	L	C
Q.1	a.	Define Calorific value. Explain about the determination of Calorific value of fuel using Bomb calorimeter.	7	L2	CO1
	b.	Calculate GCV and NCV of a fuel from the following data : Mass of fuel = 0.75g , W = 350g , t = 3.02°C , Mass of water = 1150 and % H ₂ = 2.8.	7	L3	CO1
	c.	Explain the construction and working of Lithium in Battery along with its applications.	6	L2	CO1
OR					
Q.2	a.	Explain the production of Hydrogen by Electrolysis method and mention its advantages.	6	L2	CO1
	b.	Explain Construction , Working of Photovoltaic cell along with its advantages.	7	L2	CO1
	c.	What are the principles of Green Chemistry? What is Power Alcohol? Explain in brief.	7	L2	CO1
Module – 2					
Q.3	a.	Explain the Electrochemical theory of corrosion in detail taking Iron as an example.	7	L2	CO2
	b.	Explain i) Differential Metal corrosion. ii) Differential Aeration corrosion.	6	L3	CO2
	c.	Describe Galvanizing and mention its application.	7	L2	CO2
OR					
Q.4	a.	What is Sacrificial Anodic Protection? Explain.	6	L2	CO2
	b.	What is Metal Finishing? Mention any five of its Technological importance.	7	L2	CO2
	c.	Distinguish between Electro plating and Electro less plating. Explain Electro plating of Chromium (Decorative).	7	L3	CO2
Module – 3					
Q.5	a.	What are Polymers? Explain the different methods of Polymerization.	7	L3	CO3

	b.	Explain the synthesis of CPVC and mention its applications (CPVC – Chlorinated Polyvinyl Chloride).	6	L2	CO3
	c.	Explain the synthesis , properties and industrial application of Kevlar Fiber.	7	L2	CO3
OR					
Q.6	a.	Explain the synthesis of Polystyrene and mention its applications.	7	L2	CO3
	b.	Describe the properties and applications of Lubricants.	6	L2	CO3
	c.	What are Composites? Explain the properties and application of Carbon – based Reinforced composites (Graphene / Carbon nanotube).	7	L2	CO3
Module – 4					
Q.7	a.	Define Phase , Components and Degree of Freedom and Phase rule equation.	6	L2	CO4
	b.	Explain the Principle , Instrumentation and Application of Colorimetry.	7	L2	CO4
	c.	Explain the Principle , Instrumentation and Working of Glass Electrode.	7	L2	CO4
OR					
Q.8	a.	Explain along with diagram Lead – Silver Two Components system.	7	L2	CO4
	b.	Explain the Principle , Instrumentation and Application of Potentiometry sensor.	7	L2	CO4
	c.	Explain the process of estimation of Copper in Industrial water by using Optical sensor.	6	L2	CO4
Module – 5					
Q.9	a.	What are Alloys? Explain the composition along with properties of AlNiCo.	6	L3	CO5
	b.	Explain the synthesis of Nanomaterials by Sol – Gel method.	7	L2	CO5
	c.	Explain the Chemical composition , Properties and Application of Pervoskites.	7	L2	CO5
OR					
Q.10	a.	Explain the composites along with properties of Brass and Stainless steel.	6	L3	CO5
	b.	Explain the size dependent properties of Nano materials and with respect to Catalytic , Thermal and Surface area.	7	L2	CO5
	c.	Explain the properties of application of Carbon Nano tubes and Graphene.	7	L2	CO5

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First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024

Applied Chemistry for CSE Stream

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. VTU Formula Hand Book is permitted.
 3. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	What are electrochemical sensors? Explain the principle and working of electrochemical sensor.	07	L1	CO1
	b.	Explain the principle, working and any two applications of optical sensor.	06	L1	CO1
	c.	What is Quantum Dot sensitized solar cell? Explain the construction and working of Quantum Dot sensitized solar cell.	07	L1	CO1
OR					
Q.2	a.	Explain the detection of bio-molecule ascorbic acid using disposable sensor and also write the electro oxidation reaction.	07	L1	CO1
	b.	Explain the working principle of electrochemical gas sensors for the detection of SO _x and NO _x .	06	L1	CO1
	c.	Explain the construction and working of Li-ion battery. Mention any two applications.	07	L1	CO1
Module – 2					
Q.3	a.	What are memory devices? Explain the classification of electronic memory devices.	07	L2	CO2
	b.	Define optoelectronic device. Explain the working principle of optoelectronic device.	06	L2	CO2
	c.	What are liquid crystals? Explain the classification of liquid crystals.	07	L2	CO2
OR					
Q.4	a.	Explain the types of organic memory devices by talking p-type and n-type semiconducting materials.	07	L2	CO2
	b.	Explain any three properties and applications of polythiophene (P3HT) suitable for optoelectronic devices.	06	L2	CO2
	c.	What is QLED? Mention any three properties and applications of QLED.	07	L2	CO2
Module – 3					
Q.5	a.	Define metallic corrosion. Explain electrochemical theory of corrosion.	07	L3	CO3
	b.	A thick steel sheet of area 400 inch ² is exposed to moist air. After 2 years of period, it was found to experience a weight lost of 375g due to corrosion if the density of steel is 7.9 g/cm ³ , calculate CPR in mpy and mmpy.	06	L1	CO3
	c.	What are reference electrodes? Explain the construction, working and applications of calomel electrode.	07	L1	CO3
OR					
Q.6	a.	What is galvanization? Explain galvanization of Iron. Mention its applications.	07	L1	CO3
	b.	What are concentration cells? Calculate the cell potential of the following cell at 298 K. Ag AgNO ₃ (0.005M) AgNO ₃ (0.5M) Ag	06	L1	CO3
	c.	Explain the principle and instruction of conductometry taking estimation of weak acid using a strong base as an example.	07	L2	CO3

Module – 4					
Q.7	a.	In a sample of a polymer 20% molecules have molecular mass 15,000g/mol, 35% molecules have molecular mass 20000g/mol. Calculate the number average and weight average molecular mass of the polymer.	07	L3	CO4
	b.	Explain the preparation of Kevlar. Mention any four applications.	06	L2	CO4
	c.	Explain the generation of hydrogen by Alkaline water electrolysis with a neat labelled diagram.	07	L2	CO4
OR					
Q.8	a.	What are conducting polymers? Explain the conduction mechanism in polyacetylene through oxidative doping technique. Mention any two applications.	07	L3	CO4
	b.	What are PV cells? Explain the construction and working of photovoltaic cell.	06	L2	CO4
	c.	Explain the generation of hydrogen by proton exchange membrane electrolysis.	07	L2	CO4
Module – 5					
Q.9	a.	Define E-waste. Explain the sources and composition of E-waste.	07	L2	CO5
	b.	Explain the ill effects of materials used in manufacturing electrical and electronic products.	06	L2	CO5
	c.	Explain pyrometallurgical process of extraction of E-waste.	07	L2	CO5
OR					
Q.10	a.	Explain the extraction of gold from E-waste.	07	L2	CO5
	b.	Explain direct recycling of E-waste.	06	L2	CO5
	c.	Write a brief note on role of stakeholders for example, producers consumers, recyclers and statutory bodies in management of E-waste.	07	L2	CO5

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BEMEM103/203

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024 Elements of Mechanical Engineering

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.
3. Use of thermodynamic data handbook is permitted.*

Module – 1			M	L	C
Q.1	a.	Discuss the Emerging trends in manufacturing and automotive sector.	8	L2	CO1
	b.	With neat sketch, explain the working of thermal power plant.	6	L2	CO1
	c.	Discuss the difference between renewable and non-renewable energy sources.	6	L2	CO1
OR					
Q.2	a.	Explain the formation of steam at constant pressure with suitable sketches.	8	L2	CO1
	b.	Define the following terms with respect to steam : (i) Sensible heat (ii) Latent heat (iii) Internal energy	6	L1	CO1
	c.	Find the specific volume and enthalpy of 1 kg of steam at 0.8 MPa, with $T_s = 170.4^\circ\text{C}$, $V_s = 0.2403\text{ m}^3/\text{K}$, $h_f = 720.94\text{ kJ/kg}$, $h_{fg} = 2046.5\text{ kJ/kg}$: (i) When the dryness fraction is 0.9 (ii) When the steam is super heated to temperature of 300°C . The specific heat of superheated steam is 2.25 kJ/kgK .	6	L3	CO4
Module – 2					
Q.3	a.	With neat sketch, explain taper turning by swiveling of compound rest method.	8	L2	CO2
	b.	Explain the following operations performed on drilling machine with neat sketch : (i) Reaming (ii) Tapping (iii) Counter boring	6	L2	CO2
	c.	Discuss plane milling, end milling and slot milling operation performed on milling machine.	6	L2	CO2
OR					
Q.4	a.	Define 3D printing also explain the steps involved in 3D printing with a flow chart.	7	L1	CO2
	b.	Discuss the components of CNC machine with a neat sketch.	7	L2	CO2
	c.	Discuss the advantages of CNC machine also write any three applications of 3D printing.	6	L2	CO2

Module – 3					
Q.5	a.	With neat sketch, explain the parts of IC engine.	7	L2	CO2
	b.	Explain the working of 4-stroke petrol engine with neat sketch.	8	L2	CO2
	c.	A gas engine working on four-stroke cycle has a cylinder of 250 mm diameter, length of stroke 450 mm and is running at 180 rpm. Its mechanical efficiency is 80% when the mean effective pressure is 0.65 MPa. Find (i) Indicated power (ii) Brake power (iii) Friction power.	5	L3	CO4
OR					
Q.6	a.	With neat sketch, explain the working of room air condition.	7	L2	CO2
	b.	Discuss the properties of good refrigerant.	6	L2	CO2
	c.	Explain with neat sketch, the working of Vapour Compression Refrigerator (VCR).	7	L2	CO2
Module – 4					
Q.7	a.	With a neat sketch, derive an expression for velocity ratio in Compound Gear Train.	8	L3	CO3
	b.	Discuss Open and Cross belt driver.	6	L2	CO3
	c.	The velocity ratio of a belt drive is 3 : 2. If the diameter of the driven pulley is 120 cm, which runs at 180 rpm. Find the diameter and speed of the driver pulley and linear velocity of the belt.	6	L3	CO3
OR					
Q.8	a.	With neat sketch discuss different types of flames in oxy-acetylene gas welding, also state application of each flame.	8	L2	CO3
	b.	Explain TIG welding process.	6	L2	CO3
	c.	Differentiate between Welding, Soldering and Brazing.	6	L1	CO3
Module – 5					
Q.9	a.	With neat sketch, explain the parts of electric vehicles.	8	L2	CO3
	b.	State the advantages and disadvantages of hybrid vehicles.	6	L2	CO3
	c.	Write the difference between electric and hybrid vehicles.	6	L1	CO3
OR					
Q.10	a.	List different types of Robots configuration and discuss any two configuration in detail with sketch.	8	L2	CO3
	b.	Explain open and closed loop mechatronic system with an example for each.	6	L2	CO3
	c.	Explain the elements of a Robotic system with neat sketch.	6	L2	CO3

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Question Paper Version : A

**First/Second Semester B.E./B.Tech/B.Design Degree Examination,
June/July 2024**

**Communicative English
(COMMON TO ALL BRANCHES)**

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Choose the correct option from those given in each of the sentences below : [Q.No. 1 to Q.No. 5].

1. Communication is a _____ process of understanding between two or more persons.
a) One - way b) Three - way c) Two - way d) No - way
2. Total number of sounds in English language is _____.
a) 8 b) 20 c) 12 d) 44
3. The direction in which the formal communication flow is always _____.
a) Upwards b) Downwards c) Horizontal way d) All of these
4. Which one of the following cannot become a type of written communication _____.
a) Picture and Visual b) Rules and Instructions
c) Meeting and Conference d) Letter and Suggestion
5. Which of the following is/are not included in the tools of Verbal Communication _____.
a) Graphics b) Writing c) Reading d) Listening

Choose the correct word / Phrase / Number. [Q.No. 6 to Q.No. 10].

6. One Vowel in a word usually make _____ sound.
a) Long sound b) Very long sound c) Short sound d) Very short sound
7. Long sound usually have _____ vowels.
a) 3 b) 2 c) 4 d) 6

8. Double 'OO' is pronounced as ____
 a) ai b) Za c) U d) e
9. 'G' and 'K' are always silent before ____
 a) X b) Z c) N d) S
10. STON is pronounced as ____
 a) SH b) SCH c) Shun d) ai

Silent and non silent words : Select the missing or silent letter [Q.No. 11 to Q.No. 15].

11. ____ Sychology.
 a) P b) t c) K d) Z
12. ____ rong.
 a) X b) C c) W d) F
13. Do you have a ____ ?
 a) doubt b) dot c) dout d) dought
14. I always ____ in class.
 a) lisen b) listen c) lizen d) listen
15. ____ nife.
 a) C b) K c) Z d) W

Name the parts of speech which are underlined [Q.No. 16 to Q.No. 20].

16. He walked around the part.
 a) Noun b) Preposition c) Verb d) Conjunction
17. She got a strawberry ice – cream.
 a) Noun b) Verb c) Interjection d) Adverb
18. Older people have less energy.
 a) Verb b) Adjective c) Adverb d) Preposition
19. I like Chips and Cake.
 a) Adverb b) Noun c) Conjunction d) Verb
20. My sister answered quietly.
 a) Verb b) Adverb c) Noun d) Conjunction
21. I bought ____ pair of shoes.
 a) the b) no article c) an d) a
22. I saw ____ movie last night.
 a) an b) the c) no article d) a
23. Did you get married after leaving ____ University.
 a) no article b) a c) an d) the

24. I was at _____ railway station when you called me.
a) an b) the c) a d) no article
25. _____ Mexico is a beautiful Country.
a) An b) A c) The d) no article

Choose the right preposition [Q.No. 26 to Q.No 30].

26. Could you put your idea _____ paper?
a) at b) on c) with d) from
27. Do not waste time _____ regret?
a) with b) at c) on d) above
28. He is very simple _____ heart.
a) on b) under c) at d) for
29. She was blind _____ the age of ten.
a) by b) a c) over d) at
30. The shops are _____ walking distance.
a) with in b) with c) by d) on
31. Develop _____
a) ccvcvcc b) cvcvcvc c) cvcvcvcv d) vccvccv
32. Principal _____
a) ccv cv vcc cv b) ccvc cv cv c c) cvc cvcc d) cvc cvc
33. Mathematics _____
a) cvc vcvcvcc b) cvc cv cv cv cvcc c) cvcvcvcvcc d) cvccv
34. Window _____
a) cvccvc b) cvccv c) cvcvc d) cvccv
35. Degree _____
a) cvccv b) cvccvc c) cvccvvc d) cvccvcv

Choose the appropriate form of the verb : [Q.No. 36 to Q.No. 40].

36. Every thing _____ find when its done correctly.
a) works b) working c) worked d) doesn't work
37. None of them _____ able to solve this question.
a) where b) was c) were d) wear
38. A pair of trousers _____ all that I brought along.
a) is b) was c) were d) where
39. The wind _____ furiously.
a) Blue b) blew c) blow d) blown

40. He _____ to his mother every week.
a) right b) wrote c) writes d) write

Complete the following sentences by using the correct question tag :
[Q.No. 41 to Q.No. 45]

41. Now you can make question tags _____?
a) can you b) won't you c) can't you d) will you
42. Few people know that you are an artist _____.
a) do they b) don't they c) aren't they d) haven't they
43. You are a bit late today _____.
a) don't you b) aren't you c) are you d) do you
44. We are learning English Grammar _____.
a) are we b) shall we c) aren't we d) shan't we
45. He was asleep _____.
a) was he b) wasn't he c) didn't he d) did he

Fill in the blank with the right prefix or suffix [Q.No. 46 to Q.No. 50].

46. She was acting very _____ during the performance.
a) dislike b) unhappiness c) unlikely d) unhappy
47. The magician's tricks were truly _____.
a) likeable b) dislike c) impossible d) carefully
48. The puppy showed great _____ when learning new things.
a) careful b) interest c) unhappiness d) interest
49. Sarah decided to _____ her wardrobe by adding some new accessories.
a) unlikely b) careful c) enhance d) dislike
50. The Scientist conducted a series of _____ experiments to test the hypothesis.
a) unlikely b) careful c) unprecedented d) dislike

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Question Paper Version : B

**First/Second Semester B.E./B.Tech./B.Design Degree Examination,
June/July 2024**

**Communicative English
(COMMON TO ALL BRANCHES)**

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Complete the following sentences by using the correct question tag :
[Q.No. 1 to Q.No. 5]

1. Now you can make question tags ____?
a) can you b) won't you c) can't you d) will you
2. Few people know that you are an artist ____?
a) do they b) don't they c) aren't they d) haven't they
3. You are a bit late today ____?
a) don't you b) aren't you c) are you d) do you
4. We are learning English Grammar ____?
a) are we b) shall we c) aren't we d) shan't we
5. He was asleep ____?
a) was he b) wasn't he c) didn't he d) did he

Fill in the blank with the right prefix or suffix [Q.No. 6 to Q.No. 10].

6. She was acting very ____ during the performance.
a) dislike b) unhappiness c) unlikely d) unhappy
7. The magician's tricks were truly ____?
a) likeable b) dislike c) impossible d) carefully
8. The puppy showed great ____ when learning new things.
a) careful b) interest c) unhappiness d) interest

9. Sarah decided to _____ her wardrobe by adding some new accessories.
a) unlikely b) careful c) enhance d) dislike
10. The Scientist conducted a series of _____ experiments to test the hypothesis.
a) unlikely b) careful c) unprecedented d) dislike

Choose the appropriate form of the verb : [Q.No. 11 to Q.No. 15].

11. Every thing _____ find when its done correctly.
a) works b) working c) worked d) doesn't work
12. None of them _____ able to solve this question.
a) where b) was c) were d) wear
13. A pair of trousers _____ all that I brought along.
a) is b) was c) were d) where
14. The wind _____ furiously.
a) Blue b) blew c) blow d) blown
15. He _____ to his mother every week.
a) right b) wrote c) writes d) write

Choose the right preposition [Q.No. 16 to Q.No 20].

16. Could you put your idea _____ paper?
a) at b) on c) with d) from
17. Do not waste time _____ regret?
a) with b) at c) on d) above
18. He is very simple _____ heart.
a) on b) under c) at d) for
19. She was blind _____ the age of ten.
a) by b) a c) over d) at
20. The shops are _____ walking distance.
a) with in b) with c) by d) on
21. Develop _____
a) ccvccv b) cvccvccv c) cvccvccv d) vccvccv
22. Principal _____
a) ccv cv vcc cv b) ccvc cv cv c c) cvc cvcc d) cvc cvc
23. Mathematics _____
a) cvc vccvcc b) cvc cv cv cv cvcc c) cvccvccvcc d) cvccv
24. Window _____
a) cvccv b) cvccv c) cvccv d) cvccv

25. Degree _____
 a) cvccv b) cvccvc c) cvccvvc d) cvccvcv

**Choose the correct option from those given in each of the sentences below :
 [Q.No. 26 to Q.No. 30].**

26. Communication is a _____ process of understanding between two or more persons.
 a) One - way b) Three - way c) Two - way d) No - way
27. Total number of sounds in English language is _____
 a) 8 b) 20 c) 12 d) 44
28. The direction in which the formal communication flow is always _____
 a) Upwards b) Downwards c) Horizontal way d) All of these
29. Which one of the following cannot become a type of written communication _____
 a) Picture and Visual b) Rules and Instructions
 c) Meeting and Conference d) Letter and Suggestion
30. Which of the following is/are not included in the tools of Verbal Communication _____
 a) Graphics b) Writing c) Reading d) Listening

Choose the correct word / Phrase / Number. [Q.No. 31 to Q.No. 35].

31. One Vowel in a word usually make _____ sound.
 a) Long sound b) Very long sound c) Short sound d) Very short sound
32. Long sound usually have _____ vowels.
 a) 3 b) 2 c) 4 d) 6
33. Double 'OO' is pronounced as _____
 a) ai b) Za c) U d) e
34. 'G' and 'K' are always silent before _____
 a) X b) Z c) N d) S
35. STON is pronounced as _____
 a) SH b) SCH c) Shun d) ai

**Silent and non silent words : Select the missing or silent letter
 [Q.No. 36 to Q.No. 40].**

36. _____ Sychology.
 a) P b) t c) K d) Z
37. _____ rong.
 a) X b) C c) W d) F
38. Do you have a _____ ?
 a) doubt b) dot c) dout d) dought

39. I always _____ in class.
a) lisen b) listen c) lizen d) listen
40. _____ nife.
a) C b) K c) Z d) W

Name the parts of speech which are underlined [Q.No. 41 to Q.No. 45].

41. He walked around the part.
a) Noun b) Preposition c) Verb d) Conjunction
42. She got a strawberry ice – cream.
a) Noun b) Verb c) Interjection d) Adverb
43. Older people have less energy.
a) Verb b) Adjective c) Adverb d) Preposition
44. I like Chips and Cake.
a) Adverb b) Noun c) Conjunction d) Verb
45. My sister answered quietly.
a) Verb b) Adverb c) Noun d) Conjunction
46. I bought _____ pair of shoes.
a) the b) no article c) an d) a
47. I saw _____ movie last night.
a) an b) the c) no article d) a
48. Did you get married after leaving _____ University.
a) no article b) a c) an d) the
49. I was at _____ railway station when you called me.
a) an b) the c) a d) no article
50. _____ Mexico is a beautiful Country.
a) An b) A c) The d) no article

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Question Paper Version : C

**First/Second Semester B.E./B.Tech./B.Design Degree Examination,
June/July 2024**

**Communicative English
(COMMON TO ALL BRANCHES)**

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Choose the right preposition [Q.No. 1 to Q.No 5].

1. Could you put your idea _____ paper?
a) at b) on c) with d) from
2. Do not waste time _____ regret?
a) with b) at c) on d) above
3. He is very simple _____ heart.
a) on b) under c) at d) for
4. She was blind _____ the age of ten.
a) by b) a c) over d) at
5. The shops are _____ walking distance.
a) with in b) with c) by d) on
6. Develop _____
a) ccvccv b) cvccvccv c) cvccvccv d) vccvccv
7. Principal _____
a) ccv cv vcc cv b) ccvc cv cv c c) cvc cvcc d) cvc cvc
8. Mathematics _____
a) cvc vccvcc b) cvc cv cv cv cvcc c) cvccvccvcc d) cvccv
9. Window _____
a) cvccv b) cvccv c) cvccv d) cvccv

10. Degree _____
 a) cvccv b) cvccvc c) cvccvvc d) cvccvcv

**Choose the correct option from those given in each of the sentences below :
 [Q.No. 11 to Q.No. 15].**

11. Communication is a _____ process of understanding between two or more persons.
 a) One - way b) Three - way c) Two - way d) No - way
12. Total number of sounds in English language is _____
 a) 8 b) 20 c) 12 d) 44
13. The direction in which the formal communication flow is always _____
 a) Upwards b) Downwards c) Horizontal way d) All of these
14. Which one of the following cannot become a type of written communication _____
 a) Picture and Visual b) Rules and Instructions
 c) Meeting and Conference d) Letter and Suggestion
15. Which of the following is/are not included in the tools of Verbal Communication _____
 a) Graphics b) Writing c) Reading d) Listening

Choose the correct word / Phrase / Number. [Q.No. 16 to Q.No. 20].

16. One Vowel in a word usually make _____ sound.
 a) Long sound b) Very long sound c) Short sound d) Very short sound
17. Long sound usually have _____ vowels.
 a) 3 b) 2 c) 4 d) 6
18. Double 'OO' is pronounced as _____
 a) ai b) Za c) U d) e
19. 'G' and 'K' are always silent before _____
 a) X b) Z c) N d) S
20. STON is pronounced as _____
 a) SH b) SCH c) Shun d) ai

**Silent and non silent words : Select the missing or silent letter
 [Q.No. 21 to Q.No. 25].**

21. _____ Sychology.
 a) P b) t c) K d) Z
22. _____ rong.
 a) X b) C c) W d) F
23. Do you have a _____ ?
 a) doubt b) dot c) dout d) dought

24. I always _____ in class.
a) lisen b) listen c) lizen d) listen
25. _____ nife.
a) C b) K c) Z d) W

Name the parts of speech which are underlined [Q.No. 26 to Q.No. 30].

26. He walked around the part.
a) Noun b) Preposition c) Verb d) Conjunction
27. She got a strawberry ice – cream.
a) Noun b) Verb c) Interjection d) Adverb
28. Older people have less energy.
a) Verb b) Adjective c) Adverb d) Preposition
29. I like Chips and Cake.
a) Adverb b) Noun c) Conjunction d) Verb
30. My sister answered quietly.
a) Verb b) Adverb c) Noun d) Conjunction
31. I bought _____ pair of shoes.
a) the b) no article c) an d) a
32. I saw _____ movie last night.
a) an b) the c) no article d) a
33. Did you get married after leaving _____ University.
a) no article b) a c) an d) the
34. I was at _____ railway station when you called me.
a) an b) the c) a d) no article
35. _____ Mexico is a beautiful Country.
a) An b) A c) The d) no article

Complete the following sentences by using the correct question tag :
[Q.No. 36 to Q.No. 40]

36. Now you can make question tags _____?
a) can you b) won't you c) can't you d) will you
37. Few people know that you are an artist _____?
a) do they b) don't they c) aren't they d) haven't they
38. You are a bit late today _____?
a) don't you b) aren't you c) are you d) do you

39. We are learning English Grammar _____.
 a) are we b) shall we c) aren't we d) shan't we
40. He was asleep _____.
 a) was he b) wasn't he c) didn't he d) did he

Fill in the blank with the right prefix or suffix [Q.No. 41 to Q.No. 45].

41. She was acting very _____ during the performance.
 a) dislike b) unhappiness c) unlikely d) unhappy
42. The magician's tricks were truly _____.
 a) likeable b) dislike c) impossible d) carefully
43. The puppy showed great _____ when learning new things.
 a) careful b) interest c) unhappiness d) interest
44. Sarah decided to _____ her wardrobe by adding some new accessories.
 a) unlikely b) careful c) enhance d) dislike
45. The Scientist conducted a series of _____ experiments to test the hypothesis.
 a) unlikely b) careful c) unprecedented d) dislike

Choose the appropriate form of the verb : [Q.No. 46 to Q.No. 50].

46. Every thing _____ find when its done correctly.
 a) works b) working c) worked d) doesn't work
47. None of them _____ able to solve this question.
 a) where b) was c) were d) wear
48. A pair of trousers _____ all that I brought along.
 a) is b) was c) were d) where
49. The wind _____ furiously.
 a) Blue b) blew c) blow d) blown
50. He _____ to his mother every week.
 a) right b) wrote c) writes d) write

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Question Paper Version : D

**First/Second Semester B.E./B.Tech./B.Design Degree Examination,
June/July 2024**

**Communicative English
(COMMON TO ALL BRANCHES)**

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Choose the appropriate form of the verb : [Q.No. 1 to Q.No. 5].

1. Every thing _____ find when its done correctly.
a) works b) working c) worked d) doesn't work
2. None of them _____ able to solve this question.
a) where b) was c) were d) wear
3. A pair of trousers _____ all that I brought along.
a) is b) was c) were d) where
4. The wind _____ furiously.
a) Blue b) blew c) blow d) blown
5. He _____ to his mother every week.
a) right b) wrote c) writes d) write

**Complete the following sentences by using the correct question tag :
[Q.No. 6 to Q.No. 10]**

6. Now you can make question tags _____?
a) can you b) won't you c) can't you d) will you
7. Few people know that you are an artist _____?
a) do they b) don't they c) aren't they d) haven't they
8. You are a bit late today _____?
a) don't you b) aren't you c) are you d) do you

9. We are learning English Grammar _____.
 a) are we b) shall we c) aren't we d) shan't we
10. He was asleep _____.
 a) was he b) wasn't he c) didn't he d) did he

Fill in the blank with the right prefix or suffix [Q.No. 11 to Q.No. 15].

11. She was acting very _____ during the performance.
 a) dislike b) unhappiness c) unlikely d) unhappy
12. The magician's tricks were truly _____.
 a) likeable b) dislike c) impossible d) carefully
13. The puppy showed great _____ when learning new things.
 a) careful b) interest c) unhappiness d) interest
14. Sarah decided to _____ her wardrobe by adding some new accessories.
 a) unlikely b) careful c) enhance d) dislike
15. The Scientist conducted a series of _____ experiments to test the hypothesis.
 a) unlikely b) careful c) unprecedented d) dislike

**Choose the correct option from those given in each of the sentences below :
 [Q.No. 16 to Q.No. 20].**

16. Communication is a _____ process of understanding between two or more persons.
 a) One - way b) Three - way c) Two - way d) No - way
17. Total number of sounds in English language is _____.
 a) 8 b) 20 c) 12 d) 44
18. The direction in which the formal communication flow is always _____.
 a) Upwards b) Downwards c) Horizontal way d) All of these
19. Which one of the following cannot become a type of written communication _____.
 a) Picture and Visual b) Rules and Instructions
 c) Meeting and Conference d) Letter and Suggestion
20. Which of the following is/are not included in the tools of Verbal Communication _____.
 a) Graphics b) Writing c) Reading d) Listening

Choose the correct word / Phrase / Number. [Q.No. 21 to Q.No. 25].

21. One Vowel in a word usually make _____ sound.
 a) Long sound b) Very long sound c) Short sound d) Very short sound
22. Long sound usually have _____ vowels.
 a) 3 b) 2 c) 4 d) 6

23. Double 'OO' is pronounced as ____
 a) ai b) Za c) U d) e
24. 'G' and 'K' are always silent before ____
 a) X b) Z c) N d) S
25. STON is pronounced as ____
 a) SH b) SCH c) Shun d) ai

Silent and non silent words : Select the missing or silent letter [Q.No. 26 to Q.No. 30].

26. ____ Sychology.
 a) P b) t c) K d) Z
27. ____ rong.
 a) X b) C c) W d) F
28. Do you have a ____?
 a) doubt b) dot c) dout d) dought
29. I always ____ in class.
 a) lisen b) listen c) lizen d) listen
30. ____ nife.
 a) C b) K c) Z d) W

Name the parts of speech which are underlined [Q.No. 31 to Q.No. 35].

31. He walked around the part.
 a) Noun b) Preposition c) Verb d) Conjunction
32. She got a strawberry ice – cream.
 a) Noun b) Verb c) Interjection d) Adverb
33. Older people have less energy.
 a) Verb b) Adjective c) Adverb d) Preposition
34. I like Chips and Cake.
 a) Adverb b) Noun c) Conjunction d) Verb
35. My sister answered quietly.
 a) Verb b) Adverb c) Noun d) Conjunction
36. I bought ____ pair of shoes.
 a) the b) no article c) an d) a
37. I saw ____ movie last night.
 a) an b) the c) no article d) a

38. Did you get married after leaving _____ University.
a) no article b) a c) an d) the
39. I was at _____ railway station when you called me.
a) an b) the c) a d) no article
40. _____ Mexico is a beautiful Country.
a) An b) A c) The d) no article

Choose the right preposition [Q.No. 41 to Q.No 45].

41. Could you put your idea _____ paper?
a) at b) on c) with d) from
42. Do not waste time _____ regret?
a) with b) at c) on d) above
43. He is very simple _____ heart.
a) on b) under c) at d) for
44. She was blind _____ the age of ten.
a) by b) a c) over d) at
45. The shops are _____ walking distance.
a) with in b) with c) by d) on
46. Develop _____
a) ccvcvcc b) cvcvcvc c) cvcvcvcv d) vccvccv
47. Principal _____
a) ccv cv vcc cv b) ccvc cv cv c c) cvc cvcc d) cvc cvc
48. Mathematics _____
a) cvc vcvcvcc b) cvc cv cv cv cvvc c) cvcvcvcvcc d) cvccv
49. Window _____
a) cvccvc b) cvccv c) cvcvc d) cvccv
50. Degree _____
a) cvccv b) cvccvc c) cvccvvc d) cvccvvc

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CBCS SCHEME

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BESCK204D/BESCKD204

Second Semester B.E./B.Tech. Degree Examination, June/July 2024 Introduction to Mechanical Engineering

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	With a neat sketch explain the working principle of Nuclear Power Plant.	10	L2	CO1
	b.	Write a short note on the following: i) Global Warming ii) Ozone Depletion	10	L2	CO1
OR					
Q.2	a.	Elucidate the emerging trends and technologies in the following sectors: i) Manufacturing sector ii) Energy sector	08	L2	CO1
	b.	With a neat sketch explain the working principle of Hydro Power Plant.	08	L2	CO1
	c.	What is the role of mechanical engineering in Industries and Society?	04	L1	CO1
Module – 2					
Q.3	a.	What is CNC? Explain the basic components of CNC machine with a neat sketch.	10	L1 L2	CO2
	b.	List different operations that can be performed on Lathe Machine and explain the following with a neat sketch: i) Turning ii) Knurling	10	L1 L2	CO2
OR					
Q.4	a.	List the operations that are performed on Drilling machine and explain the below: i) Boring ii) Drilling	10	L1 L2	CO2
	b.	List the advantages and applications of CNC.	06	L1	CO2
	c.	Write a short note on 3D printing.	04	L2	CO2
Module – 3					
Q.5	a.	With a neat sketch explain the working principle of 4 stroke petrol engine along with P.V. diagram.	10	L2	CO3
	b.	Explain the components of Electric and Hybrid vehicle with a neat sketch.	10	L2	CO3
OR					
Q.6	a.	With a neat sketch explain the working principle of 4 stroke CI engine along with P.V diagram.	10	L2	CO3
	b.	Discuss the concept of Electric and Hybrid vehicles. Also list the advantages and disadvantages of EVs and Hybrid vehicles.	10	L2	CO3
Module – 4					
Q.7	a.	Classify engineering materials.	04	L2	CO4
	b.	Differentiate between Soldering, Brazing and Welding.	10	L2	CO4
	c.	Write a short note on Diamond and Silica materials.	06	L2	CO4
OR					
Q.8	a.	Explain the working principle of Electric Arc Welding with a neat sketch.	10	L2	CO4
	b.	Explain different types of Flames used in Gas welding.	06	L2	CO4
	c.	Write a short note on Shape Memory Alloys.	04	L2	CO4
Module – 5					
Q.9	a.	Define Automation. Explain the types of automation.	10	L1 L2	CO5
	b.	With an example explain open and closed loop mechatronic system.	10	L2	CO5
OR					
Q.10	a.	Define IoT and explain the characteristics of IoT.	10	L1 L2	CO5
	b.	Explain the functional blocks of IoT with a neat sketch.	10	L2	CO5

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BETCK205B/BETCKB205

Second Semester B.E./B.Tech Degree Examination, June/July 2024 Green Buildings

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the manufacturing process of concrete block with proper advantages.	10	L2	CO1
	b.	Discuss the environmental issues related to building material.	10	L2	CO1
OR					
Q.2	a.	Discuss the properties and advantage of gypsum board.	10	L2	CO1
	b.	Explain the necessary of recycling process of any two building materials.	10	L2	CO1
Module – 2					
Q.3	a.	Explain the contribution of Nirmithi Kendra and Halrital towards environmental friendly and cost effective building technologies.	10	L2	CO2
	b.	Explain the following with neat sketch. i) Rat – trap bond ii) Cavity walls.	10	L2	CO2
OR					
Q.4	a.	Differentiate between Pre engineered building and conventional building.	10	L3	CO2
	b.	Explain Casting techniques and properties of Ferro cement along with its applications.	10	L3	CO2
Module – 3					
Q.5	a.	Summarize the causes and effects of global warming. And also explain the contribution of buildings towards global warming.	10	L3	CO3
	b.	What is green building? Explain the features and benefits of green buildings.	10	L2	CO3
OR					
Q.6	a.	Explain the term Carbon foot print and also discuss the global efforts to reduce Carbon Emission.	10	L3	CO3
	b.	Discuss the process of evaluating the economic performance of building over its entire life with flow chart.	10	L3	CO3
Module – 4					
Q.7	a.	Explain the basic principle of sustainable development in building design and also mention the characteristic of sustainable buildings.	10	L3	CO4
	b.	Explain the following : i) GRIHA ii) GREEN STAR.	10	L2	CO4
OR					
Q.8	a.	List the various systems used for rating the green building in India and explain any one in detail.	10	L2	CO4
	b.	Differentiate between LEED and BREEM rating systems of green building.	10	L2	CO4
Module – 5					
Q.9	a.	Explain the methods of manufacturing and advantages of green composites.	10	L2	CO5
	b.	Define municipal solid waste and discuss the types of solid waste management.	10	L2	CO5
OR					
Q.10	a.	Explain the following terms : i) Solar Air heaters ii) Low energy approach to water management.	10	L3	CO5
	b.	Explain the sewage treatment methods with neat flow chart.	10	L3	CO5

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CBCS SCHEME

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BETCK205F/BETCKF205

Second Semester B.E./B.Tech. Degree Examination, June/July 2024 Waste Management

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the different sources and types of solid wastes.	10	L1	CO1
	b.	Explain briefly the functional elements of solid waste management with a flow chart and define each component.	10	L1	CO1
OR					
Q.2	a.	Define municipal solid waste. Enumerate the progress of solid waste management in India.	10	L1	CO1
	b.	Write short notes on : i) Environmentally Sound Technologies (EST) ii) Factors Affecting Solid Waste Management.	10	L1	CO1
Module – 2					
Q.3	a.	Briefly explain physical and chemical characteristics of waste.	10	L1	CO1
	b.	Explain how Waste Stream Assessment (WSA) is applied in solid waste management.	10	L1	CO1
OR					
Q.4	a.	Briefly explain the health and environmental effects due to solid waste generation.	10	L1	CO1
	b.	Briefly explain comparative assessment to waste generation and composition of developing nations.	10	L2	CO2
Module – 3					
Q.5	a.	Briefly describe Hauled container and stationary container systems.	10	L2	CO2
	b.	Enumerate factors to be considered for route optimization technique in designing waste collection routes.	10	L2	CO2
OR					
Q.6	a.	Define the term landfill and draw a neat sketch of a modern land fill and name of important components.	10	L2	CO2
	b.	Explain the process of leachate formation in a landfill. Give the composition of a typical leachate.	10	L2	CO2
Module – 4					
Q.7	a.	Explain in brief i) Mechanical volume reduction ii) Mechanical size reduction	10	L2	CO3
	b.	List the commonly recycled materials. Explain any one process involved in recycling materials with flow chart.	10	L2	CO3

OR					
Q.8	a.	Explain the following with a neat sketch i) Air separation ii) Magnetic separation	10	L2	CO3
	b.	Explain in brief drying and dewatering.	10	L2	CO3
Module – 5					
Q.9	a.	List and explain the different types of hazardous wastes.	10	L2	CO4
	b.	Explain the factors involved in minimization of hazardous water.	10	L2	CO4
OR					
Q.10	a.	List the hazardous waste treatment strategies and explain any one in brief.	10	L2	CO4
	b.	Write a note on E-waste recycling with a special emphasis on Indian – Scenario.	10	L2	CO4

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Question Paper Version : A

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024
Indian Constitution

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Our Constitution was adopted on and came into effect on:

a) 26 th Nov 1949 and 26 th Jan 1950	b) 26 th Jan 1949 and 26 th Jan 1950
c) 26 th Feb 1946 and 26 th Jan 1949	d) 15 th Jan 1949 and 15 th Jan 1950
 2. The idea of the Constitution of India was flashed for the first time by

a) Dr. B. R. Ambedkar	b) Dr. Rajendra Prasad
c) Mahatma Gandhiji	d) Jawaharlal Nehru
 3. The main pillars or organs of our Constitution are

a) Executive and Judiciary	b) Legislature, Executive and Judiciary
c) Legislature and Judiciary	d) Executive and Legislature
 4. One of the salient features of our Constitution.

a) It is fully Rigid	b) None of these
c) It is fully flexible	d) It is partly rigid and partly flexible
 5. Which Institution is called as 'Interpretor to the Indian Constitution'?

a) Supreme Court	b) Parliament	c) President	d) Prime Minister
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 6. The concept of Fundamental Rights are in the nature of _____

a) Enforceable	b) Justiceable	c) Exhaustive	d) All of these
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 7. Which part of the Constitution aims at establishing a welfare state in the country?

a) Preamble	b) Fundamental Duties
c) Fundamental Rights	d) Directive principles of state policy
 8. A 14 of the Indian Constitution allows:

a) Class legislation	b) Classification
c) Division of people based on geography	d) Both 'a' and 'b'
 9. Which article of the Constitution provides protection to the Civil servants?

a) 288	b) 315	c) 312	d) 311
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10. What is contained in the tenth schedule of the Constitution?
 - a) Languages recognized by Constitution
 - b) Forms of oath or affirmation
 - c) Laws that cannot be challenged in any court of law
 - d) Provision regarding disqualification on grounds of defection.
11. The Fundamental Rights granted by the Constitution of India to its citizens cannot be suspended.
 - a) Except by an order of the Supreme Court
 - b) Under any circumstances
 - c) Except by an order of the president during National Emergency
 - d) Except through an order of the President during war.
12. Our Constitution grants to the citizens _____ fundamental rights.
 - a) Six
 - b) Five
 - c) Ten
 - d) Seven
13. The fundamental duties of Indian citizen are contained in
 - a) Part III of Constitution
 - b) Part IV of Constitution
 - c) The seventh schedule of the Constitution
 - d) Part IV (a) of the Constitution
14. Preamble declares the objectives of Constitution as
 - a) Secularism
 - b) Justice, Liberty, Equality and Fraternity
 - c) Democratic Socialist
 - d) Liberalism
15. Secularism means
 - a) Absence of State Religion
 - b) Right of religious freedom
 - c) Equality of all religions
 - d) All of these
16. Free legal assistance for economically backward people is a _____
 - a) Fundamental Right
 - b) Fundamental duty
 - c) Directive principles of state policy
 - d) Special provision
17. Indian Constitution has
 - a) 12 schedules
 - b) 7 schedules
 - c) 9 schedules
 - d) 10 schedules
18. Which is not a Fundamental Right
 - a) Right to freedom
 - b) Right to Equality
 - c) Right to Constitutional Remedies
 - d) Right to property.
19. Three types of Justice referred in our preamble are.
 - a) Social, Economic and Political
 - b) Economic, International and Political
 - c) Economic, Religious and Social
 - d) Religious, Social and Political
20. The preamble of the Constitution of India has been amended so far
 - a) Four times
 - b) Thrice
 - c) Twice
 - d) Once
21. "Equal pay for Equal work" is
 - a) Gandhian principle
 - b) Sonolastic principle
 - c) Liberal principle
 - d) All of these
22. Article 22 of the Indian Constitution protects:
 - a) Senior citizens
 - b) Women
 - c) Children
 - d) Rights of arrested persons.

23. This is not a Fundamental Duty
 a) To develop scientific temper
 b) To protect natural environment
 c) Not to indulge in corrupt practice
 d) To abide by the constitution
24. Which one of the following directive principles can be described as Gandhian in nature?
 a) Providing equal pay for equal work for both men and women.
 b) Worker's participation in management.
 c) Organization of village panchayats as units of self-government.
 d) Separation of Judiciary from the executive.
25. Directive principles of state policies are
 a) Non-Justiciable
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26. The president gives his resignation to the
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 b) Parliament
 c) Vice President
 d) Prime Minister
27. The governor of a state is appointed by the president on the advice of the
 a) Prime Minister
 b) Vice-President
 c) Chief Minister
 d) Chief Justice
28. Who appoints the Prime Minister of India?
 a) Lok Sabha
 b) President
 c) Parliament
 d) Citizens of India
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 a) 23 yrs
 b) 21 yrs
 c) 35 yrs
 d) 30 yrs
30. Who among the following, has the right to sanction the expenditure of public money in India?
 a) Speaker
 b) President
 c) Prime Minister
 d) Parliament
31. Who can initiate impeachment proceedings against the president of India?
 a) Either the Houses of Parliament
 b) Any Vidhan Sabha
 c) Rajya Sabha
 d) Only Lok Sabha
32. The impeachment proceedings against the Vice-President can be initiated.
 a) Only in Lok Sabha
 b) In neither House of Parliament
 c) In either House of Parliament
 d) Only in Rajya Sabha
33. Who appoints the ambassadors to the other nations in India
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 b) President of India
 c) Minister for External affairs
 d) Prime Minister
34. The Vice President is elected by an electoral college consisting of members of
 a) Both Houses of Parliament and State Legislative
 b) Both the Houses of Parliament
 c) Lok Sabha
 d) Rajya Sabha
35. Which court is called as 'Custodian of Indian Constitution'?
 a) Supreme Court
 b) High Court
 c) Both 'a' and 'b'
 d) All the Courts
36. Who administers the oath of office to the President of India before he enters upon the office?
 a) Chief Justice of India
 b) Speaker
 c) Vice President
 d) Prime Minister

37. One feature distinguishing the Rajya Sabha from the Vidhan Parishad is
 a) Power of Impeachment b) Indirect Election
 c) Nomination of Members d) Tenure of Membership
38. Who acts on President of India when neither the President nor the Vice-President is available?
 a) Seniormost Governor of a State b) Chief Justice of India
 c) Speaker of Lok Sabha d) Auditor General of India
39. The retirement age of High Court Judges is:
 a) 58 Years b) 62 Years c) 65 Years d) 64 Years
40. What is the minimum age in years for becoming the MP at Lok Sabha and Rajya Sabha?
 a) 18 and 25 b) 25 and 18 c) 25 and 30 d) 30 and 25
41. Who among the following holds office during the pleasure of President
 a) Governor b) Election Commissioner
 c) Speaker of Lok Sabha d) Prime Minister
42. A proclamation of emergency issued under Article 352 must be approached by the Parliament within.
 a) 3 Months b) 2 Months c) 1 Month d) 6 Weeks
43. The President's Rule in a state means that the state is ruled by
 a) A caretaker Government
 b) The Chief Minister nominated by President
 c) The Governor of the state
 d) The President directly
44. The control of the preparation of electoral rolls for parliament and legislature rests with the
 a) President b) Election Commission
 c) Cabinet d) Prime Minister
45. In India, Political parties are given recognition by
 a) Election Commission b) Speaker of Lok Sabha
 c) President d) Law Commission
46. Which article deals with Constitutional amendment procedures in India
 a) A 370 b) A 368 c) A 350 d) A 320
47. Center can declare Constitutional emergency in a state under article
 a) A 152 b) A 365 c) A 360 d) A356
48. The Lengthiest Constitutional amendment in India was
 a) 44th amendment b) 49th amendment
 c) 42nd amendment d) 92nd amendment
49. Election Commission conducts the election as per
 a) People's Representative Act b) Parliament act
 c) Judicial act d) All of these
50. In terms of election laws of India, electioneering ceases in a constituency at least _____ hours before the commencement of the polling.
 a) 48 b) 36 c) 24 d) 12

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Question Paper Version : B

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024 Indian Constitution

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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-
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c) President
d) Law Commission
 6. Which article deals with Constitutional amendment procedures in India

a) A 370
b) A 368
c) A 350
d) A 320
 7. Center can declare Constitutional emergency in a state under article

a) A 152
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c) A 360
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a) It is fully Rigid b) None of these
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35. Which Institution is called as 'Interpretor to the Indian Constitution'?
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36. The concept of Fundamental Rights are in the nature of _____
a) Enforceable b) Justiceable c) Exhaustive d) All of these

Version – B – 4 of 4

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Question Paper Version : C

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024
Indian Constitution

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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-
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c) Economic, Religious and Social d) Religious, Social and Political
50. The preamble of the Constitution of India has been amended so far
a) Four times b) Thrice c) Twice d) Once

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Question Paper Version : D

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024 Indian Constitution

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
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3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
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-
1. "Equal pay for Equal work" is

a) Gandhian principle

c) Liberal principle

b) Sonolastic principle

d) All of these
 2. Article 22 of the Indian Constitution protects:

a) Senior citizens

c) Children

b) Women

d) Rights of arrested persons.
 3. This is not a Fundamental Duty

a) To develop scientific temper

c) Not to indulge in corrupt practice

b) To protect natural environment

d) To abide by the constitution
 4. Which one of the following directive principles can be described as Gandhian in nature?

a) Providing equal pay for equal work for both men and women.

b) Worker's participation in management.

c) Organization of village panchayats as units of self-government.

d) Seperation of Judiciary from the executive.
 5. Directive principles of state policies are

a) Non-Justiciable

c) Only some are Justiciable

b) Justiciable

d) None of these
 6. The president gives his resignation to the

a) Chief Justice

b) Parliament

c) Vice President

d) Prime Minister
 7. The governor of a state is appointed by the president on the advice of the

a) Prime Minister

b) Vice-President

c) Chief Minister

d) Chief Justice
 8. Who appoints the Prime Minister of India?

a) Lok Sabha

b) President

c) Parliament

d) Citizens of India

9. Minimum age required to contest for office of the president
a) 23 yrs b) 21 yrs c) 35 yrs d) 30 yrs
10. Who among the following, has the right to sanction the expenditure of public money in India?
a) Speaker b) President c) Prime Minister d) Parliament
11. Who among the following holds office during the pleasure of President
a) Governor b) Election Commissioner
c) Speaker of Lok Sabha d) Prime Minister
12. A proclamation of emergency issued under Article 352 must be approached by the Parliament within.
a) 3 Months b) 2 Months c) 1 Month d) 6 Weeks
13. The President's Rule in a state means that the state is ruled by
a) A caretaker Government
b) The Chief Minister nominated by President
c) The Governor of the state
d) The President directly
14. The control of the preparation of electoral rolls for parliament and legislature rests with the
a) President b) Election Commission
c) Cabinet d) Prime Minister
15. In India, Political parties are given recognition by
a) Election Commission b) Speaker of Lok Sabha
c) President d) Law Commission
16. Which article deals with Constitutional amendment procedures in India
a) A 370 b) A 368 c) A 350 d) A 320
17. Center can declare Constitutional emergency in a state under article
a) A 152 b) A 365 c) A 360 d) A356
18. The Lengthiest Constitutional amendment in India was
a) 44th amendment b) 49th amendment
c) 42nd amendment d) 92nd amendment
19. Election Commission conducts the election as per
a) People's Representative Act b) Parliament act
c) Judicial act d) All of these
20. In terms of election laws of India, electioneering ceases in a constituency at least _____ hours before the commencement of the polling.
a) 48 b) 36 c) 24 d) 12
21. Who can initiate impeachment proceedings against the president of India?
a) Either the Houses of Parliament b) Any Vidham Sabha
c) Rajya Sabha d) Only Lok Sabha
22. The impeachment proceedings against the Vice-President can be initiated.
a) Only in Lok Sabha b) In neither House of Parliament
c) In either House of Parliament d) Only in Rajya Sabha

23. Who appoints the ambassadors to the other nations in India
 - a) Foreign Minister
 - b) President of India
 - c) Minister for External affairs
 - d) Prime Minister
24. The Vice President is elected by an electoral college consisting of members of
 - a) Both Houses of Parliament and State Legislative
 - b) Both the Houses of Parliament
 - c) Lok Sabha
 - d) Rajya Sabha
25. Which court is called as 'Custodian of Indian Constitution'?
 - a) Supreme Court
 - b) High Court
 - c) Both 'a' and 'b'
 - d) All the Courts
26. Who administers the oath of office to the President of India before he enters upon the office?
 - a) Chief Justice of India
 - b) Speaker
 - c) Vice President
 - d) Prime Minister
27. One feature distinguishing the Rajya Sabha from the Vidhan Parishad is
 - a) Power of Impeachment
 - b) Indirect Election
 - c) Nomination of Members
 - d) Tenure of Membership
28. Who acts on President of India when neither the President nor the Vice-President is available?
 - a) Seniormost Governor of a State
 - b) Chief Justice of India
 - c) Speaker of Lok Sabha
 - d) Auditor General of India
29. The retirement age of High Court Judges is:
 - a) 58 Years
 - b) 62 Years
 - c) 65 Years
 - d) 64 Years
30. What is the minimum age in years for becoming the MP at Lok Sabha and Rajya Sabha?
 - a) 18 and 25
 - b) 25 and 18
 - c) 25 and 30
 - d) 30 and 25
31. The Fundamental Rights granted by the Constitution of India to its citizens cannot be suspended.
 - a) Except by an order of the Supreme Court
 - b) Under any circumstances
 - c) Except by an order of the president during National Emergency
 - d) Except through an order of the President during war.
32. Our Constitution grants to the citizens _____ fundamental rights.
 - a) Six
 - b) Five
 - c) Ten
 - d) Seven
33. The fundamental duties of Indian citizen are contained in
 - a) Part III of Constitution
 - b) Part IV of Constitution
 - c) The seventh schedule of the Constitution
 - d) Part IV (a) of the Constitution
34. Preamble declares the objectives of Constitution as
 - a) Secularism
 - b) Justice, Liberty, Equality and Fraternity
 - c) Democratic Socialist
 - d) Liberalism
35. Secularism means
 - a) Absence of State Religion
 - b) Right of religious freedom
 - c) Equality of all religions
 - d) All of these
36. Free legal assistance for economically backward people is a _____
 - a) Fundamental Right
 - b) Fundamental duty
 - c) Directive principles of state policy
 - d) Special provision

37. Indian Constitution has
a) 12 schedules b) 7 schedules c) 9 schedules d) 10 schedules
38. Which is not a Fundamental Right
a) Right to freedom b) Right to Equality
c) Right to Constitutional Remedies d) Right to property.
39. Three types of Justice referred in our preamble are.
a) Social, Economic and Political b) Economic, International and Political
c) Economic, Religious and Social d) Religious, Social and Political
40. The preamble of the Constitution of India has been amended so far
a) Four times b) Thrice c) Twice d) Once
41. Our Constitution was adopted on and came into effect on:
a) 26th Nov 1949 and 26th Jan 1950 b) 26th Jan 1949 and 26th Jan 1950
c) 26th Feb 1946 and 26th Jan 1949 d) 15th Jan 1949 and 15th Jan 1950
42. The idea of the Constitution of India was flashed for the first time by
a) Dr. B. R. Ambedkar b) Dr. Rajendra Prasad
c) Mahatma Gandhiji d) Jawaharlal Nehru
43. The main pillars or organs of our Constitution are
a) Executive and Judiciary b) Legislature, Executive and Judiciary
c) Legislature and Judiciary d) Executive and Legislature
44. One of the salient features of our Constitution.
a) It is fully Rigid b) None of these
c) It is fully flexible d) It is partly rigid and partly flexible
45. Which Institution is called as 'Interpretor to the Indian Constitution'?
a) Supreme Court b) Parliament c) President d) Prime Minister
46. The concept of Fundamental Rights are in the nature of ____
a) Enforceable b) Justiceable c) Exhaustive d) All of these
47. Which part of the Constitution aims at establishing a welfare state in the country?
a) Preamble b) Fundamental Duties
c) Fundamental Rights d) Directive principles of state policy
48. A 14 of the Indian Constitution allows:
a) Class legislation b) Classification
c) Division of people based on geography d) Both 'a' and 'b'
49. Which article of the Constitution provides protection to the Civil servants?
a) 288 b) 315 c) 312 d) 311
50. What is contained in the tenth schedule of the Constitution?
a) Languages recognized by Constitution
b) Forms of oath or affirmation
c) Laws that cannot be challenged in any court of law
d) Provision regarding disqualification on grounds of defection.

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Question Paper Version : A

**First/Second Semester B.E./B.Tech./B.Design Degree Examination,
June/July 2024**

Innovation and Design Thinking

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Design thinking is _____
a) A process that allows engineers and designers to create new and innovative solutions to business challenges.
b) An iterative, non-linear process which focuses on a collaboration between designers and users
c) a and b
d) None of these
 2. Select the correct order of the different stages of design thinking.
i) Empathize ii) Define iii) Ideate iv) Prototype v) Test
a) ii – iii – iv – v – i b) i – ii – iii – iv – v
c) iii – iv – v – i – ii d) iv – v – i – ii – iii
 3. Which of the following principles are not considered for design thinking?
a) Embrace experimentation b) Human centric design
c) Profit centric d) Pattern identification for problem solving
 4. To empathize one has to
a) Observe b) Engage c) Listen d) All of these
 5. Which of the following are not tools of visualization?
a) Maps b) Images c) Stories d) Videos
 6. Journey mapping is also called _____ mapping
a) Path b) Experience c) Conduct d) Feedback
 7. Which of the following are not tools of design thinking?
a) Co creation b) Proto typing c) Mind mapping d) Online mapping
 8. Which design thinking phase is closely related to the creation of an minimum viable product (MVP)?
a) Empathize b) Prototype c) Ideate d) Test

9. Which one is the minimum viable product?

a)



b)



c) Both a and b

d) None of these

10. How can the theory and practice of design thinking work together?

a) Theory provides a set of rules to follow, while practice ensures the rules are applied effectively.

b) Theory and practice are two separate processes that do not overlap.

c) Theory provides a framework for creative problem solving, while practice develops the skills and knowledge to apply the framework effectively.

d) Theory and practice are interchangeable and can be used in any order.

11. What is real-time design interaction capture and analysis?

a) A process of capturing user feedback after the design process is complete.

b) A method of analyzing design interactions as they happen.

c) A technique for capturing user interactions with a product after it is released.

d) A way to analyze design interactions after they have occurred.

12. What types of interactions can be captured and analyzed in real-time design interaction capture and analysis?

a) User feed back and comments

b) User interface interactions

c) User demographic and preferences

d) User purchase behavior.

13. What is the importance of collaboration in design thinking?

a) It speeds up the design process

b) It eliminates the need for user research

c) It ensures all design decisions are final

d) It brings diverse perspectives and expertise to the design process.

14. What are some digital tools that can be enable efficient collaboration in design thinking?

a) E Mail

b) Social media platforms

c) Video conferencing and collaboration software

d) Design software

15. What is empathy in design?

a) Understanding the emotions and experiences of users

b) Creating products that appeal to users emotions

c) Limiting the number of user persons in the design process

d) Only designing products based on market research.

16. Why is empathy important in the design process?

a) It ensures all design decisions are final

b) It brings perspectives and expertise to the design process

c) It speeds up the design process

d) It results in product that meet users needs and desires.

17. What is the benefit of empathy in design?
 - a) It guarantees success in the market
 - b) It speeds up the design process
 - c) It ensures all design decisions are final
 - d) It leads to more meaningful and impactful products.
18. What is distributed design collaboration?
 - a) Collaboration between designers who are geographically separated.
 - b) Collaboration between designers and machines.
 - c) Collaboration between designers and machines
 - d) Collaboration between designers from different industries.
19. What are some challenges of distributed design collaboration?
 - a) Limited access to digital tools
 - b) Technical difficulties and connectivity issues
 - c) A lack of clear communication channels
 - d) A limited pool of expertise and perspectives
20. What are some benefits of real-time interaction and analysis in design thinking?
 - a) It allows for quick and agile decision making
 - b) It limits the number of team members involved in the design process
 - c) It eliminates the need for user research and testing
 - d) It reduces the amount of feed back received from users.
21. How can real time interaction and analysis be integrated into the design process?
 - a) By conducting user research and testing at the end of the design process only.
 - b) By eliminating user feedback and relying solely on personal preferences.
 - c) By analyzing market trends and sales data only.
 - d) By incorporating user feedback throughout the entire design process.
22. What are some common tools used in the empathy phase of design thinking?
 - a) Persona development and user interviews
 - b) Brain storming and ideation sessions
 - c) Sketching and prototyping
 - d) User testing and feedback analysis.
23. What is the role of empathy in design thinking in IT?
 - a) To understand the needs and challenges of IT users and stake holders.
 - b) To create visually appealing IT products and services.
 - c) To ensure compliance with industry standards and regulations.
 - d) To increase profits for the IT organization.
24. What is the importance of prototyping in design thinking in IT?
 - a) To test and refine IT products and services before launch.
 - b) To show case IT capabilities to stake holders.
 - c) To impress user with cutting edge technology.
 - d) To ensure compliance with industry standards and regulations.
25. What is the goal of using design thinking in business process modeling?
 - a) To create visually appealing process diagrams.
 - b) To stream line business operations and increase efficiency.
 - c) To reduce costs for the organization.
 - d) To comply with industry regulations and standards.

26. What is the importance of prototyping in design thinking in business process modeling?
- a) To test and refine process models before implementation.
 - b) To show case the organizations capabilities to stakeholders.
 - c) To impress users with cutting edge technology.
 - d) To ensure compliance with industry standards and regulations.
27. What are some common challenges that can be addressed using design thinking in business process modeling?
- a) Inefficient processes that waste time and resources.
 - b) Poor communication and collaboration between departments.
 - c) Resistance to change from process users and stakeholders.
 - d) All of these
28. What is the goal of using agile methodology in design thinking?
- a) To create aesthetically pleasing designs
 - b) To increase efficiency and productivity in the design process
 - c) To reduce costs for the organization
 - d) To comply with Industry regulations and standards.
29. What is the importance of prototyping in agile design thinking?
- a) To quickly test and refine design ideas
 - b) To showcase design capabilities to stoke holders
 - c) To impress users with cutting edge technology
 - d) To ensure compliance with industry standards and regulations
30. What are some common challenges that can be addressed using agile design thinking?
- a) Inefficient design processes that waste time and resources.
 - b) Poor communication and collaboration between team members.
 - c) Resistance to change from users and stakeholders.
 - d) All of these
31. What is the goal of using agile technology in virtual collaboration environments for design thinking?
- a) To create aesthetically pleasing designs
 - b) To increase efficiency and productivity in the design process
 - c) To reduce costs for the organization
 - d) To comply with industry regulations and standards
32. Why is scenario based prototyping important in innovation design thinking?
- a) It allows for faster development of prototypes
 - b) It helps to identify potential usability issues before the product is released
 - c) It reduces the cost of creating prototypes
 - d) None of these
33. What is the first step in scenario based prototyping?
- a) Creating a detailed technical specification
 - b) identifying potential users and use cases
 - c) Building a physical mock up of the product
 - d) None of these

34. What are the key stages of the design thinking process?
- a) Ideation, testing, and implementation
 - b) Empathy, define, ideate, prototype and test
 - c) Planning, execution and evaluation
 - d) None of these
35. What is storytelling in the context of design thinking?
- a) It is the process of creating a narrative around a design solution.
 - b) It is the process of creating a story board to communicate design ideas.
 - c) It is the process of creating a visual representation of a design concept.
 - d) None of these
36. What are some common tools and techniques used in strategic foresight?
- a) Scenario planning, trend analysis, and expert interviews
 - b) Prototyping, user testing and iterative design
 - c) Market research, customer feedback, and competitive analysis
 - d) None of these
37. What are some common tools and techniques used in sensemaking?
- a) Empathy mapping, customer journey mapping and data visualization.
 - b) Prototyping, user testing and iterative design
 - c) Market research, customer feedback and competitive analysis
 - d) None of these
38. What is maintenance in the context of design thinking?
- a) The process of repairing and upgrading existing design solutions.
 - b) The process of creating new design solutions from scratch.
 - c) The process of evaluating the effectiveness of existing design solutions.
 - d) None of these
39. What is value redefinition in the context of design thinking?
- a) The process of identifying and redefining the value proposition of a design solution.
 - b) The process of creating a detailed technical specification for a design solution.
 - c) The process of evaluating the effectiveness of existing design solutions.
 - d) None of these
40. What is extreme competition in the context of design thinking?
- a) The process of competing against other design teams to create the best solution.
 - b) The process of pushing design teams to their limits to create innovative solutions.
 - c) The process of collaborating with competitors to create a joint solution.
 - d) None of these
41. What is experience design in the context of design thinking?
- a) The process of designing physical products and services.
 - b) The process of designing digital interfaces and interactions.
 - c) The process of designing holistic experiences for users across all touch points.
 - d) None of these
42. What is standardization in the context of design thinking?
- a) The process of creating standardized design solutions.
 - b) The process of following a standard set of design principles.
 - c) The process of establishing standards for design processes and methodologies.
 - d) None of these

43. What is humanization in the context of design thinking?
- a) The process of making designs more human centered and empathetic.
 - b) The process of making designs more technically advanced.
 - c) The process of making designs more aesthetically pleasing.
 - d) None of these
44. How can a creative culture be fastened in the design thinking process?
- a) By encouraging risk-taking and experimentation
 - b) By establishing rigid design processes and guide lines
 - c) By prioritizing cost effectiveness over innovation
 - d) None of these
45. What are some common tools and techniques used in rapid prototyping?
- a) Sketching, wire framing and paper prototyping
 - b) 3D printing, laser cutting and CNC machining
 - c) User research, market analysis and competitive analysis
 - d) None of these
46. How can a business model be designed using design thinking?
- a) By identifying customer needs and pain points and designing solutions that address them.
 - b) By following a set of established business practices and industry standards.
 - c) By prioritizing cost effective over customer value.
 - d) None of these
47. Which phase is referred to as an experimental phase where continuous iterations can take place?
- a) Define
 - b) Empathise
 - c) Prototype
 - d) None of these
48. What is the term used to describe the process of narrowing down thoughts to reach the final solution?
- a) Convergent thinking
 - b) Divergent thinking
 - c) None of these
 - d) Both a and b
49. Design thinking is typically used to provide a solution based approach to problem solving
- a) True
 - b) False
50. Can design thinking be applied in professions outside of design?
- a) True
 - b) False

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Question Paper Version : B

**First/Second Semester B.E./B.Tech./B.Design Degree Examination,
June/July 2024**

Innovation and Design Thinking

Time: 1 hr.]

[Max. Marks: 50

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 - d) Design software

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31. Design thinking is _____
- A process that allows engineers and designers to create new and innovative solutions to business challenges.
 - An iterative, non-linear process which focuses on a collaboration between designers and users
 - a and b
 - None of these
32. Select the correct order of the different stages of design thinking.
- | | | | | |
|--------------------------|--------------------------|-------------|---------------|---------|
| i) Empathize | ii) Define | iii) Ideate | iv) Prototype | v) Test |
| a) ii – iii – iv – v – i | b) i – ii – iii – iv – v | | | |
| c) iii – iv – v – i – ii | d) iv – v – i – ii – iii | | | |
33. Which of the following principles are not considered for design thinking?
- Embrace experimentation
 - Human centric design
 - Profit centric
 - Pattern identification for problem solving

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a) Empathize b) Prototype c) Ideate d) Test
39. Which one is the minimum viable product?
a)  b) 
c) Both a and b d) None of these
40. How can the theory and practice of design thinking work together?
a) Theory provides a set of rules to follow, while practice ensures the rules are applied effectively.
b) Theory and practice are two separate processes that do not overlap.
c) Theory provides a framework for creative problem solving, while practice develops the skills and knowledge to apply the framework effectively.
d) Theory and practice are interchangeable and can be used in any order.
41. How can real time interaction and analysis be integrated into the design process?
a) By conducting user research and testing at the end of the design process only.
b) By eliminating user feedback and relying solely on personal preferences.
c) By analyzing market trends and sales data only.
d) By incorporating user feedback throughout the entire design process.
42. What are some common tools used in the empathy phase of design thinking?
a) Persona development and user interviews
b) Brain storming and ideation sessions
c) Sketching and prototyping
d) User testing and feedback analysis.
43. What is the role of empathy in design thinking in IT?
a) To understand the needs and challenges of IT users and stake holders.
b) To create visually appealing IT products and services.
c) To ensure compliance with industry standards and regulations.
d) To increase profits for the IT organization.

44. What is the importance of prototyping in design thinking in IT?
- a) To test and refine IT products and services before launch.
 - b) To show case IT capabilities to stake holders.
 - c) To impress user with cutting edge technology.
 - d) To ensure compliance with industry standards and regulations.
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 - d) To comply with industry regulations and standards.
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 - b) To show case the organizations capabilities to stakeholders.
 - c) To impress users with cutting edge technology.
 - d) To ensure compliance with industry standards and regulations.
47. What are some common challenges that can be addressed using design thinking in business process modeling?
- a) Inefficient processes that waste time and resources.
 - b) Poor communication and collaboration between departments.
 - c) Resistance to change from process users and stakeholders.
 - d) All of these
48. What is the goal of using agile methodology in design thinking?
- a) To create aesthetically pleasing designs
 - b) To increase efficiency and productivity in the design process
 - c) To reduce costs for the organization
 - d) To comply with Industry regulations and standards.
49. What is the importance of prototyping in agile design thinking?
- a) To quickly test and refine design ideas
 - b) To showcase design capabilities to stoke holders
 - c) To impress users with cutting edge technology
 - d) To ensure compliance with industry standards and regulations
50. What are some common challenges that can be addressed using agile design thinking?
- a) Inefficient design processes that waste time and resources.
 - b) Poor communication and collaboration between team members.
 - c) Resistance to change from users and stakeholders.
 - d) All of these

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Question Paper Version : C

**First/Second Semester B.E./B.Tech./B.Design Degree Examination,
June/July 2024**

Innovation and Design Thinking

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. What is the goal of using agile technology in virtual collaboration environments for design thinking?
 - a) To create aesthetically pleasing designs
 - b) To increase efficiency and productivity in the design process
 - c) To reduce costs for the organization
 - d) To comply with industry regulations and standards
 2. Why is scenario based prototyping important in innovation design thinking?
 - a) It allows for faster development of prototypes
 - b) It helps to identify potential usability issues before the product is released
 - c) It reduces the cost of creating prototypes
 - d) None of these
 3. What is the first step in scenario based prototyping?
 - a) Creating a detailed technical specification
 - b) identifying potential users and use cases
 - c) Building a physical mock up of the product
 - d) None of these
 4. What are the key stages of the design thinking process?
 - a) Ideation, testing, and implementation
 - b) Empathy, define, ideate, prototype and test
 - c) Planning, execution and evaluation
 - d) None of these
 5. What is storytelling in the context of design thinking?
 - a) It is the process of creating a narrative around a design solution.
 - b) It is the process of creating a story board to communicate design ideas.
 - c) It is the process of creating a visual representation of a design concept.
 - d) None of these

6. What are some common tools and techniques used in strategic foresight?
 - a) Scenario planning, trend analysis, and expert interviews
 - b) Prototyping, user testing and iterative design
 - c) Market research, customer feed back, and competitive analysis
 - d) None of these
7. What are some common tools and techniques used in sensemaking?
 - a) Empathy mapping, customer journey mapping and data visualization.
 - b) Prototyping, user testing and iterative design
 - c) Market research, customer feedback and competitive analysis
 - d) None of these
8. What is maintenance in the context of design thinking?
 - a) The process of repairing and upgrading existing design solutions.
 - b) The process of creating new design solutions from scratch.
 - c) The process of evaluating the effectiveness of existing design solutions.
 - d) None of these
9. What is value redefinition in the context of design thinking?
 - a) The process of identifying and redefining the value proposition of a design solution.
 - b) The process of creating a detailed technical specification for a design solution.
 - c) The process of evaluating the effectiveness of existing design solutions.
 - d) None of these
10. What is extreme competition in the context of design thinking?
 - a) The process of competing against other design teams to create the best solution.
 - b) The process of pushing design teams to their limits to create innovative solutions.
 - c) The process of collaborating with competitors to create a joint solution.
 - d) None of these
11. Design thinking is _____
 - a) A process that allows engineers and designers to create new and innovative solutions to business challenges.
 - b) An iterative, non-linear process which focuses on a collaboration between designers and users
 - c) a and b
 - d) None of these
12. Select the correct order of the different stages of design thinking.

i) Empathize	ii) Define	iii) Ideate	iv) Prototype	v) Test
a) ii – iii – iv – v – i	b) i – ii – iii – iv – v			
c) iii – iv – v – i – ii	d) iv – v – i – ii – iii			
13. Which of the following principles are not considered for design thinking?

a) Embrace experimentation	b) Human centric design
c) Profit centric	d) Pattern identification for problem solving
14. To empathize one has to

a) Observe	b) Engage	c) Listen	d) All of these
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15. Which of the following are not tools of visualization?

a) Maps	b) Images	c) Stories	d) Videos
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16. Journey mapping is also called _____ mapping
 a) Path b) Experience c) Conduct d) Feedback
17. Which of the following are not tools of design thinking?
 a) Co creation b) Proto typing c) Mind mapping d) Online mapping
18. Which design thinking phase is closely related to the creation of an minimum viable product (MVP)?
 a) Empathize b) Prototype c) Ideate d) Test
19. Which one is the minimum viable product?
 a)  b) 
 c) Both a and b d) None of these
20. How can the theory and practice of design thinking work together?
 a) Theory provides a set of rules to follow, while practice ensures the rules are applied effectively.
 b) Theory and practice are two separate processes that do not overlap.
 c) Theory provides a framework for creative problem solving, while practice develops the skills and knowledge to apply the framework effectively.
 d) Theory and practice are interchangeable and can be used in any order.
21. What is experience design in the context of design thinking?
 a) The process of designing physical products and services.
 b) The process of designing digital interfaces and interactions.
 c) The process of designing holistic experiences for users across all touch points.
 d) None of these
22. What is standardization in the context of design thinking?
 a) The process of creating standardized design solutions.
 b) The process of following a standard set of design principles.
 c) The process of establishing standards for design processes and methodologies.
 d) None of these
23. What is humanization in the context of design thinking?
 a) The process of making designs more human centered and empathetic.
 b) The process of making designs more technically advanced.
 c) The process of making designs more aesthetically pleasing.
 d) None of these
24. How can a creative culture be fastened in the design thinking process?
 a) By encouraging risk-taking and experimentation
 b) By establishing rigid design processes and guide lines
 c) By prioritizing cost effectiveness over innovation
 d) None of these

25. What are some common tools and techniques used in rapid prototyping?
- a) Sketching, wire framing and paper prototyping
 - b) 3D printing, laser cutting and CNC machining
 - c) User research, market analysis and competitive analysis
 - d) None of these
26. How can a business model be designed using design thinking?
- a) By identifying customer needs and pain points and designing solutions that address them.
 - b) By following a set of established business practices and industry standards.
 - c) By prioritizing cost effective over customer value.
 - d) None of these
27. Which phase is referred to as an experimental phase where continuous iterations can take place?
- a) Define
 - b) Empathise
 - c) Prototype
 - d) None of these
28. What is the term used to describe the process of narrowing down thoughts to reach the final solution?
- a) Convergent thinking
 - b) Divergent thinking
 - c) None of these
 - d) Both a and b
29. Design thinking is typically used to provide a solution based approach to problem solving
- a) True
 - b) False
30. Can design thinking be applied in professions outside of design?
- a) True
 - b) False
31. How can real time interaction and analysis be integrated into the design process?
- a) By conducting user research and testing at the end of the design process only.
 - b) By eliminating user feedback and relying solely on personal preferences.
 - c) By analyzing market trends and sales data only.
 - d) By incorporating user feedback throughout the entire design process.
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- a) Persona development and user interviews
 - b) Brain storming and ideation sessions
 - c) Sketching and prototyping
 - d) User testing and feedback analysis.
33. What is the role of empathy in design thinking in IT?
- a) To understand the needs and challenges of IT users and stake holders.
 - b) To create visually appealing IT products and services.
 - c) To ensure compliance with industry standards and regulations.
 - d) To increase profits for the IT organization.
34. What is the importance of prototyping in design thinking in IT?
- a) To test and refine IT products and services before launch.
 - b) To show case IT capabilities to stake holders.
 - c) To impress user with cutting edge technology.
 - d) To ensure compliance with industry standards and regulations.

35. What is the goal of using design thinking in business process modeling?
- a) To create visually appealing process diagrams.
 - b) To stream line business operations and increase efficiency.
 - c) To reduce costs for the organization.
 - d) To comply with industry regulations and standards.
36. What is the importance of prototyping in design thinking in business process modeling?
- a) To test and refine process models before implementation.
 - b) To show case the organizations capabilities to stakeholders.
 - c) To impress users with cutting edge technology.
 - d) To ensure compliance with industry standards and regulations.
37. What are some common challenges that can be addressed using design thinking in business process modeling?
- a) Inefficient processes that waste time and resources.
 - b) Poor communication and collaboration between departments.
 - c) Resistance to change from process users and stakeholders.
 - d) All of these
38. What is the goal of using agile methodology in design thinking?
- a) To create aesthetically pleasing designs
 - b) To increase efficiency and productivity in the design process
 - c) To reduce costs for the organization
 - d) To comply with Industry regulations and standards.
39. What is the importance of prototyping in agile design thinking?
- a) To quickly test and refine design ideas
 - b) To showcase design capabilities to stoke holders
 - c) To impress users with cutting edge technology
 - d) To ensure compliance with industry standards and regulations
40. What are some common challenges that can be addressed using agile design thinking?
- a) Inefficient design processes that waste time and resources.
 - b) Poor communication and collaboration between team members.
 - c) Resistance to change from users and stakeholders.
 - d) All of these
41. What is real-time design interaction capture and analysis?
- a) A process of capturing user feedback after the design process is complete.
 - b) A method of analyzing design interactions as they happen.
 - c) A technique for capturing user interactions with a product after it is released.
 - d) A way to analyze design interactions after they have occurred.
42. What types of interactions can be captured and analyzed in real-time design interaction capture and analysis?
- a) User feed back and comments
 - b) User interface interactions
 - c) User demographic and preferences
 - d) User purchase behavior.
43. What is the importance of collaboration in design thinking?
- a) It speeds up the design process
 - b) It eliminates the need for user research
 - c) It ensures all design decisions are final
 - d) It brings diverse perspectives and expertise to the design process.

44. What are some digital tools that can enable efficient collaboration in design thinking?
- a) E Mail
 - b) Social media platforms
 - c) Video conferencing and collaboration software
 - d) Design software
45. What is empathy in design?
- a) Understanding the emotions and experiences of users
 - b) Creating products that appeal to users emotions
 - c) Limiting the number of user persons in the design process
 - d) Only designing products based on market research.
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- a) It ensures all design decisions are final
 - b) It brings perspectives and expertise to the design process
 - c) It speeds up the design process
 - d) It results in product that meet users needs and desires.
47. What is the benefit of empathy in design?
- a) It guarantees success in the market
 - b) It speeds up the design process
 - c) It ensures all design decisions are final
 - d) It leads to more meaningful and impactful products.
48. What is distributed design collaboration?
- a) Collaboration between designers who are geographically separated.
 - b) Collaboration between designers and machines.
 - c) Collaboration between designers and machines
 - d) Collaboration between designers from different industries.
49. What are some challenges of distributed design collaboration?
- a) Limited access to digital tools
 - b) Technical difficulties and connectivity issues
 - c) A lack of clear communication channels
 - d) A limited pool of expertise and perspectives
50. What are some benefits of real-time interaction and analysis in design thinking?
- a) It allows for quick and agile decision making
 - b) It limits the number of team members involved in the design process
 - c) It eliminates the need for user research and testing
 - d) It reduces the amount of feed back received from users.

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Question Paper Version : D

**First/Second Semester B.E./B.Tech./B.Design Degree Examination,
June/July 2024**

Innovation and Design Thinking

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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 5. What is the goal of using design thinking in business process modeling?
 - a) To create visually appealing process diagrams.
 - b) To stream line business operations and increase efficiency.
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 d) None of these
42. Select the correct order of the different stages of design thinking.
 i) Empathize ii) Define iii) Ideate iv) Prototype v) Test
 a) ii – iii – iv – v – i b) i – ii – iii – iv – v
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43. Which of the following principles are not considered for design thinking?
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 c) Profit centric d) Pattern identification for problem solving
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 a) Observe b) Engage c) Listen d) All of these
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 a) Empathize b) Prototype c) Ideate d) Test
49. Which one is the minimum viable product?
 a)  b) 
 c) Both a and b d) None of these
50. How can the theory and practice of design thinking work together?
 a) Theory provides a set of rules to follow, while practice ensures the rules are applied effectively.
 b) Theory and practice are two separate processes that do not overlap.
 c) Theory provides a framework for creative problem solving, while practice develops the skills and knowledge to apply the framework effectively.
 d) Theory and practice are interchangeable and can be used in any order.

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CBCS SCHEME

BKBKK107/207

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Question Paper Version : A

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024

Balake Kannada (COMMON TO ALL BRANCHES)

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
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3. For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Write the Kannada vocabulary for the following English words.

1. Who _____
a) Enu b) Eke c) Yaaru d) Elli
2. This _____
a) Idu b) Adu c) Avu d) Iyu
3. She _____
a) avanu b) avaLu c) avaru d) Ivaru
4. His _____
a) avariage b) avaLu c) avana d) Nanage
5. When _____
a) Yaaru b) Hege c) Estu d) Yavaga

Fill in the blank for the given English words into Kannada meaning.

6. Long _____
a) Chikka b) Dappa c) agala d) Udda
7. New _____
a) gaTTi b) Hosa c) HaLeya d) Taaja
8. Hard _____
a) SaNNA b) gaTTi c) hagura d) dubaari
9. Salty _____
a) Uppu b) huLi c) Khaara d) Sihi
10. Dirty _____
a) Subhra b) BiLi c) ONa d) galiiju

Translate the following English sentence into Kannada sentence.

11. Who are you?
a) Naanu Yaaru b) avaLu Yaaru c) niinu Yaaru d) adu Yaaru
12. What is your name?
a) Ninna Hesaru Enu? b) Nanna Hesaru Enu
c) Idara Hesaru Enu? d) AvaLa Hesaru Enu
13. Where is your House?
a) Avana Mane Elli Ide? b) Ninna Mane Elli Ide?
c) AvaLa Mane Elli Ide? d) Adara Mane Elli Ide?
14. Who is he?
a) Adu Yaaru? b) Avanu Yaaru? c) Idu Yaaru? d) Idu Elli?
15. Where is your younger sister?
a) Ninna Tamma Elli Iddale? b) Ninna Akka Elli Iddale?
c) Ninna Tangi Elli Iddale? d) Ninna Anna Elli Iddale?

Match the following using the Table given below?

a)	Student	i)	Vidyalaya
b)	Younger brother	ii)	Vidyarthi
c)	Teacher	iii)	Vaidya
d)	Doctor	iv)	Tamma
e)	College	v)	Shikshaka

16. Student _____
a) = i b) = ii c) = iii d) = iv
17. Younger Brother _____
a) = i b) = ii c) = iii d) = iv
18. Teacher _____
a) = i b) = ii c) = iv d) = v
19. Doctor _____
a) = i b) = ii c) = iii d) = iv
20. College _____
a) = i b) = ii c) = iii d) = iv

Write the English Word for the following :

21. Mane _____
a) Hotel b) House c) Shop d) Street
22. Mara _____
a) Tree b) Leaf c) Flower d) Fruit
23. Maga _____
a) Father b) Mother c) Uncle d) Son
24. Amma _____
a) Younger Brother b) Elder Brother c) Mother d) Elder Sister
25. HaNNu _____
a) Fruit b) Flower c) Seed d) Plant

Transform the following Kannada words as per the given model (example) :

Example : angaDi – angaDiyalli

26. Kacheri _____
a) Maneyalli b) Kacheriyalli c) Shaleyalli d) Halliyalli
27. Batte _____
a) Batteyalli b) Angiyalli c) BaNNadalli d) Batteyinda
28. How would you write “his Mother” in Kannada?
a) Avana Tande b) Avana Tamma c) Avana Taayi d) Avana Tangi
29. How would you write “This is My College” in Kannada?
a) Adu Nmma Vidyalaya
b) Idu Nanna Vidyalaya
c) Adu Avara Vidyalaya
d) Idu Ivana Vidyalaya
30. How would you write “Who is She?” in Kannada?
a) AvaLu Yaaru? b) Avanu Yaaru? c) Idu Yaaru d) Ivaru Yaaru?
31. What is the meaning of “Don’t – go”?
a) Bara Beda b) Hoga Beda c) Tinna Beda d) Kudiya Beda
32. What is the meaning of “My Book” in Kannada?
a) Nanna Mitra b) Nanna Pustaka c) Nanna Angi d) Nama Mane
33. What is the meaning of “Flower” in Kannada?
a) HaNNu b) Hoovu c) Ele d) Mane
34. Which one of the following means the color “Red” in Kannada?
a) Kempu b) BiLi c) NeeLi d) HaLadi

Write appropriate words for the following :

35. Where _____
a) ELLi b) Yaake c) Yaaru d) Estu
36. Teacher _____
a) GeLathi b) Shikshaka c) Vidyarti d) Huduga
37. Girl _____
a) Huduga b) Mitra c) Hudugi d) Snehita
38. Bitter _____
a) Sihi b) Uppu c) Kahi d) Khara
39. Library _____
a) Shale b) Vidyakaya c) AngaDi d) Granthalaya

Match the following using the table given below :

a)	Green	i)	Huduga
b)	Fruit	ii)	Sihi
c)	Boy	iii)	HaNNu
d)	Son	iv)	Hasiru
e)	Sweet	v)	Maga

40. Green _____
a) = i b) = ii c) = iii d) = iv
41. Fruit _____
a) = i b) = ii c) = iii d) = v
42. Boy _____
a) = i b) = ii c) = iii d) = iv
43. Son _____
a) = i b) = ii c) = iv d) = v
44. Sweet _____
a) = i b) = ii c) = iii d) = v

Translate the following English words into Kannada.

45. Book _____
a) Buk b) Pustaka c) Byag d) Pencil
46. Her House _____
a) IvaLa Mane b) AvaLa Mane c) Avana Mane d) Adara Mane
47. Big Tree _____
a) Chikka Mara b) doDDa Mara c) AgaLa Mara d) SaNNa Mara

Translate the following Kannada words into English.

48. MagaLu _____
a) Son b) Mother c) Daughter d) Sister
49. Kappu BaNNa _____
a) Black color b) Red color c) White color d) Green color
50. Obba Vidyarthi _____
a) One Teacher b) One Servant c) One Brother d) One Student.

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CBCS SCHEME

BKBBK107/207

USN

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Question Paper Version : B

First/Second Semester B.E. /B.Tech. Degree Examination, June/July 2024

Balake Kannada

(COMMON TO ALL BRANCHES)

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners on the OMR sheets are strictly prohibited.**

Translate the following English sentence into Kannada sentence.

- 1 Who are you?
a) Naanu Yaaru b) avaLu Yaaru c) niinu Yaaru d) adu Yaaru
- 2 What is your name?
a) Ninna Hesaru Enu? b) Nanna Hesaru Enu
c) Idara Hesaru Enu? d) AvaLa Hesaru Enu
- 3 Where is your House?
a) Avana Mane Elli Ide? b) Ninna Mane Elli Ide?
c) AvaLa Mane Elli Ide? d) Adara Mane Elli Ide?
- 4 Who is he?
a) Adu Yaaru? b) Avanu Yaaru? c) Idu Yaaru? d) Idu Elli?
- 5 Where is your younger sister?
a) Ninna Tamma Elli Iddale? b) Ninna Akka Elli Iddale?
c) Ninna Tangi Elli Iddale? d) Ninna Anna Elli Iddale?

Transform the following Kannada words as per the given model (example) :

Example : angaDi – angaDiyalli

- 6 Kacheri
a) Maneyalli b) Kacheriyalli c) Shaleyalli d) Halliyalli

- 7 Batte _____
a) Batteyalli b) Angiyalli c) BaNNadalli d) Batteyinda
- 8 How would you write "his Mother" in Kannada?
a) Avana Tande b) Avana Tamma c) Avana Taayi d) Avana Tangi
- 9 How would you write "This is My College" in Kannada?
a) Adu Nmma Vidyalaya
b) Idu Nanna Vidyalaya
c) Adu Avara Vidyalaya
d) Idu Ivana Vidyalaya
- 10 How would you write "Who is She?" in Kannada?
a) AvaLu Yaaru? b) Avanu Yaaru? c) Idu Yaaru d) Ivaru Yaaru?
- 11 What is the meaning of "Don't – go"?
a) Bara Beda b) Hoga Beda c) Tinna Beda d) Kudiya Beda
- 12 What is the meaning of "My Book" in Kannada?
a) Nanna Mitra b) Nanna Pustaka c) Nanna Angi d) Nama Mane
- 13 What is the meaning of "Flower" in Kannada?
a) HaNNu b) Hoovu c) Ele d) Mane
- 14 Which one of the following means the color "Red" in Kannada?
a) Kempu b) BiLi c) NeeLi d) HaLadi

Translate the following English words into Kannada.

- 15 Book _____
a) Buk b) Pustaka c) Byag d) Pencil
- 16 Her House
a) IvaLa Mane b) AvaLa Mane c) Avana Mane d) Adara Mane
- 17 Big Tree
a) Chikka Mara b) doDDa Mara c) AgaLa Mara d) SaNNa Mara

Fill in the blank for the given English words into Kannada meaning.

- 18 Long _____
a) Chikka b) Dappa c) agala d) Udda
- 19 New _____
a) gaTTi b) Hosa c) HaLeya d) Taaja
- 20 Hard _____
a) SaNNA b) gaTTi c) hagura d) dubaari
- 21 Salty _____
a) Uppu b) huLi c) Khaara d) Sihi
- 22 Dirty _____
a) Subhra b) BiLi c) ONa d) galiiju

Match the following using the table given below :

a)	Green	i)	Huduga
b)	Fruit	ii)	Sihi
c)	Boy	iii)	HaNNu
d)	Son	iv)	Hasiru
e)	Sweet	v)	Maga

- 23 Green _____
a) = i b) = ii c) = iii d) = iv
- 24 Fruit _____
a) = i b) = ii c) = iii d) = v
- 25 Boy _____
a) = i b) = ii c) = iii d) = iv
- 26 Son _____
a) = i b) = ii c) = iv d) = v
- 27 Sweet _____
a) = i b) = ii c) = iii d) = v

Write the Kannada vocabulary for the following English words.

- 28 Who _____
a) Enu b) Eke c) Yaaru d) Elli
- 29 This _____
a) Idu b) Adu c) Avu d) Ivu
- 30 She _____
a) avanu b) avaLu c) avaru d) Ivaru
- 31 His _____
a) avariage b) avaLu c) avana d) Nanage
- 32 When _____
a) Yaaru b) Hege c) Estu d) Yavaga

Translate the following Kannada words into English.

- 33 MagaLu
a) Son b) Mother c) Daughter d) Sister
- 34 Kappu BaNNa
a) Black color b) Red color c) White color d) Green color
- 35 Obba Vidyarthi
a) One Teacher b) One Servant c) One Brother d) One Student.

Write the English Word for the following :

- 36 Mane _____
a) Hotel _____ b) House _____ c) Shop _____ d) Street _____
- 37 Mara _____
a) Tree _____ b) Leaf _____ c) Flower _____ d) Fruit _____
- 38 Maga _____
a) Father _____ b) Mother _____ c) Uncle _____ d) Son _____
- 39 Amma _____
a) Younger Brother _____ b) Elder Brother _____ c) Mother _____ d) Elder Sister _____
- 40 HaNNu _____
a) Fruit _____ b) Flower _____ c) Seed _____ d) Plant _____

Match the following using the Table given below?

a)	Student	i)	Vidyalaya
b)	Younger brother	ii)	Vidyarthi
c)	Teacher	iii)	Vaidya
d)	Doctor	iv)	Tamma
e)	College	v)	Shikshaka

- 41 Student _____
a) = i _____ b) = ii _____ c) = iii _____ d) = iv _____
- 42 Younger Brother _____
a) = i _____ b) = ii _____ c) = iii _____ d) = iv _____
- 43 Teacher _____
a) = i _____ b) = ii _____ c) = iv _____ d) = v _____
- 44 Doctor _____
a) = i _____ b) = ii _____ c) = iii _____ d) = iv _____
- 45 College _____
a) = i _____ b) = ii _____ c) = iii _____ d) = iv _____

Write appropriate words for the following :

- 46 Where _____
a) ELLi _____ b) Yaake _____ c) Yaaru _____ d) Estu _____
- 47 Teacher _____
a) GeLathi _____ b) Shikshaka _____ c) Vidyarti _____ d) Huduga _____
- 48 Girl _____
a) Huduga _____ b) Mitra _____ c) Hudugi _____ d) Snehita _____
- 49 Bitter _____
a) Sihi _____ b) Uppu _____ c) Kahi _____ d) Khara _____
- 50 Library _____
a) Shale _____ b) Vidyakaya _____ c) AngaDi _____ d) Granthalaya _____

CBCS SCHEME

BKBKK107/207

USN

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Question Paper Version : C

First/Second Semester B.E. /B.Tech. Degree Examination, June/July 2024

Balake Kannada

(COMMON TO ALL BRANCHES)

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners on the OMR sheets are strictly prohibited.**

Translate the following Kannada words into English.

- 1 MagaLu
a) Son b) Mother c) Daughter d) Sister
- 2 Kappu BaNNa
a) Black color b) Red color c) White color d) Green color
- 3 Obba Vidyarthi
a) One Teacher b) One Servant c) One Brother d) One Student.

Translate the following English words into Kannada.

- 4 Book _____
a) Buk b) Pustaka c) Byag d) Pencil
- 5 Her House
a) IvaLa Mane b) AvaLa Mane c) Avana Mane d) Adara Mane
- 6 Big Tree
a) Chikka Mara b) doDDa Mara c) AgaLa Mara d) SaNNa Mara

Match the following using the table given below :

a)	Green	i)	Huduga
b)	Fruit	ii)	Sihi
c)	Boy	iii)	HaNNu
d)	Son	iv)	Hasiru
e)	Sweet	v)	Maga

- 7 Green _____
a) = i b) = ii c) = iii d) = iv
- 8 Fruit _____
a) = i b) = ii c) = iii d) = v
- 9 Boy _____
a) = i b) = ii c) = iii d) = iv
- 10 Son _____
a) = i b) = ii c) = iv d) = v
- 11 Sweet _____
a) = i b) = ii c) = iii d) = v

Write appropriate words for the following :

- 12 Where _____
a) ELLi b) Yaake c) Yaaru d) Estu
- 13 Teacher _____
a) GeLathi b) Shikshaka c) Vidyarti d) Huduga
- 14 Girl _____
a) Huduga b) Mitra c) Hudugi d) Snehita
- 15 Bitter _____
a) Sihi b) Uppu c) Kahi d) Khara
- 16 Library _____
a) Shale b) Vidyakaya c) AngaDi d) Granthalaya

Transform the following Kannada words as per the given model (example) :

Example : angaDi – angaDiyalli

- 17 Kacheri _____
a) Maneyalli b) Kacheriyalli c) Shaleyalli d) Halliyalli
- 18 Batte _____
a) Batteyalli b) Angiyalli c) BaNNadalli d) Batteyinda

- 19 How would you write "his Mother" in Kannada?
a) Avana Tande b) Avana Tamma c) Avana Taayi d) Avana Tangi
- 20 How would you write "This is My College" in Kannada?
a) Adu Nmna Vidyalaya b) Idu Nanna Vidyalaya
c) Adu Avara Vidyalaya d) Idu Ivana Vidyalaya
- 21 How would you write "Who is She?" in Kannada?
a) AvaLu Yaaru? b) Avanu Yaaru? c) Idu Yaaru d) Ivaru Yaaru?
- 22 What is the meaning of "Don't – go"?
a) Bara Beda b) Hoga Beda c) Tinna Beda d) Kudiya Beda
- 23 What is the meaning of "My Book" in Kannada?
a) Nanna Mitra b) Nanna Pustaka c) Nanna Angi d) Nama Mane
- 24 What is the meaning of "Flower" in Kannada?
a) HaNNu b) Hoovu c) Ele d) Mane
- 25 Which one of the following means the color "Red" in Kannada?
a) Kempu b) BiLi c) NeeLi d) HaLadi

Write the English Word for the following :

- 26 Mane _____
a) Hotel b) House c) Shop d) Street
- 27 Mara _____
a) Tree b) Leaf c) Flower d) Fruit
- 28 Maga _____
a) Father b) Mother c) Uncle d) Son
- 29 Amma _____
a) Younger Brother b) Elder Brother c) Mother d) Elder Sister
- 30 HaNNu _____
a) Fruit b) Flower c) Seed d) Plant

Match the following using the Table given below?

a)	Student	i)	Vidyalaya
b)	Younger brother	ii)	Vidyarthi
c)	Teacher	iii)	Vaidya
d)	Doctor	iv)	Tamma
e)	College	v)	Shikshaka

- 31 Student _____
a) = i b) = ii c) = iii d) = iv
- 32 Younger Brother _____
a) = i b) = ii c) = iii d) = iv
- 33 Teacher _____
a) = i b) = ii c) = iv d) = v

- 34 Doctor _____
a) = i b) = ii c) = iii d) = iv
- 35 College _____
a) = i b) = ii c) = iii d) = iv

Translate the following English sentence into Kannada sentence.

- 36 Who are you?
a) Naanu Yaaru b) avaLu Yaaru c) niinu Yaaru d) adu Yaaru
- 37 What is your name?
a) Ninna Hesaru Enu? b) Nanna Hesaru Enu
c) Idara Hesaru Enu? d) AvaLa Hesaru Enu
- 38 Where is your House?
a) Avana Mane Elli Ide? b) Ninna Mane Elli Ide?
c) AvaLa Mane Elli Ide? d) Adara Mane Elli Ide?
- 39 Who is he?
a) Adu Yaaru? b) Avanu Yaaru? c) Idu Yaaru? d) Idu Elli?
- 40 Where is your younger sister?
a) Ninna Tamma Elli Iddale? b) Ninna Akka Elli Iddale?
c) Ninna Tangi Elli Iddale? d) Ninna Anna Elli Iddale?

Fill in the blank for the given English words into Kannada meaning.

- 41 Long _____
a) Chikka b) Dappa c) agala d) Udda
- 42 New _____
a) gaTTi b) Hosa c) HaLeya d) Taaja
- 43 Hard _____
a) SaNNA b) gaTTi c) hagura d) dubaari
- 44 Salty _____
a) Uppu b) huLi c) Khaara d) Sihi
- 45 Dirty _____
a) Subhra b) BiLi c) ONa d) galiiju

Write the Kannada vocabulary for the following English words.

- 46 Who _____
a) Enu b) Eke c) Yaaru d) Elli
- 47 This _____
a) Idu b) Adu c) Avu d) Ivu
- 48 She _____
a) avanu b) avaLu c) avaru d) Ivaru
- 49 His _____
a) avariage b) avaLu c) avana d) Nanage
- 50 When _____
a) Yaaru b) Hege c) Estu d) Yavaga

CBGS SCHEME

BKBBK107/207

USN

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Question Paper Version : D

First/Second Semester B.E. /B.Tech. Degree Examination, June/July 2024

Balake Kannada

(COMMON TO ALL BRANCHES)

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Fill in the blank for the given English words into Kannada meaning.

- 1 Long _____
a) Chikka b) Dappa c) agala d) Udda
- 2 New _____
a) gaTTi b) Hosa c) HaLeya d) Taaja
- 3 Hard _____
a) SaNNA b) gaTTi c) hagura d) dubaari
- 4 Salty _____
a) Uppu b) huLi c) Khaara d) Sihi
- 5 Dirty _____
a) Subhra b) BiLi c) ONa d) galiiju

Match the following using the Table given below?

a)	Student	i)	Vidyalaya
b)	Younger brother	ii)	Vidyarthi
c)	Teacher	iii)	Vaidya
d)	Doctor	iv)	Tamma
e)	College	v)	Shikshaka

- 6 Student _____
a) = i b) = ii c) = iii d) = iv
- 7 Younger Brother _____
a) = i b) = ii c) = iii d) = iv

8 Teacher _____

a) = i

b) = ii

c) = iv

d) = v

9 Doctor _____

a) = i

b) = ii

c) = iii

d) = iv

10 College _____

a) = i

b) = ii

c) = iii

d) = iv

Transform the following Kannada words as per the given model (example) :

Example : angaDi – angaDiyalli

11 Kacheri _____

a) Maneyalli

b) Kacheriyalli

c) Shaleyalli

d) Halliyalli

12 Batte _____

a) Batteyalli

b) Angiyalli

c) BaNNadalli

d) Batteyinda

13 How would you write “his Mother” in Kannada?

a) Avana Tande

b) Avana Tamma

c) Avana Taayi

d) Avana Tangi

14 How would you write “This is My College” in Kannada?

a) Adu Nmna Vidyalya

b) Idu Nanna Vidyalya

c) Adu Avara Vidyalya

d) Idu Ivana Vidyalya

15 How would you write “Who is She?” in Kannada?

a) AvaLu Yaaru?

b) Avanu Yaaru?

c) Idu Yaaru

d) Ivaru Yaaru?

16 What is the meaning of “Don’t – go”?

a) Bara Beda

b) Hoga Beda

c) Tinna Beda

d) Kudiya Beda

17 What is the meaning of “My Book” in Kannada?

a) Nanna Mitra

b) Nanna Pustaka

c) Nanna Angi

d) Nama Mane

18 What is the meaning of “Flower” in Kannada?

a) HaNNu

b) Hoovu

c) Ele

d) Mane

19 Which one of the following means the color “Red” in Kannada?

a) Kempu

b) BiLi

c) NeeLi

d) HaLadi

Match the following using the table given below :

a)	Green	i)	Huduga
b)	Fruit	ii)	Sihi
c)	Boy	iii)	HaNNu
d)	Son	iv)	Hasiru
e)	Sweet	v)	Maga

20 Green _____

a) = i

b) = ii

c) = iii

d) = iv

21 Fruit _____

a) = i

b) = ii

c) = iii

d) = v

- 22 Boy _____
a) = i b) = ii c) = iii d) = iv
- 23 Son _____
a) = i b) = ii c) = iv d) = v
- 24 Sweet _____
a) = i b) = ii c) = iii d) = v

Translate the following Kannada words into English.

- 25 MagaLu
a) Son b) Mother c) Daughter d) Sister
- 26 Kappu BaNNa
a) Black color b) Red color c) White color d) Green color
- 27 Obba Vidyarthi
a) One Teacher b) One Servant c) One Brother d) One Student.

Translate the following English words into Kannada.

- 28 Book _____
a) Buk b) Pustaka c) Byag d) Pencil
- 29 Her House
a) IvaLa Mane b) AvaLa Mane c) Avana Mane d) Adara Mane
- 30 Big Tree
a) Chikka Mara b) doDDa Mara c) AgaLa Mara d) SaNNa Mara

Write the Kannada vocabulary for the following English words.

- 31 Who _____
a) Enu b) Eke c) Yaaru d) Elli
- 32 This _____
a) Idu b) Adu c) Avu d) Ivu
- 33 She _____
a) avanu b) avaLu c) avaru d) Ivaru
- 34 His _____
a) avariage b) avaLu c) avana d) Nanage
- 35 When _____
a) Yaaru b) Hege c) Estu d) Yavaga

Write appropriate words for the following :

- 36 Where _____
a) ELLi b) Yaake c) Yaaru d) Estu
- 37 Teacher _____
a) GeLathi b) Shikshaka c) Vidyarti d) Huduga
- 38 Girl _____
a) Huduga b) Mitra c) Hudugi d) Snehita
- 39 Bitter _____
a) Sihi b) Uppu c) Kahi d) Khara
- 40 Library _____
a) Shale b) Vidyakaya c) AngaDi d) Granthalaya

Write the English Word for the following :

- 41 Mane _____
a) Hotel b) House c) Shop d) Street
- 42 Mara _____
a) Tree b) Leaf c) Flower d) Fruit
- 43 Maga _____
a) Father b) Mother c) Uncle d) Son
- 44 Amma _____
a) Younger Brother b) Elder Brother c) Mother d) Elder Sister
- 45 HaNNu _____
a) Fruit b) Flower c) Seed d) Plant

Translate the following English sentence into Kannada sentence.

- 46 Who are you?
a) Naanu Yaaru b) avaLu Yaaru c) niinu Yaaru d) adu Yaaru
- 47 What is your name?
a) Ninna Hesaru Enu? b) Nanna Hesaru Enu
c) Idara Hesaru Enu? d) AvaLa Hesaru Enu
- 48 Where is your House?
a) Avana Mane Elli Ide? b) Ninna Mane Elli Ide?
c) AvaLa Mane Elli Ide? d) Adara Mane Elli Ide?
- 49 Who is he?
a) Adu Yaaru? b) Avanu Yaaru? c) Idu Yaaru? d) Idu Elli?
- 50 Where is your younger sister?
a) Ninna Tamma Elli Iddale? b) Ninna Akka Elli Iddale?
c) Ninna Tangi Elli Iddale? d) Ninna Anna Elli Iddale?

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First/Second Semester B.E/B.Tech. Degree Examination, June/July 2024

ಸಾಂಸ್ಕೃತಿಕ ಕನ್ನಡ

(COMMON TO ALL BRANCHES)

Time: 1 hrs.]

[Max. Marks: 50

ಸೂಚನೆಗಳು

1. ಎಲ್ಲ ಜಿಂ ಪ್ರಶ್ನೆಗಳಿಗೂ ಉತ್ತರಿಸಿರಿ. ಪ್ರತಿ ಪ್ರಶ್ನೆಗೆ ಒಂದು ಅಂಕ.
2. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಯು.ಎಸ್.ಎನ್ ಸಂಖ್ಯೆ ಹಾಗೂ ಪಶ್ಚಿಮ ಪತ್ರಿಕೆಯ ಶ್ರೇಣಿಯನ್ನು ಅಂದರೆ A, B, C ಅಥವಾ D ಯನ್ನು ತಪ್ಪಿಲ್ಲದಂತೆ ಕಡ್ಡಾಯವಾಗಿ ಗುರುತಿಸುವುದು ಅಭ್ಯರ್ಥಿಯ ಜವಾಬ್ದಾರಿಯಾಗಿರುತ್ತದೆ.
3. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ನಿಗದಿಪಡಿಸಿರುವ ಸ್ಥಳದಲ್ಲಿ ಭರ್ತಿಮಾಡದೆ ಹಾಗೆಯೇ ಬಿಟ್ಟಲ್ಲಿ ಅಥವಾ ಭರ್ತಿಮಾಡಿದ ಮಾಹಿತಿಯಲ್ಲಿ ಯಾವುದೇ ವ್ಯತ್ಯಾಸವಿದ್ದಲ್ಲಿ ಅಂತಹ ಉತ್ತರ ಪತ್ರಿಕೆಗಳನ್ನು ರದ್ದು ಪಡಿಸಲಾಗುವುದು.
4. ಕೇವಲ ಒಂದು ಉತ್ತರವನ್ನು ಮಾತ್ರ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಗುರುತಿಸತಕ್ಕದ್ದು. ಒಂದೆ ಪ್ರಶ್ನೆಗೆ ಎರಡು ಉತ್ತರವನ್ನು ಗುರುತಿಸುವುದು ಅಮಾನ್ಯ.
5. ಎಲ್ಲಾ ಉತ್ತರಗಳನ್ನು ನಿಮಗೆ ಒದಗಿಸಲಾದ ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯ ಹಾಳೆಯ ಮೇಲೆ ಕಪ್ಪು ಅಥವಾ ನೀಲಿ ಶಾಹಿಯ ಬಾಲ್‌ಪಾಯಿಂಟ್ ಪೆನ್ನಿನಿಂದ ಗುರುತು ಮಾಡಬೇಕು.

1. ತಾಳಗುಪ್ಪ ಗಿರಿಜನ ಶ್ರಮ ಶಾಲೆಯ ಮಾಸ್ತರರು ಯಾರು?
a) ಸಮಪಾಲುಗೋಪಣ್ಣ b) ಪ್ರಹ್ಲಾದ c) ಹುಚ್ಚಪ್ಪ d) ಕರೀಂಖಾನ
2. ಗೋಪಣ್ಣ ಮಾಸ್ತರು ಕಣಗಲಿ ಗಿಡದಲ್ಲಿ ಏನು ನೋಡಿದರು?
a) ಹೂ b) ಹಣ್ಣು c) ಬಟ್ಟೆ d) ಜನಿವಾರ
3. ಹಾಡುವಳ್ಳಿಗೆ ಹಿಂದೆ -----ಹೆಸರಿತ್ತು.
a) ಸಂಗೀತಪುರ b) ಭಟ್ಟಳ c) ಸಾಗರ d) ಕಾರವಾರ
4. ಹೊಸಚಿಗುರು ----- ಕೂಡಿರಲು ಮರಸೊಬಗು
a) ಹೊಸ ಎಲೆ b) ಹೊಸ ಹಸಿರು c) ಹೊಸಕಾಂಡ d) ಹಳಬೇರು
5. “ಪರಿಸರ, ಪರಂಪರೆ ಹಾಗೂ ಪ್ರತ್ಯಕ್ಷ ಜೀವನದ ಅನೇಕ ಸಂಸ್ಕಾರಗಳ ಮೂಲಕ ಮನುಷ್ಯನ ಅಂತರಂಗ ಪಡೆಯುವ ಪರಿಪಕ್ವತೆ” ಯನ್ನು ----- ಎನ್ನಬಹುದು.
a) ಕೆಲೆ b) ಸಂಸ್ಕಾರ c) ಸಂಸ್ಕೃತಿ d) ಕೆಂಪು

6. ಕುಣಬಿ ಜನರ ಮುಖ್ಯವಾದ ಹಬ್ಬ ಯಾವುದು?
a) ಯುಗಾದಿ b) ದಸರಾ c) ಹೋಳಿ ಹುಣ್ಣಿಮೆ d) ದೀಪಾವಳಿ
7. ಬಣ್ಣಗಳನ್ನು ನೀಡುವ ಗಿಡಗಳು ಭಾರತದಲ್ಲಿ ಎಷ್ಟವೆ.
a) 400 b) 300 c) 500 d) 600.
8. ಕರಕುಶಲ ಕಲೆಗಳಿಗೆ ಪೆಟ್ಟು ಬಿದ್ದಿದ್ದು ಯಾವುದರಿಂದ?
a) ಜಾಗತೀಕರಣ b) ಖಾಸಗೀಕರಣ
c) ಕೈಗಾರಿಕರಣ d) ಸೋಂಪೇರಿತನ.
9. ಸತಿ ಪದದ ಅರ್ಥವೇನು?
a) ಹೆಂಡತಿ b) ಪತಿ c) ಮಕ್ಕಳು d) ತಾಯಿ.
10. ವಿದ್ಯಾವರ್ಧಕ ಸಂಘ ಎಲ್ಲಿ ಸ್ಥಾಪನೆಯಾಯಿತು?
a) ಬೆಂಗಳೂರು b) ಧಾರವಾಡ
c) ಶಿವಮೊಗ್ಗ d) ಮೈಸೂರು.
11. ಕಾಲಿಗೆ ಬಿದ್ದವರ ಯಾವುದು ತುಳಿಯುತ್ತಲಿತ್ತು?
a) ಹಣ b) ಕಾಂಚಾನ c) ಬಂಗಾರ d) ಕುರುಡ ಕಾಂಚಾಣ.
12. ಭಾಷೆಗೆ ಎಷ್ಟು ಪ್ರಮುಖ ಕೌಶಲ್ಯಗಳು ಇರುತ್ತವೆ?
a) 10 b) 20 c) 08 d) 04
13. ಆಡಳಿತ ಭಾಷೆ ಯಾವಾಗಲೂ ----- ಇರಬೇಕು.
a) ಕಷ್ಟಕರವಾಗಿ b) ಅರ್ಥವಾಗದಂತೆ
c) ಸರಳ ಮತ್ತು ಸ್ಪಷ್ಟವಾಗಿ d) ಗೊಂದಲದಿಂದ.
14. ಸರ್ಕಾರಿ ಆಸ್ಪತ್ರೆಯಲ್ಲಿ ಮಾಸ್ತರಿಗೆ ಸಹಾಯ ಮಾಡಿದ ದಾದಿ ಯಾರು?
a) ಚಾಂದಿನಿ b) ರಾಧ c) ಫಾತಿಮಾ d) ರುಕ್ಮಿಣಿ.
15. ----- ಭಾರತದ ಕೆಲವು ನೇಕಾರರು ಚಿಂದಿ ಬಟ್ಟೆಯ ಕಂಬಳಿಗಳನ್ನು ತಮ್ಮ ಮಗ್ಗಗಳಲ್ಲಿ ನೇಯುವರು.
a) ಉತ್ತರ b) ಪೂರ್ವ c) ಪಶ್ಚಿಮ d) ದಕ್ಷಿಣ.
16. ಎಲ್ಲರನು ಸಲಹುವವನು ಯಾರು?
a) ಪುರಂದರ ವಿಠಲ b) ಕಾಗಿನೆಲೆ ಆದಿಕೇಶವ
c) ಮಲ್ಲಿಕಾರ್ಜುನ d) ಕೂಡಲಸಂಗಮದೇವ.

17. ಶ್ರೀಮಂತರು ಯಾರ ಕಣ್ಣಿನ ಕವಡೆಯನ್ನು ಜೋಮಾಲೆ ಮಾಡಿಕೊಂಡು ಹಾಕಿಕೊಳ್ಳುವರು?
a) ಬಾಣಂತಿ b) ನಣ್ಣ ಕಂದಮ್ಮ c) ಮುದುಕರು d) ಹೆಂಗಸರು.
18. ಯಾವ ಜನಾಂಗದ ಚಿತ್ರೀಕರಣ ಲೇಖಕರ ಮನದಲ್ಲಿತ್ತು?
a) ಗೊಂಡ b) ತೊಡವ c) ನಾಗ d) ಅಲೆಮಾರಿ.
19. ಕುಂಬಾರಕಿ ಯಾವುದರ ಮೇಲೆ ಮುಗಿಯನ್ನು ಇಡುವಳು?
a) ಕಲ್ಲು b) ಗಡಿಗೆ c) ವಸುಧೆ d) ಕಟ್ಟೆ.
20. ವಿಷನರ್ವಳಿಗೆ ಯಾವುದರ ಗೂಡು ಹುತ್ತವಾಗುವುದು?
a) ಗೆದ್ದಲಿರುವೆ b) ಉಡ c) ಇಲಿ d) ಗೀಜಗ.
21. ಎಲ್ಲರೊಳಗೆ ನಾವು ಏನಾಗಬೇರೆಂದು ಡ.ವಿ.ಜಿ ತಿಳಿಸಿದ್ದಾರೆ?
a) ಬೇರೆಯಾಗಬೇಕು b) ಒಂದಾಗಬೇಕು
c) ದೂರವಾಗಬೇಕು d) ಮಾತಾಡಬೇಕು.
22. ಕಬ್ಬಿಗರ ಕಾವ್ಯ ಇದರ ಕತ್ಯ ಯಾರು?
a) ಪಂಪ b) ಆಂಡಯ್ಯ c) ಬಸವಣ್ಣ d) ಹಂಪನಾ.
23. ಹುಲಿಗಂಜಿ ಹುತ್ತವ ಹೊಕ್ಕಡೆ ----- ತಿಂಬುದ ಮಾಬುದ?
a) ನರ್ಪ b) ಆನೆ c) ಸಿಂಹ d) ಚಿರತೆ.
24. ಪುರಂದರ ದಾಸರ ವಚನಗಳ ಅಂಕಿತನಾಮ ಯಾವುದು?
a) ರಾಮನಾಥ b) ವಿಠಲ
c) ಗುಹೇಶ್ವರ d) ಪುರಂದರ ವಿಠಲ.
25. ತಲ್ಲಣಿಸಿದಿರು ಕಂಡ್ಯ ತಾಳು ಮನವೇ ಕೀರ್ತನೆ ಕೀರ್ತನಕಾರರು ಯಾರು?
a) ಪುರಂದರದಾಸರು b) ಕನಕದಾಸರು
c) ಶಿವಯೋಗಿ d) ಶಿಶುನಾಳ ಶರೀಫರು.
26. ಕುಂಬಾರಕಿ ಆರು ಕಾಸಿಗೊಂದು ಏನನ್ನು ಮಾರುತ್ತಾಳೆ?
a) ಕುಡಕಿ b) ಮಡಿಕೆ
c) ಗುಡಾಣ d) ಗಡಿಗೆ.

27. ಕನ್ನಡ ಭಾಷೆಯ ಲಿಪಿಯನ್ನು ಲಿಪಿಗಳ ರಾಣಿ ಎಂದು ಕರೆದವರು ಯಾರು?
 a) ಸರೋಜಿನಿ ನಾಯ್ಡು
 b) ಮದರ ತರೆಸಾ
 c) ವಿನೋಬಾ ಭಾವೆ
 d) ಒನಕೆ ಓಬವ್ವ.
28. ಮೋಹಿನಿಗೆ ಮರುಳಾದವರು ಯಾರು?
 a) ಮಾನವನು
 b) ಮಹಿಳೆಯರು
 c) ಮೂರ್ಖದಾನವರ
 d) ಯಾರು ಇಲ್ಲ.
29. ಹೊಸಬಾಳಿನ ಗೀತೆ ಕವನದ ಕವಿ ಯಾರು?
 a) ಕುವೆಂಪು
 b) ಬೇಂದ್ರೆ
 c) ಶಿವರುದ್ರಪ್ಪ
 d) ಡಿ.ವಿ.ಜಿ.
30. ಅವರವರ ಭಕ್ತಿಗೆ ಅವರವರ ಭಾವಕ್ಕೆ ಎಂದು ಹಾಡಿದವರು ಯಾರು?
 a) ಶಿಶುನಾಳ ಶರೀಫ
 b) ಕುವೆಂಪು
 c) ಡಿ.ವಿ.ಜಿ
 d) ಬಾಲಲೀಲಾ ಮಹಾಂತ ಶಿವಯೋಗಿ.
31. ಮೈಸೂರು ರಾಜ್ಯಕ್ಕೆ ಕರ್ನಾಟಕ ಎಂದು ನಾಮಕರಣ ಮಾಡಿದ ವರ್ಷ -----.
 a) 1973
 b) 1964
 c) 1955
 d) 1947.
32. ಸರ್. ಎಂ. ವಿಶ್ವೇಶ್ವರಯ್ಯನವರು ಪಡೆದಿರುವ ದೇಶದ ಅತ್ಯುತ್ತಮ ಪ್ರಶಸ್ತಿ ಯಾವುದು?
 a) ಪದ್ಮಶ್ರೀ
 b) ಪದ್ಮಭೂಷಣ
 c) ಪದ್ಮವಿಭೂಷಣ
 d) ಭಾರತ ರತ್ನ.
33. ಕರ್ನಾಟಕ ಸರ್ಕಾರದ ಆಡಳಿತ ಭಾಷೆ ಯಾವುದು?
 a) ಕನ್ನಡ
 b) ತಮಿಳು
 c) ತೆಲಗು
 d) ಹಿಂದಿ.
34. ಕರ್ನಾಟಕದ ರಾಜಧಾನಿ ಯಾವುದು?
 a) ಮೈಸೂರು
 b) ಬೆಂಗಳೂರು
 c) ಧಾರವಾಡ
 d) ಬೆಳಗಾವಿ.
35. ಸಂತ ಶಿಶುನಾಳ ಶರೀಫರು ಹೆಸರುವಾಸಿಯಾಗಿದ್ದು ----- ಗಳಿಗೆ.
 a) ಜನಪದ
 b) ಕೀರ್ತನೆ
 c) ವಚನ
 d) ತತ್ವಪದ.
36. ಬಟ್ಟೆ ಮೇಲಿನ ಮುದ್ರಣ ಕಲೆಗೆ ಯಾವ ದೇಶವು ಮೂಲ ನೆಲೆಯಾಗಿದೆ.
 a) ಇಂಗ್ಲೆಂಡ್
 b) ಪ್ರಾನ್ಸ್
 c) ಭಾರತ
 d) ಅಮೇರಿಕಾ.

37. ಪ್ರಹ್ಲಾದನ ತಾಯಿಯ ಹೆಸರೇನು?
 a) ರುದ್ರಮ್ಮ b) ಕಾಳಮ್ಮ c) ರುಕ್ಮಿಣಿ d) ತಾಯಮ್ಮ.
38. ಸಂವಿಧಾನದ ಯಾವ ಅನುಚ್ಛೇದದ ಅನುಸಾರ ರಾಜ್ಯಗಳಲ್ಲಿ ಬಳಕೆಯಾಗುವ ಭಾಷೆಗಳನ್ನೆ ಆಡಳಿತ ಭಾಷೆಯೆಂದು ತೀರ್ಮಾನಿಸಲಾಗಿದೆ.
 a) 343 b) 443 c) 543 d) 643.
39. ಮೇಗಾನೆ ಗ್ರಾಮದಲ್ಲಿ ವಾಸವಿರುವ ಬಡಕಟ್ಟು ಜನಾಂಗದ ಹೆಸರೇನು?
 a) ಕುಣಬಿ b) ನಾಗ c) ಭದ್ರ d) ಮಲಯ.
40. "ಕರ್ನಾಟಕ ಸಂಸ್ಕೃತಿ" ಲೇಖನದ ಕರ್ತೃ ಯಾರು?
 a) ಕುವೆಂಪು b) ಹಂಪ ನಾಗರಾಜಯ್ಯ
 c) ಗೊವಿಂದ ಪೈ d) ಬೇಂದ್ರೆ.
41. ಅಲ್ಲಮಪ್ರಭು ಹೆಸರುವಾಸಿಯಾಗಿದ್ದು ----- ಗಳಿಗೆ.
 a) ಜನಪದ b) ಕೀರ್ತನೆ c) ವಚನ d) ತತ್ವಪದ.
42. ಅಕ್ಕಮಹಾದೇವಿಯವರ ಅಂಕಿತನಾಮ ಯಾವುದು?
 a) ಚೆನ್ನಮಲ್ಲಿಕಾರ್ಜುನ b) ಅಲ್ಲಮಪ್ರಭು
 c) ಕೊಡಲಸಂಗಮದೇವ d) ಗುಹೇಶ್ವರ.
43. ಕುಪ್ಪಳ್ಳಿ ವೆಂಕಟಪ್ಪ ಪುಟ್ಟಪ್ಪ ಇವರ ಕಾವ್ಯನಾಮ ತಿಳಿಸಿ.
 a) ದ.ರಾ.ಬೇಂದ್ರೆ b) ಕುವೆಂಪು c) ಡಿ.ವಿ.ಜಿ d) ತ.ರಾ.ಸು.
44. ಕರ್ನಾಟಕ ಗತವೈಭವ ಗ್ರಂಥದ ಲೇಖಕರು ಯಾರು?
 a) ಕುವೆಂಪು b) ಮಾಸ್ತಿ
 c) ಜಿ.ಎಸ್.ಎಸ್ d) ಆಲೂರು ವೆಂಕಟರಾಯರು
45. 'ಅಟ್ಟಕೇರಿಸು' ಈ ನುಡಿಗಟ್ಟಿನ ವಿಶೇಷಾರ್ಥ-----
 a) ಅಣರಿಸು b) ತೆಗಳು
 c) ಹೊಗಳು d) ನಗಿಸು.
46. ಸಂಗೀತಪುರವು ಈ ಹಿಂದೆ ಯಾರ ಉರಾಗಿತ್ತು?
 a) ಬೌದ್ಧರ b) ತಮಿಳರ
 c) ಪಾರ್ಸಿಗಳ d) ಜೈನರ.

47. 1955ರಲ್ಲಿ ರಾಯಚೂರಿನಲ್ಲಿ ನಡೆದ ಕನ್ನಡ ಸಾಹಿತ್ಯ ಪರಿಷತ್ತಿನ ಸಮ್ಮೇಳನದ ಅಧ್ಯಕ್ಷರು ಆಗಿದ್ದವರು ಯಾರು?
- a) ಶ್ರೀರಂಗರು b) ದಾ.ರಾ.ಬೇಂದ್ರೆ
c) ಚಂದ್ರಶೇಖರ ಕಂಬಾರ d) ಡಿ.ವಿ.ಜಿ.
48. ಅಲ್ಲಮಪ್ರಭುಗಳ ವಚನದ ಕಾವ್ಯನಾಮ ಯಾವುದು?
- a) ಗುಹೇಶ್ವರ b) ಸಂಗಮದೇವ
c) ಚೆನ್ನಮಲ್ಲಿಕಾರ್ಜುನ d) ಶಿಮೇಶ್ವರ.
49. ಕುರುಡು ಕಾಂಚಾಣವನ್ನು ಯಾವ ಕವನ ಸಂಕಲನದಿಂದ ಆಯ್ದುಕೊಳ್ಳಲಾಗಿದೆ.
- a) ನಾಕುತಂತಿ b) ಮರಳಿ ಮಣ್ಣಿಗೆ
c) ಕಲ್ಲು ಕರಗುವ ಸಮಯ d) ನಾದಲೀಲೆ.
50. “ನಾಡ ಗೀತೆ”ಯನ್ನು ರಚಿಸಿರುವ ಕವಿ ಯಾರು?
- a) ಬೇಂದ್ರೆ b) ಕುವೆಂಪು
c) ರಾಜರತ್ನಂ d) ಪಿ. ಲಂಕೇಶ್.

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First/Second Semester B.E/B.Tech. Degree Examination, June/July 2024

ಸಾಂಸ್ಕೃತಿಕ ಕನ್ನಡ

(COMMON TO ALL BRANCHES)

Time: 1 hrs.]

[Max. Marks: 50

ಸೂಚನೆಗಳು

1. ಎಲ್ಲ ಜಿಂ ಪ್ರಶ್ನೆಗಳಿಗೂ ಉತ್ತರಿಸಿರಿ. ಪ್ರತಿ ಪ್ರಶ್ನೆಗೆ ಒಂದು ಅಂಕ.
2. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಯು.ಎಸ್.ಎನ್ ಸಂಖ್ಯೆ ಹಾಗೂ ಪಶ್ಚಿಮ ಪತ್ರಿಕೆಯ ಶ್ರೇಣಿಯನ್ನು ಅಂದರೆ A, B, C ಅಥವಾ D ಯನ್ನು ತಪ್ಪಿಲ್ಲದಂತೆ ಕಡ್ಡಾಯವಾಗಿ ಗುರುತಿಸುವುದು ಅಭ್ಯರ್ಥಿಯ ಜವಾಬ್ದಾರಿಯಾಗಿರುತ್ತದೆ.
3. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ನಿಗದಿಪಡಿಸಿರುವ ಸ್ಥಳದಲ್ಲಿ ಭರ್ತಿಮಾಡದೆ ಹಾಗೆಯೇ ಬಿಟ್ಟಲ್ಲಿ ಅಥವಾ ಭರ್ತಿಮಾಡಿದ ಮಾಹಿತಿಯಲ್ಲಿ ಯಾವುದೇ ವ್ಯತ್ಯಾಸವಿದ್ದಲ್ಲಿ ಅಂತಹ ಉತ್ತರ ಪತ್ರಿಕೆಗಳನ್ನು ರದ್ದು ಪಡಿಸಲಾಗುವುದು.
4. ಕೇವಲ ಒಂದು ಉತ್ತರವನ್ನು ಮಾತ್ರ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಗುರುತಿಸತಕ್ಕದ್ದು. ಒಂದೆ ಪ್ರಶ್ನೆಗೆ ಎರಡು ಉತ್ತರವನ್ನು ಗುರುತಿಸುವುದು ಅಮಾನ್ಯ.
5. ಎಲ್ಲಾ ಉತ್ತರಗಳನ್ನು ನಿಮಗೆ ಒದಗಿಸಲಾದ ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯ ಹಾಳೆಯ ಮೇಲೆ ಕಪ್ಪು ಅಥವಾ ನೀಲಿ ಶಾಹಿಯ ಬಾಲ್‌ಪಾಯಿಂಟ್ ಪೆನ್ನಿನಿಂದ ಗುರುತು ಮಾಡಬೇಕು.

1. ಅಲ್ಲಮಪ್ರಭು ಹೆಸರುವಾಸಿಯಾಗಿದ್ದು ----- ಗಳಿಗೆ.
a) ಜನಪದ b) ಕೀರ್ತನೆ c) ವಚನ d) ತತ್ವಪದ.
2. ಅಕ್ಕಮಹಾದೇವಿಯವರ ಅಂಕಿತನಾಮ ಯಾವುದು?
a) ಚನ್ನಮಲ್ಲಿಕಾರ್ಜುನ b) ಅಲ್ಲಮಪ್ರಭು
c) ಕೊಡಲಸಂಗಮದೇವ d) ಗುಹೇಶ್ವರ.
3. ಕುಪ್ಪಳ್ಳಿ ವೆಂಕಟಪ್ಪ ಪುಟ್ಟಪ್ಪ ಇವರ ಕಾವ್ಯನಾಮ ತಿಳಿಸಿ.
a) ದ.ರಾ.ಬೇಂದ್ರೆ b) ಕುವೆಂಪು c) ಡಿ.ವಿ.ಜಿ d) ತ.ರಾ.ಸು.
4. ಕರ್ನಾಟಕ ಗತವೈಭವ ಗ್ರಂಥದ ಲೇಖಕರು ಯಾರು?
a) ಕುವೆಂಪು b) ಮಾಸ್ತಿ
c) ಜಿ.ಎಸ್.ಎಸ್ d) ಆಲೂರು ವೆಂಕಟರಾಯರು

5. 'ಅಟ್ಟಕೇರಿಸು' ಈ ನುಡಿಗಟ್ಟಿನ ವಿಶೇಷಾರ್ಥ-----.
- a) ಅಣರೆಸು
b) ತೆಗಳು
c) ಹೊಗಳು
d) ನಗಿಸು.
6. ಸಂಗೀತಪುರವು ಈ ಹಿಂದೆ ಯಾರ ಉರಾಗಿತ್ತು?
- a) ಬೌದ್ಧರ
b) ತಮಿಳರ
c) ಪಾರ್ಸಿಗಳ
d) ಜೈನರ.
7. 1955ರಲ್ಲಿ ರಾಯಚೂರಿನಲ್ಲಿ ನಡೆದ ಕನ್ನಡ ಸಾಹಿತ್ಯ ಪರಿಷತ್ತಿನ ಸಮ್ಮೇಳನದ ಅಧ್ಯಕ್ಷರು ಆಗಿದ್ದವರು ಯಾರು?
- a) ಶ್ರೀರಂಗರು
b) ದಾ.ರಾ.ಬೇಂದ್ರೆ
c) ಚಂದ್ರಶೇಖರ ಕಂಬಾರ
d) ಡಿ.ವಿ.ಜಿ.
8. ಅಲ್ಲಮಪ್ರಭುಗಳ ವಚನದ ಕಾವ್ಯನಾಮ ಯಾವುದು?
- a) ಗುಹೇಶ್ವರ
b) ಸಂಗಮದೇವ
c) ಚೆನ್ನಮಲ್ಲಿಕಾರ್ಜುನ
d) ಶಿಮೇಶ್ವರ.
9. ಕುರುಡು ಕಾಂಚಾಣವನ್ನು ಯಾವ ಕವನ ಸಂಕಲನದಿಂದ ಆಯ್ದುಕೊಳ್ಳಲಾಗಿದೆ.
- a) ನಾಕುತಂತಿ
b) ಮರಳಿ ಮಣ್ಣಿಗೆ
c) ಕಲ್ಲು ಕರಗುವ ಸಮಯ
d) ನಾದಲೀಲೆ.
10. "ನಾಡ ಗೀತೆ"ಯನ್ನು ರಚಿಸಿರುವ ಕವಿ ಯಾರು?
- a) ಬೇಂದ್ರೆ
b) ಕುವೆಂಪು
c) ರಾಜರತ್ನಂ
d) ಪಿ. ಲಂಕೇಶ್.
11. ಮೈಸೂರು ರಾಜ್ಯಕ್ಕೆ ಕರ್ನಾಟಕ ಎಂದು ನಾಮಕರಣ ಮಾಡಿದ ವರ್ಷ -----.
- a) 1973
b) 1964
c) 1955
d) 1947.
12. ಸರ್. ಎಂ. ವಿಶ್ವೇಶ್ವರಯ್ಯನವರು ಪಡೆದಿರುವ ದೇಶದ ಅತ್ಯುತ್ತಮ ಪ್ರಶಸ್ತಿ ಯಾವುದು?
- a) ಪದ್ಮಶ್ರೀ
b) ಪದ್ಮಭೂಷಣ
c) ಪದ್ಮವಿಭೂಷಣ
d) ಭಾರತ ರತ್ನ.
13. ಕರ್ನಾಟಕ ಸರ್ಕಾರದ ಆಡಳಿತ ಭಾಷೆ ಯಾವುದು?
- a) ಕನ್ನಡ
b) ತಮಿಳು
c) ತೆಲಗು
d) ಹಿಂದಿ.

14. ಕರ್ನಾಟಕದ ರಾಜಧಾನಿ ಯಾವುದು?
 a) ಮೈಸೂರು
 b) ಬೆಂಗಳೂರು
 c) ಧಾರವಾಡ
 d) ಬೆಳಗಾವಿ.
15. ಸಂತ ಶಿಶುನಾಳ ಶರೀಫರು ಹೆಸರುವಾಸಿಯಾಗಿದ್ದು ----- ಗಳಿಗೆ.
 a) ಜನಪದ
 b) ಕೀರ್ತನೆ
 c) ವಚನ
 d) ತತ್ವಪದ.
16. ಬಟ್ಟೆ ಮೇಲಿನ ಮುದ್ರಣ ಕಲೆಗೆ ಯಾವ ದೇಶವು ಮೂಲ ನೆಲೆಯಾಗಿದೆ.
 a) ಇಂಗ್ಲೆಂಡ್
 b) ಪ್ರಾನ್ಸ್
 c) ಭಾರತ
 d) ಅಮೇರಿಕಾ.
17. ಪ್ರಹ್ಲಾದನ ತಾಯಿಯ ಹೆಸರೇನು?
 a) ರುದ್ರಮ್ಮ
 b) ಕಾಳಮ್ಮ
 c) ರುಕ್ಮಿಣಿ
 d) ತಾಯಮ್ಮ.
18. ಸಂವಿಧಾನದ ಯಾವ ಅನುಚ್ಛೇದದ ಅನುಸಾರ ರಾಜ್ಯಗಳಲ್ಲಿ ಬಳಕೆಯಾಗುವ ಭಾಷೆಗಳನ್ನು ಆಡಳಿತ ಭಾಷೆಯೆಂದು ತೀರ್ಮಾನಿಸಲಾಗಿದೆ.
 a) 343
 b) 443
 c) 543
 d) 643.
19. ಮೇಗಾನೆ ಗ್ರಾಮದಲ್ಲಿ ವಾಸವಿರುವ ಬುಡಕಟ್ಟು ಜನಾಂಗದ ಹೆಸರೇನು?
 a) ಕುಣಬಿ
 b) ನಾಗ
 c) ಭದ್ರ
 d) ಮಲಯ.
20. "ಕರ್ನಾಟಕ ಸಂಸ್ಕೃತಿ" ಲೇಖನದ ಕರ್ತೃ ಯಾರು?
 a) ಕುವೆಂಪು
 b) ಹಂಪ ನಾಗರಾಜಯ್ಯ
 c) ಗೊವಿಂದ ಪೈ
 d) ಬೇಂದ್ರೆ.
21. ಕಾಲಿಗೆ ಬಿದ್ದವರ ಯಾವುದು ತುಳಿಯುತ್ತಲಿತ್ತು?
 a) ಹಣ
 b) ಕಾಂಚಾನ
 c) ಬಂಗಾರ
 d) ಕುರುಡ ಕಾಂಚಾಣ.
22. ಭಾಷೆಗೆ ಎಷ್ಟು ಪ್ರಮುಖ ಕೌಶಲ್ಯಗಳು ಇರುತ್ತವೆ?
 a) 10
 b) 20
 c) 08
 d) 04
23. ಆಡಳಿತ ಭಾಷೆ ಯಾವಾಗಲೂ ----- ಇರಬೇಕು.
 a) ಕಷ್ಟಕರವಾಗಿ
 b) ಅರ್ಥವಾಗದಂತೆ
 c) ಸರಳ ಮತ್ತು ಸ್ಪಷ್ಟವಾಗಿ
 d) ಗೊಂದಲದಿಂದ.
24. ಸರ್ಕಾರಿ ಆಸ್ಪತ್ರೆಯಲ್ಲಿ ಮಾಸ್ತರಿಗೆ ಸಹಾಯ ಮಾಡಿದ ದಾದಿ ಯಾರು?
 a) ಚಾಂದಿನಿ
 b) ರಾಧ
 c) ಫಾತಿಮಾ
 d) ರುಕ್ಮಿಣಿ.

25. ----- ಭಾರತದ ಕೆಲವು ನೇಕಾರರು ಚಿಂದಿ ಬಟ್ಟೆಯ ಕಂಬಳಿಗಳನ್ನು ತಮ್ಮ ಮಗ್ಗಗಳಲ್ಲಿ ನೇಯುವರು.
a) ಉತ್ತರ b) ಪೂರ್ವ c) ಪಶ್ಚಿಮ d) ದಕ್ಷಿಣ.
26. ಎಲ್ಲರನು ಸಲಹುವವನು ಯಾರು?
a) ಪುರಂದರ ವಿಠಲ b) ಕಾಗಿನೆಲೆ ಆದಿಕೇಶವ
c) ಮಲ್ಲಿಕಾರ್ಜುನ d) ಕೂಡಲನಂಗಮದೇವ.
27. ಶ್ರೀಮಂತರು ಯಾರ ಕಣ್ಣಿನ ಕವಡೆಯನ್ನು ಜೋಮಾಲೆ ಮಾಡಿಕೊಂಡು ಹಾಕಿಕೊಳ್ಳುವರು?
a) ಬಾಣಂತಿ b) ಸಣ್ಣ ಕೆಂದಮ್ಮ c) ಮುದುಕರು d) ಹೆಂಗಸರು.
28. ಯಾವ ಜನಾಂಗದ ಚಿತ್ರಕರಣ ಲೇಖಕರ ಮನದಲ್ಲಿತ್ತು?
a) ಗೊಂಡ b) ತೊಡವ c) ನಾಗ d) ಅಲೆಮಾರಿ.
29. ಕುಂಬಾರಕಿ ಯಾವುದರ ಮೇಲೆ ಮುಗಿಯನ್ನು ಇಡುವಳು?
a) ಕಲ್ಲು b) ಗಡಿಗೆ c) ವಸುಧೆ d) ಕಟ್ಟೆ.
30. ವಿಷನರ್ವಳಿಗೆ ಯಾವುದರ ಗೂಡು ಹುತ್ತವಾಗುವುದು?
a) ಗೆದ್ದಲಿರುವೆ b) ಉಡ c) ಇಲಿ d) ಗೀಜಗ.
31. ತಾಳಗುಪ್ಪ ಗಿರಿಜನ ಶ್ರಮ ಶಾಲೆಯ ಮಾಸ್ತರರು ಯಾರು?
a) ಸಮಪಾಲುಗೋಪಣ್ಣ b) ಪ್ರಹ್ಲಾದ c) ಹುಚ್ಚಪ್ಪ d) ಕರೀಂಖಾನ
32. ಗೋಪಣ್ಣ ಮಾಸ್ತರು ಕಣಗಲಿ ಗಿಡದಲ್ಲಿ ಏನು ನೋಡಿದರು?
a) ಹೂ b) ಹಣ್ಣು c) ಬಟ್ಟೆ d) ಜನಿವಾರ
33. ಹಾಡುವಳಿಗೆ ಹಿಂದೆ -----ಹೆಸರಿತ್ತು.
a) ಸಂಗೀತಪುರ b) ಭಟ್ಟಳ c) ಸಾಗರ d) ಕಾರವಾರ
34. ಹೊಸಚಿಗುರು ----- ಕೂಡಿರಲು ಮರಸೊಬಗು
a) ಹೊಸ ಎಲೆ b) ಹೊಸ ಹಸಿರು c) ಹೊಸಕಾಂಡ d) ಹಳಬೇರು
35. “ಪರಿಸರ, ಪರಂಪರೆ ಹಾಗೂ ಪ್ರತ್ಯಕ್ಷ ಜೀವನದ ಅನೇಕ ಸಂಸ್ಕಾರಗಳ ಮೂಲಕ ಮನುಷ್ಯನ ಅಂತರಂಗ ಪಡೆಯುವ ಪರಿಪಕ್ವತೆ” ಯನ್ನು ----- ಎನ್ನಬಹುದು.
a) ಕೆಲೆ b) ಸಂಸ್ಕಾರ c) ಸಂಸ್ಕೃತಿ d) ಕೆಂಪು

36. ಕುಣಬಿ ಜನರ ಮುಖ್ಯವಾದ ಹಬ್ಬ ಯಾವುದು?
 a) ಯುಗಾದಿ b) ದಸರಾ c) ಹೋಳಿ ಹುಣ್ಣಿಮೆ d) ದೀಪಾವಳಿ
37. ಬಣ್ಣಗಳನ್ನು ನೀಡುವ ಗಿಡಗಳು ಭಾರತದಲ್ಲಿ ಎಷ್ಟವೆ.
 a) 400 b) 300 c) 500 d) 600.
38. ಕರಕುಶಲ ಕಲೆಗಳಿಗೆ ಪೆಟ್ಟು ಬಿದ್ದಿದ್ದು ಯಾವುದರಿಂದ?
 a) ಜಾಗತೀಕರಣ b) ಖಾಸಗೀಕರಣ
 c) ಕೈಗಾರಿಕರಣ d) ಸೋಂಚೇರಿತನ.
39. ಸತಿ ಪದದ ಅರ್ಥವೇನು?
 a) ಹೆಂಡತಿ b) ಪತಿ c) ಮಕ್ಕಳು d) ತಾಯಿ.
40. ವಿದ್ಯಾವರ್ಧಕ ಸಂಘ ಎಲ್ಲಿ ಸ್ಥಾಪನೆಯಾಯಿತು?
 a) ಬೆಂಗಳೂರು b) ಧಾರವಾಡ
 c) ಶಿವಮೊಗ್ಗ d) ಮೈಸೂರು.
41. ಎಲ್ಲರೊಳಗೆ ನಾವು ಏನಾಗಬೇರೆಂದು ಡ.ವಿ.ಜಿ ತಿಳಿಸಿದ್ದಾರೆ?
 a) ಬೇರೆಯಾಗಬೇಕು b) ಒಂದಾಗಬೇಕು
 c) ದೂರವಾಗಬೇಕು d) ಮಾತಾಡಬೇಕು.
42. ಕಬ್ಬಿಗರ ಕಾವ್ಯ ಇದರ ಕತೃ ಯಾರು?
 a) ಪಂಪ b) ಆಂಡಯ್ಯ c) ಬಸವಣ್ಣ d) ಹಂಪನಾ.
43. ಹುಲಿಗಂಜಿ ಹುತ್ತವ ಹೊಕ್ಕಡೆ ----- ತಿಂಬುದ ಮಾಬುದೆ?
 a) ಸರ್ಪ b) ಆನೆ c) ಸಿಂಹ d) ಚಿರತೆ.
44. ಪುರಂದರ ದಾಸರ ವಚನಗಳ ಅಂಕಿತನಾಮ ಯಾವುದು?
 a) ರಾಮನಾಥ b) ವಿಠಲ
 c) ಗುಹೇಶ್ವರ d) ಪುರಂದರ ವಿಠಲ.
45. ತಲ್ಲಣಿಸಿದಿರು ಕಂಡ್ಯ ತಾಳು ಮನವೇ ಕೀರ್ತನೆ ಕೀರ್ತನಕಾರರು ಯಾರು?
 a) ಪುರಂದರದಾಸರು b) ಕನಕದಾಸರು
 c) ಶಿವಯೋಗಿ d) ಶಿಶುನಾಳ ಶರೀಫರು.

46. ಕುಂಬಾರಕಿ ಆರು ಕಾಸಿಗೊಂದು ಏನನ್ನು ಮಾರುತ್ತಾಳೆ?
 a) ಕುಡಕಿ b) ಮಡಿಕೆ
 c) ಗುಡಾಣ d) ಗಡಿಗೆ.
47. ಕನ್ನಡ ಭಾಷೆಯ ಲಿಪಿಯನ್ನು ಲಿಪಿಗಳ ರಾಣಿ ಎಂದು ಕರೆದವರು ಯಾರು?
 a) ಸರೋಜಿನಿ ನಾಯ್ಡು b) ಮದರ ತರಸಾ
 c) ವಿನೋಬಾ ಭಾವೆ d) ಒನಕೆ ಓಬವ್ವ.
48. ಮೋಹಿನಿಗೆ ಮರುಳಾದವರು ಯಾರು?
 a) ಮಾನವನು b) ಮಹಿಳೆಯರು
 c) ಮೂರ್ಖದಾನವರ d) ಯಾರು ಇಲ್ಲ.
49. ಹೊಸಬಾಳಿನ ಗೀತೆ ಕವನದ ಕವಿ ಯಾರು?
 a) ಕುವೆಂಪು b) ಬೇಂದ್ರೆ c) ಶಿವರುದ್ರಪ್ಪ d) ಡಿ.ವಿ.ಜಿ.
50. ಅವರವರ ಭಕ್ತರಿಗೆ ಅವರವರ ಭಾವಕ್ಕೆ ಎಂದು ಹಾಡಿದವರು ಯಾರು?
 a) ಶಿಶುನಾಳ ಶರೀಫ b) ಕುವೆಂಪು
 c) ಡಿ.ವಿ.ಜಿ d) ಬಾಲಲೀಲಾ ಮಹಾಂತ ಶಿವಯೋಗಿ.

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First/Second Semester B.E/B.Tech. Degree Examination, June/July 2024

ಸಾಂಸ್ಕೃತಿಕ ಕನ್ನಡ

(COMMON TO ALL BRANCHES)

Time: 1 hrs.]

[Max. Marks: 50

ಸೂಚನೆಗಳು

1. ಎಲ್ಲಾ ೫೦ ಪ್ರಶ್ನೆಗಳಿಗೂ ಉತ್ತರಿಸಿರಿ. ಪ್ರತಿ ಪ್ರಶ್ನೆಗೆ ಒಂದು ಅಂಕ.
2. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಯು.ಎಸ್.ಎನ್ ಸಂಖ್ಯೆ ಹಾಗೂ ಪಶ್ಚಿಮ ಪ್ರಶ್ನೆಯ ಶ್ರೇಣಿಯನ್ನು ಅಂದರೆ A, B, C ಅಥವಾ D ಯನ್ನು ತಪ್ಪಿಲ್ಲದಂತೆ ಕಡ್ಡಾಯವಾಗಿ ಗುರುತಿಸುವುದು ಅಭ್ಯರ್ಥಿಯ ಜವಾಬ್ದಾರಿಯಾಗಿರುತ್ತದೆ.
3. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ನಿಗದಿಪಡಿಸಿರುವ ಸ್ಥಳದಲ್ಲಿ ಭರ್ತಿಮಾಡದೆ ಹಾಗೆಯೇ ಬಿಟ್ಟಲ್ಲಿ ಅಥವಾ ಭರ್ತಿಮಾಡಿದ ಮಾಹಿತಿಯಲ್ಲಿ ಯಾವುದೇ ವ್ಯತ್ಯಾಸವಿದ್ದಲ್ಲಿ ಅಂತಹ ಉತ್ತರ ಪತ್ರಿಕೆಗಳನ್ನು ರದ್ದು ಪಡಿಸಲಾಗುವುದು.
4. ಕೇವಲ ಒಂದು ಉತ್ತರವನ್ನು ಮಾತ್ರ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಗುರುತಿಸತಕ್ಕದ್ದು. ಒಂದೆ ಪ್ರಶ್ನೆಗೆ ಎರಡು ಉತ್ತರವನ್ನು ಗುರುತಿಸುವುದು ಅಮಾನ್ಯ.
5. ಎಲ್ಲಾ ಉತ್ತರಗಳನ್ನು ನಿಮಗೆ ಒದಗಿಸಲಾದ ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯ ಹಾಳೆಯ ಮೇಲೆ ಕಪ್ಪು ಅಥವಾ ನೀಲಿ ಶಾಹಿಯ ಬಾಲ್‌ಪಾಯಿಂಟ್ ಪೆನ್ನಿನಿಂದ ಗುರುತು ಮಾಡಬೇಕು.

1. ಮೈಸೂರು ರಾಜ್ಯಕ್ಕೆ ಕರ್ನಾಟಕ ಎಂದು ನಾಮಕರಣ ಮಾಡಿದ ವರ್ಷ -----.
a) 1973 b) 1964 c) 1955 d) 1947.
2. ಸರ್. ಎಂ. ವಿಶ್ವೇಶ್ವರಯ್ಯನವರು ಪಡೆದಿರುವ ದೇಶದ ಅತ್ಯುತ್ತಮ ಪ್ರಶಸ್ತಿ ಯಾವುದು?
a) ಪದ್ಮಶ್ರೀ b) ಪದ್ಮಭೂಷಣ
c) ಪದ್ಮವಿಭೂಷಣ d) ಭಾರತ ರತ್ನ.
3. ಕರ್ನಾಟಕ ಸರ್ಕಾರದ ಆಡಳಿತ ಭಾಷೆ ಯಾವುದು?
a) ಕನ್ನಡ b) ತಮಿಳು c) ತೆಲಗು d) ಹಿಂದಿ.
4. ಕರ್ನಾಟಕದ ರಾಜಧಾನಿ ಯಾವುದು?
a) ಮೈಸೂರು b) ಬೆಂಗಳೂರು
c) ಧಾರವಾಡ d) ಬೆಳಗಾವಿ.

5. ಸಂತ ಶಿಶುನಾಳ ಶರೀಫರು ಹೆಸರುವಾಸಿಯಾಗಿದ್ದು ----- ಗಳಿಗೆ.
a) ಜನಪದ b) ಕೀರ್ತನೆ c) ವಚನ d) ತತ್ವಪದ.
6. ಬಟ್ಟೆ ಮೇಲಿನ ಮುದ್ರಣ ಕಲೆಗೆ ಯಾವ ದೇಶವು ಮೂಲ ನೆಲೆಯಾಗಿದೆ.
a) ಇಂಗ್ಲೆಂಡ್ b) ಪ್ರಾನ್ಸ್ c) ಭಾರತ d) ಅಮೇರಿಕಾ.
7. ಪ್ರಹ್ಲಾದನ ತಾಯಿಯ ಹೆಸರೇನು?
a) ರುದ್ರಮ್ಮ b) ಕಾಳಮ್ಮ c) ರುಕ್ಮಿಣಿ d) ತಾಯಮ್ಮ.
8. ಸಂವಿಧಾನದ ಯಾವ ಅನುಚ್ಛೇದದ ಅನುಸಾರ ರಾಜ್ಯಗಳಲ್ಲಿ ಬಳಕೆಯಾಗುವ ಭಾಷೆಗಳನ್ನೆ ಆಡಳಿತ ಭಾಷೆಯೆಂದು ತೀರ್ಮಾನಿಸಲಾಗಿದೆ.
a) 343 b) 443 c) 543 d) 643.
9. ಮೇಗಾನೆ ಗ್ರಾಮದಲ್ಲಿ ವಾಸವಿರುವ ಬುಡಕಟ್ಟು ಜನಾಂಗದ ಹೆಸರೇನು?
a) ಕುಣಬಿ b) ನಾಗ c) ಭದ್ರ d) ಮಲಯ.
10. “ಕರ್ನಾಟಕ ಸಂಸ್ಕೃತಿ” ಲೇಖನದ ಕರ್ತೃ ಯಾರು?
a) ಕುವೆಂಪು b) ಹಂಪ ನಾಗರಾಜಯ್ಯ
c) ಗೊವಿಂದ ಪೈ d) ಬೇಂದ್ರೆ.
11. ತಾಳಗುಪ್ಪ ಗಿರಿಜನ ಶ್ರಮ ಶಾಲೆಯ ಮಾಸ್ತರರು ಯಾರು?
a) ಸಮಪಾಲುಗೋಪಣ್ಣ b) ಪ್ರಹ್ಲಾದ c) ಹುಚ್ಚಪ್ಪ d) ಕರೀಂಖಾನ
12. ಗೋಪಣ್ಣ ಮಾಸ್ತರು ಕಣಗಲಿ ಗಿಡದಲ್ಲಿ ಏನು ನೋಡಿದರು?
a) ಹೂ b) ಹಣ್ಣು c) ಬಟ್ಟೆ d) ಜನಿವಾರ
13. ಹಾಡುವಳ್ಳಿಗೆ ಹಿಂದೆ -----ಹೆಸರಿತ್ತು.
a) ಸಂಗೀತಪುರ b) ಭಟ್ಟಳ c) ಸಾಗರ d) ಕಾರವಾರ
14. ಹೊಸಚಿಗುರು ----- ಕೂಡಿರಲು ಮರಸೊಬಗು
a) ಹೊಸ ಎಲೆ b) ಹೊಸ ಹಸಿರು c) ಹೊಸಕಾಂಡ d) ಹಳಬೇರು
15. “ಪರಿಸರ, ಪರಂಪರೆ ಹಾಗೂ ಪ್ರತ್ಯಕ್ಷ ಜೀವನದ ಅನೇಕ ಸಂಸ್ಕಾರಗಳ ಮೂಲಕ ಮನುಷ್ಯನ ಅಂತರಂಗ ಪಡೆಯುವ ಪರಿಪಕ್ವತೆ” ಯನ್ನು ----- ಎನ್ನಬಹುದು.
a) ಕೆಲೆ b) ಸಂಸ್ಕಾರ c) ಸಂಸ್ಕೃತಿ d) ಕೆಂಪು

16. ಕುಣಬಿ ಜನರ ಮುಖ್ಯವಾದ ಹಬ್ಬ ಯಾವುದು?
 a) ಯುಗಾದಿ b) ದಸರಾ c) ಹೋಳಿ ಹುಣ್ಣಿಮೆ d) ದೀಪಾವಳಿ
17. ಬಣ್ಣಗಳನ್ನು ನೀಡುವ ಗಿಡಗಳು ಭಾರತದಲ್ಲಿ ಎಷ್ಟವೆ.
 a) 400 b) 300 c) 500 d) 600.
18. ಕರಕುಶಲ ಕಲೆಗಳಿಗೆ ಪೆಟ್ಟು ಬಿದ್ದಿದ್ದು ಯಾವುದರಿಂದ?
 a) ಜಾಗತೀಕರಣ b) ಖಾಸಗೀಕರಣ
 c) ಕೈಗಾರಿಕರಣ d) ಸೋಂಚೇರಿತನ.
19. ಸತಿ ಪದದ ಅರ್ಥವೇನು?
 a) ಹೆಂಡತಿ b) ಪತಿ c) ಮಕ್ಕಳು d) ತಾಯಿ.
20. ವಿದ್ಯಾವರ್ಧಕ ಸಂಘ ಎಲ್ಲಿ ಸ್ಥಾಪನೆಯಾಯಿತು?
 a) ಬೆಂಗಳೂರು b) ಧಾರವಾಡ
 c) ಶಿವಮೊಗ್ಗ d) ಮೈಸೂರು.
21. ಅಲ್ಲಮಪ್ರಭು ಹೆಸರುವಾಸಿಯಾಗಿದ್ದು ----- ಗಳಿಗೆ.
 a) ಜನಪದ b) ಕೀರ್ತನೆ c) ವಚನ d) ತತ್ವಪದ.
22. ಅಕ್ಕಮಹಾದೇವಿಯವರ ಅಂಕಿತನಾಮ ಯಾವುದು?
 a) ಚೆನ್ನಮಲ್ಲಿಕಾರ್ಜುನ b) ಅಲ್ಲಮಪ್ರಭು
 c) ಕೊಡಲಸಂಗಮದೇವ d) ಗುಹೇಶ್ವರ.
23. ಕುಪ್ಪಳ್ಳಿ ವೆಂಕಟಪ್ಪ ಪುಟ್ಟಪ್ಪ ಇವರ ಕಾವ್ಯನಾಮ ತಿಳಿಸಿ.
 a) ದ.ರಾ.ಬೇಂದ್ರೆ b) ಕುವೆಂಪು c) ಡಿ.ವಿ.ಜಿ d) ತ.ರಾ.ಸು.
24. ಕರ್ನಾಟಕ ಗತವೈಭವ ಗ್ರಂಥದ ಲೇಖಕರು ಯಾರು?
 a) ಕುವೆಂಪು b) ಮಾಸ್ತಿ
 c) ಜಿ.ಎಸ್.ಎಸ್ d) ಆಲೂರು ವೆಂಕಟರಾಯರು
25. 'ಅಟ್ಟಕೇರಿಸು' ಈ ನುಡಿಗಟ್ಟಿನ ವಿಶೇಷಾರ್ಥ-----
 a) ಅಣರೆಸು b) ತೆಗಳು
 c) ಹೊಗಳು d) ನಗಿಸು.

35. ತಲ್ಲಣಿಸಿದಿರು ಕಂಡ್ಯ ತಾಳು ಮನವೇ ಕೀರ್ತನೆ ಕೀರ್ತನಕಾರರು ಯಾರು?
 a) ಪುರಂದರದಾಸರು b) ಕನಕದಾಸರು
 c) ಶಿವಯೋಗಿ d) ಶಿಶುನಾಳ ಶರೀಫರು.
36. ಕುಂಬಾರಕಿ ಆರು ಕಾಸಿಗೊಂದು ಏನನ್ನು ಮಾರುತ್ತಾಳೆ?
 a) ಕುಡಕಿ b) ಮಡಿಕೆ
 c) ಗುಡಾಣ d) ಗಡಿಗೆ.
37. ಕನ್ನಡ ಭಾಷೆಯ ಲಿಪಿಯನ್ನು ಲಿಪಿಗಳ ರಾಣಿ ಎಂದು ಕರೆದವರು ಯಾರು?
 a) ಸರೋಜಿನಿ ನಾಯ್ಡು b) ಮದರ ತರೆಸಾ
 c) ವಿನೋಬಾ ಭಾವೆ d) ಒನಕೆ ಓಬವ್ವ.
38. ಮೋಹಿನಿಗೆ ಮರುಳಾದವರು ಯಾರು?
 a) ಮಾನವನು b) ಮಹಿಳೆಯರು
 c) ಮೂರ್ಖದಾನವರ d) ಯಾರು ಇಲ್ಲ.
39. ಹೊಸಬಾಳಿನ ಗೀತೆ ಕವನದ ಕವಿ ಯಾರು?
 a) ಕುವೆಂಪು b) ಬೇಂದ್ರೆ c) ಶಿವರುದ್ರಪ್ಪ d) ಡಿ.ವಿ.ಜಿ.
40. ಅವರವರ ಭಕುತಿಗೆ ಅವರವರ ಭಾವಕ್ಕೆ ಎಂದು ಹಾಡಿದವರು ಯಾರು?
 a) ಶಿಶುನಾಳ ಶರೀಫ b) ಕುವೆಂಪು
 c) ಡಿ.ವಿ.ಜಿ d) ಬಾಲಲೀಲಾ ಮಹಾಂತ ಶಿವಯೋಗಿ.
41. ಕಾಲಿಗೆ ಬಿದ್ದವರ ಯಾವುದು ತುಳಿಯುತ್ತಲಿತ್ತು?
 a) ಹಣ b) ಕಾಂಚಾನ c) ಬಂಗಾರ d) ಕುರುಡ ಕಾಂಚಾಣ.
42. ಭಾಷೆಗೆ ಎಷ್ಟು ಪ್ರಮುಖ ಕೌಶಲ್ಯಗಳು ಇರುತ್ತವೆ?
 a) 10 b) 20 c) 08 d) 04
43. ಆಡಳಿತ ಭಾಷೆ ಯಾವಾಗಲೂ ----- ಇರಬೇಕು.
 a) ಕಷ್ಟಕರವಾಗಿ b) ಅರ್ಥವಾಗದಂತೆ
 c) ಸರಳ ಮತ್ತು ಸ್ಪಷ್ಟವಾಗಿ d) ಗೊಂದಲದಿಂದ.
44. ಸರ್ಕಾರಿ ಆಸ್ಪತ್ರೆಯಲ್ಲಿ ಮಾಸ್ತರಿಗೆ ಸಹಾಯ ಮಾಡಿದ ದಾದಿ ಯಾರು?
 a) ಚಾಂದಿನಿ b) ರಾಧ c) ಫಾತಿಮಾ d) ರುಕ್ಮಿಣಿ.

45. ----- ಭಾರತದ ಕೆಲವು ನೇಕಾರರು ಚಿಂದಿ ಬಟ್ಟೆಯ ಕಂಬಳಿಗಳನ್ನು ತಮ್ಮ ಮಗ್ಗಗಳಲ್ಲಿ ನೇಯುವರು.
a) ಉತ್ತರ b) ಪೂರ್ವ c) ಪಶ್ಚಿಮ d) ದಕ್ಷಿಣ.
46. ಎಲ್ಲರನು ಸಲಹುವವನು ಯಾರು?
a) ಪುರಂದರ ವಿಠಲ b) ಕಾಗಿನೆಲೆ ಆದಿಕೇಶವ
c) ಮಲ್ಲಿಕಾರ್ಜುನ d) ಕೂಡಲಸಂಗಮದೇವ.
47. ಶ್ರೀಮಂತರು ಯಾರ ಕಣ್ಣಿನ ಕವಡೆಯನ್ನು ಜೋಮಾಲೆ ಮಾಡಿಕೊಂಡು ಹಾಕಿಕೊಳ್ಳುವರು?
a) ಬಾಣಂತಿ b) ಸಣ್ಣ ಕಂದಮ್ಮ c) ಮುದುಕರು d) ಹೆಂಗಸರು.
48. ಯಾವ ಜನಾಂಗದ ಚಿತ್ರಿಕರಣ ಲೇಖಕರ ಮನದಲ್ಲಿತ್ತು?
a) ಗೊಂಡ b) ತೊಡವ c) ನಾಗ d) ಅಲೆಮಾರಿ.
49. ಕುಂಬಾರಕಿ ಯಾವುದರ ಮೇಲೆ ಮುಗಿಯನ್ನು ಇಡುವಳು?
a) ಕಲ್ಲು b) ಗಡಿಗೆ c) ವಸುಧೆ d) ಕಟ್ಟೆ.
50. ವಿಷಸರ್ಪಗಳಿಗೆ ಯಾವುದರ ಗೂಡು ಹುತ್ತವಾಗುವುದು?
a) ಗೆದ್ದಲಿರುವೆ b) ಉಡ c) ಇಲಿ d) ಗೀಜಗ.

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First/Second Semester B.E/B.Tech. Degree Examination, June/July 2024

ಸಾಂಸ್ಕೃತಿಕ ಕನ್ನಡ

(COMMON TO ALL BRANCHES)

Time: 1 hrs.]

[Max. Marks: 50

ಸೂಚನೆಗಳು

1. ಎಲ್ಲ ಜಿಂ ಪ್ರಶ್ನೆಗಳಿಗೂ ಉತ್ತರಿಸಿರಿ. ಪ್ರತಿ ಪ್ರಶ್ನೆಗೆ ಒಂದು ಅಂಕ.
2. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಯು.ಎಸ್.ಎನ್ ಸಂಖ್ಯೆ ಹಾಗೂ ಪಶ್ಚಿಮ ಪತ್ರಿಕೆಯ ಶ್ರೇಣಿಯನ್ನು ಅಂದರೆ A, B, C ಅಥವಾ D ಯನ್ನು ತಪ್ಪಿಲ್ಲದಂತೆ ಕಡ್ಡಾಯವಾಗಿ ಗುರುತಿಸುವುದು ಅಭ್ಯರ್ಥಿಯ ಜವಾಬ್ದಾರಿಯಾಗಿರುತ್ತದೆ.
3. ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ನಿಗದಿಪಡಿಸಿರುವ ಸ್ಥಳದಲ್ಲಿ ಭರ್ತಿಮಾಡದೆ ಹಾಗೆಯೇ ಬಿಟ್ಟಲ್ಲಿ ಅಥವಾ ಭರ್ತಿಮಾಡಿದ ಮಾಹಿತಿಯಲ್ಲಿ ಯಾವುದೇ ವ್ಯತ್ಯಾಸವಿದ್ದಲ್ಲಿ ಅಂತಹ ಉತ್ತರ ಪತ್ರಿಕೆಗಳನ್ನು ರದ್ದು ಪಡಿಸಲಾಗುವುದು.
4. ಕೇವಲ ಒಂದು ಉತ್ತರವನ್ನು ಮಾತ್ರ ಉತ್ತರ ಪತ್ರಿಕೆಯಲ್ಲಿ ಗುರುತಿಸತಕ್ಕದ್ದು. ಒಂದೆ ಪ್ರಶ್ನೆಗೆ ಎರಡು ಉತ್ತರವನ್ನು ಗುರುತಿಸುವುದು ಅಮಾನ್ಯ.
5. ಎಲ್ಲಾ ಉತ್ತರಗಳನ್ನು ನಿಮಗೆ ಒದಗಿಸಲಾದ ಓ.ಎಂ.ಆರ್ ಉತ್ತರ ಪತ್ರಿಕೆಯ ಹಾಳೆಯ ಮೇಲೆ ಕಪ್ಪು ಅಥವಾ ನೀಲಿ ಶಾಹಿಯ ಬಾಲ್‌ಪಾಯಿಂಟ್ ಪೆನ್ನಿನಿಂದ ಗುರುತು ಮಾಡಬೇಕು.

1. ಎಲ್ಲರೊಳಗೆ ನಾವು ಏನಾಗಬೇಕೆಂದು ಡ.ವಿ.ಜಿ ತಿಳಿಸಿದ್ದಾರೆ?
a) ಬೇರೆಯಾಗಬೇಕು
b) ಒಂದಾಗಬೇಕು
c) ದೂರವಾಗಬೇಕು
d) ಮಾತಾಡಬೇಕು.
2. ಕಬ್ಬಿಗರ ಕಾವ್ಯ ಇದರ ಕತೆ ಯಾರು?
a) ಪಂಪ
b) ಆಂಡಯ್ಯ
c) ಬಸವಣ್ಣ
d) ಹಂಪನಾ.
3. ಹುಲಿಗಂಜಿ ಹುತ್ತವ ಹೊಕ್ಕಡೆ ----- ತಿಂಬುದ ಮಾಬುದೆ?
a) ಸರ್ಪ
b) ಆನೆ
c) ಸಿಂಹ
d) ಚಿರತೆ.
4. ಪುರಂದರ ದಾಸರ ವಚನಗಳ ಅಂಕಿತನಾಮ ಯಾವುದು?
a) ರಾಮನಾಥ
b) ವಿಠಲ
c) ಗುಹೇಶ್ವರ
d) ಪುರಂದರ ವಿಠಲ.

5. ತಲ್ಲಣಿಸಿದಿರು ಕಂಡ್ಯ ತಾಳು ಮನವೇ ಕೀರ್ತನೆ ಕೀರ್ತನಕಾರರು ಯಾರು?
 a) ಪುರಂದರದಾಸರು
 b) ಕನಕದಾಸರು
 c) ಶಿವಯೋಗಿ
 d) ಶಿಶುನಾಳ ಶರೀಫರು.
6. ಕುಂಬಾರಕಿ ಆರು ಕಾಸಿಗೊಂದು ಏನನ್ನು ಮಾರುತ್ತಾಳೆ?
 a) ಕುಡಕಿ
 b) ಮಡಿಕೆ
 c) ಗುಡಾಣ
 d) ಗಡಿಗೆ.
7. ಕನ್ನಡ ಭಾಷೆಯ ಲಿಪಿಯನ್ನು ಲಿಪಿಗಳ ರಾಣಿ ಎಂದು ಕರೆದವರು ಯಾರು?
 a) ಸರೋಜಿನಿ ನಾಯ್ಡು
 b) ಮದರ ತರೆಸಾ
 c) ವಿನೋಬಾ ಭಾವೆ
 d) ಒನಕೆ ಓಬವ್ವ.
8. ಮೋಹಿನಿಗೆ ಮರುಳಾದವರು ಯಾರು?
 a) ಮಾನವನು
 b) ಮಹಿಳೆಯರು
 c) ಮೂರ್ಖದಾನವರ
 d) ಯಾರು ಇಲ್ಲ.
9. ಹೊಸಬಾಳಿನ ಗೀತೆ ಕವನದ ಕವಿ ಯಾರು?
 a) ಕುವೆಂಪು
 b) ಬೇಂದ್ರೆ
 c) ಶಿವರುದ್ರಪ್ಪ
 d) ಡಿ.ವಿ.ಜಿ.
10. ಅವರವರ ಭಕುತಿಗೆ ಅವರವರ ಭಾವಕ್ಕೆ ಎಂದು ಹಾಡಿದವರು ಯಾರು?
 a) ಶಿಶುನಾಳ ಶರೀಫ
 b) ಕುವೆಂಪು
 c) ಡಿ.ವಿ.ಜಿ
 d) ಬಾಲಲೀಲಾ ಮಹಾಂತ ಶಿವಯೋಗಿ.
11. ಅಲ್ಲಮಪ್ರಭು ಹೆಸರುವಾಸಿಯಾಗಿದ್ದು ----- ಗಳಿಗೆ.
 a) ಜನಪದ
 b) ಕೀರ್ತನೆ
 c) ವಚನ
 d) ತತ್ವಪದ.
12. ಅಕ್ಕಮಹಾದೇವಿಯವರ ಅಂಕಿತನಾಮ ಯಾವುದು?
 a) ಚೆನ್ನಮಲ್ಲಿಕಾರ್ಜುನ
 b) ಅಲ್ಲಮಪ್ರಭು
 c) ಕೊಡಲಸಂಗಮದೇವ
 d) ಗುಹೇಶ್ವರ.
13. ಕುಪ್ಪಳ್ಳಿ ವೆಂಕಟಪ್ಪ ಪುಟ್ಟಪ್ಪ ಇವರ ಕಾವ್ಯನಾಮ ತಿಳಿಸಿ.
 a) ದ.ರಾ.ಬೇಂದ್ರೆ
 b) ಕುವೆಂಪು
 c) ಡಿ.ವಿ.ಜಿ
 d) ತ.ರಾ.ಸು.

14. ಕರ್ನಾಟಕ ಗತವೈಭವ ಗ್ರಂಥದ ಲೇಖಕರು ಯಾರು?
 a) ಕುವೆಂಪು
 b) ಮಾಸ್ತಿ
 c) ಜಿ.ಎಸ್.ಎಸ್
 d) ಅಲೂರು ವೆಂಕಟರಾಯರು
15. 'ಅಟ್ಟಕೇರಿಸು' ಈ ನುಡಿಗಟ್ಟಿನ ವಿಶೇಷಾರ್ಥ-----
 a) ಅಣರೆಸು
 b) ತೆಗಳು
 c) ಹೊಗಳು
 d) ನಗಿಸು.
16. ಸಂಗೀತಪುರವು ಈ ಹಿಂದೆ ಯಾರ ಉರಾಗಿತ್ತು?
 a) ಬೌದ್ಧರ
 b) ತಮಿಳರ
 c) ಪಾರ್ಸಿಗಳ
 d) ಜೈನರ.
17. 1955ರಲ್ಲಿ ರಾಯಚೂರಿನಲ್ಲಿ ನಡೆದ ಕನ್ನಡ ಸಾಹಿತ್ಯ ಪರಿಷತ್ತಿನ ಸಮ್ಮೇಳನದ ಅಧ್ಯಕ್ಷರು ಆಗಿದ್ದವರು ಯಾರು?
 a) ಶ್ರೀರಂಗರು
 b) ದಾ.ರಾ.ಬೇಂದ್ರೆ
 c) ಚಂದ್ರಶೇಖರ ಕಂಬಾರ
 d) ಡಿ.ವಿ.ಜಿ.
18. ಅಲ್ಲಮಪ್ರಭುಗಳ ವಚನದ ಕಾವ್ಯನಾಮ ಯಾವುದು?
 a) ಗುಹೇಶ್ವರ
 b) ಸಂಗಮದೇವ
 c) ಚೆನ್ನಮಲ್ಲಿಕಾರ್ಜುನ
 d) ಶಿಮೇಶ್ವರ.
19. ಕುರುಡು ಕಾಂಚಾಣವನ್ನು ಯಾವ ಕವನ ಸಂಕಲನದಿಂದ ಆಯ್ದುಕೊಳ್ಳಲಾಗಿದೆ.
 a) ನಾಕುತಂತಿ
 b) ಮರಳಿ ಮಣ್ಣಿಗೆ
 c) ಕಲ್ಲು ಕರಗುವ ಸಮಯ
 d) ನಾದಲೀಲೆ.
20. "ನಾಡ ಗೀತೆ"ಯನ್ನು ರಚಿಸಿರುವ ಕವಿ ಯಾರು?
 a) ಬೇಂದ್ರೆ
 b) ಕುವೆಂಪು
 c) ರಾಜರತ್ನಂ
 d) ಪಿ. ಲಂಕೇಶ್.
21. ಮೈಸೂರು ರಾಜ್ಯಕ್ಕೆ ಕರ್ನಾಟಕ ಎಂದು ನಾಮಕರಣ ಮಾಡಿದ ವರ್ಷ -----
 a) 1973
 b) 1964
 c) 1955
 d) 1947.
22. ಸರ್. ಎಂ. ವಿಶ್ವೇಶ್ವರಯ್ಯನವರು ಪಡೆದಿರುವ ದೇಶದ ಅತ್ಯುತ್ತಮ ಪ್ರಶಸ್ತಿ ಯಾವುದು?
 a) ಪದ್ಮಶ್ರೀ
 b) ಪದ್ಮಭೂಷಣ
 c) ಪದ್ಮವಿಭೂಷಣ
 d) ಭಾರತ ರತ್ನ.

23. ಕರ್ನಾಟಕ ಸರ್ಕಾರದ ಆಡಳಿತ ಭಾಷೆ ಯಾವುದು?
a) ಕನ್ನಡ b) ತಮಿಳು c) ತೆಲಗು d) ಹಿಂದಿ.
24. ಕರ್ನಾಟಕದ ರಾಜಧಾನಿ ಯಾವುದು?
a) ಮೈಸೂರು b) ಬೆಂಗಳೂರು
c) ಧಾರವಾಡ d) ಬೆಳಗಾವಿ.
25. ಸಂತ ಶಿಶುನಾಳ ಶರೀಫರು ಹೆಸರುವಾಸಿಯಾಗಿದ್ದು ----- ಗಳಿಗೆ.
a) ಜನಪದ b) ಕೀರ್ತನೆ c) ವಚನ d) ತತ್ವಪದ.
26. ಬಟ್ಟೆ ಮೇಲಿನ ಮುದ್ರಣ ಕಲೆಗೆ ಯಾವ ದೇಶವು ಮೂಲ ನೆಲೆಯಾಗಿದೆ.
a) ಇಂಗ್ಲೆಂಡ್ b) ಫ್ರಾನ್ಸ್ c) ಭಾರತ d) ಅಮೇರಿಕಾ.
27. ಪ್ರಹ್ಲಾದನ ತಾಯಿಯ ಹೆಸರೇನು?
a) ರುದ್ರಮ್ಮ b) ಕಾಳಮ್ಮ c) ರುಕ್ಮಿಣಿ d) ತಾಯಮ್ಮ.
28. ಸಂವಿಧಾನದ ಯಾವ ಅನುಚ್ಛೇದದ ಅನುಸಾರ ರಾಜ್ಯಗಳಲ್ಲಿ ಬಳಕೆಯಾಗುವ ಭಾಷೆಗಳನ್ನು ಆಡಳಿತ ಭಾಷೆಯೆಂದು ತೀರ್ಮಾನಿಸಲಾಗಿದೆ.
a) 343 b) 443 c) 543 d) 643.
29. ಮೇಗಾನೆ ಗ್ರಾಮದಲ್ಲಿ ವಾಸವಿರುವ ಬುಡಕಟ್ಟು ಜನಾಂಗದ ಹೆಸರೇನು?
a) ಕುಣಬಿ b) ನಾಗ c) ಭದ್ರ d) ಮಲಯ.
30. “ಕರ್ನಾಟಕ ಸಂಸ್ಕೃತಿ” ಲೇಖನದ ಕರ್ತೃ ಯಾರು?
a) ಕುವೆಂಪು b) ಹಂಪ ನಾಗರಾಜಯ್ಯ
c) ಗೊವಿಂದ ಪೈ d) ಬೇಂದ್ರೆ.
31. ಕಾಲಿಗೆ ಬಿದ್ದವರ ಯಾವುದು ತುಳಿಯುತ್ತಲಿತ್ತು?
a) ಹಣ b) ಕಾಂಚಾನ c) ಬಂಗಾರ d) ಕುರುಡ ಕಾಂಚಾಣ.
32. ಭಾಷೆಗೆ ಎಷ್ಟು ಪ್ರಮುಖ ಕೌಶಲ್ಯಗಳು ಇರುತ್ತವೆ?
a) 10 b) 20 c) 08 d) 04
33. ಆಡಳಿತ ಭಾಷೆ ಯಾವಾಗಲೂ ----- ಇರಬೇಕು.
a) ಕಷ್ಟಕರವಾಗಿ b) ಅರ್ಥವಾಗದಂತೆ
c) ಸರಳ ಮತ್ತು ಸ್ಪಷ್ಟವಾಗಿ d) ಗೊಂದಲದಿಂದ.

34. ಸರ್ಕಾರಿ ಆಸ್ಪತ್ರೆಯಲ್ಲಿ ಮಾಸ್ತರಿಗೆ ಸಹಾಯ ಮಾಡಿದ ದಾದಿ ಯಾರು?
a) ಚಾಂದಿನಿ b) ರಾಧ c) ಫಾತಿಮಾ d) ರುಕ್ಮಿಣಿ.
35. ----- ಭಾರತದ ಕೆಲವು ನೇಕಾರರು ಚಿಂದಿ ಬಟ್ಟೆಯ ಕಂಬಳಿಗಳನ್ನು ತಮ್ಮ ಮಗ್ಗಗಳಲ್ಲಿ ನೇಯುವರು.
a) ಉತ್ತರ b) ಪೂರ್ವ c) ಪಶ್ಚಿಮ d) ದಕ್ಷಿಣ.
36. ಎಲ್ಲರನು ಸಲಹುವವನು ಯಾರು?
a) ಪುರಂದರ ವಿಠಲ b) ಕಾಗಿನೆಲೆ ಆದಿಕೇಶವ
c) ಮಲ್ಲಿಕಾರ್ಜುನ d) ಕೂಡಲಸಂಗಮದೇವ.
37. ಶ್ರೀಮಂತರು ಯಾರ ಕಣ್ಣಿನ ಕವಡೆಯನ್ನು ಜೋಮಾಲೆ ಮಾಡಿಕೊಂಡು ಹಾಕಿಕೊಳ್ಳುವರು?
a) ಬಾಣಂತಿ b) ಸಣ್ಣ ಕಂದಮ್ಮ c) ಮುದುಕರು d) ಹೆಂಗಸರು.
38. ಯಾವ ಜನಾಂಗದ ಚಿತ್ರಕರಣ ಲೇಖಕರ ಮನದಲ್ಲಿತ್ತು?
a) ಗೊಂಡ b) ತೊಡವ c) ನಾಗ d) ಅಲೆಮಾರಿ.
39. ಕುಂಬಾರಕಿ ಯಾವುದರ ಮೇಲೆ ಮುಗಿಯನ್ನು ಇಡುವಳು?
a) ಕಲ್ಲು b) ಗಡಿಗೆ c) ವಸುಧೆ d) ಕಟ್ಟೆ.
40. ವಿಷಸರ್ಪಗಳಿಗೆ ಯಾವುದರ ಗೂಡು ಹುತ್ತವಾಗುವುದು?
a) ಗೆದ್ದಲಿರುವೆ b) ಉಡ c) ಇಲಿ d) ಗೀಜಗ.
41. ತಾಳಗುಪ್ಪ ಗಿರಿಜನ ಶ್ರಮ ಶಾಲೆಯ ಮಾಸ್ತರರು ಯಾರು?
a) ಸಮಪಾಲುಗೋಪಣ್ಣ b) ಪ್ರಹ್ಲಾದ c) ಹುಚ್ಚಪ್ಪ d) ಕರೀಂಖಾನ
42. ಗೋಪಣ್ಣ ಮಾಸ್ತರು ಕಣಗಲಿ ಗಿಡದಲ್ಲಿ ಏನು ನೋಡಿದರು?
a) ಹೂ b) ಹಣ್ಣು c) ಬಟ್ಟೆ d) ಜನಿವಾರ
43. ಹಾಡುವಳಿಗೆ ಹಿಂದೆ -----ಹೆಸರಿತ್ತು.
a) ಸಂಗೀತಪುರ b) ಭಟ್ಟಳ c) ಸಾಗರ d) ಕಾರವಾರ
44. ಹೊಸಚಿಗುರು ----- ಕೂಡಿರಲು ಮರಸೊಬಗು
a) ಹೊಸ ಎಲೆ b) ಹೊಸ ಹಸಿರು c) ಹೊಸಕಾಂಡ d) ಹಳಬೇರು

45. "ಪರಿಸರ, ಪರಂಪರೆ ಹಾಗೂ ಪ್ರತ್ಯಕ್ಷ ಜೀವನದ ಅನೇಕ ಸಂಸ್ಕಾರಗಳ ಮೂಲಕ ಮನುಷ್ಯನ ಅಂತರಂಗ ಪಡೆಯುವ ಪರಿಪಕ್ವತೆ" ಯನ್ನು ----- ಎನ್ನಬಹುದು.
 a) ಕೆಲೆ b) ಸಂಸ್ಕಾರ c) ಸಂಸ್ಕೃತಿ d) ಕೆಂಪು
46. ಕುಣಬಿ ಜನರ ಮುಖ್ಯವಾದ ಹಬ್ಬ ಯಾವುದು?
 a) ಯುಗಾದಿ b) ದಸರಾ c) ಹೋಳಿ ಹುಣ್ಣಿಮೆ d) ದೀಪಾವಳಿ
47. ಬಣ್ಣಗಳನ್ನು ನೀಡುವ ಗಿಡಗಳು ಭಾರತದಲ್ಲಿ ಎಷ್ಟವೆ.
 a) 400 b) 300 c) 500 d) 600.
48. ಕರಕುಶಲ ಕಲೆಗಳಿಗೆ ಪೆಟ್ಟು ಬಿದ್ದಿದ್ದು ಯಾವುದರಿಂದ?
 a) ಜಾಗತೀಕರಣ b) ಖಾಸಗೀಕರಣ
 c) ಕೈಗಾರಿಕರಣ d) ಸೋಂಬೇರಿತನ.
49. ಸತಿ ಪದದ ಅರ್ಥವೇನು?
 a) ಹೆಂಡತಿ b) ಪತಿ c) ಮಕ್ಕಳು d) ತಾಯಿ.
50. ವಿದ್ಯಾವರ್ಧಕ ಸಂಘ ಎಲ್ಲಿ ಸ್ಥಾಪನೆಯಾಯಿತು?
 a) ಬೆಂಗಳೂರು b) ಧಾರವಾಡ
 c) ಶಿವಮೊಗ್ಗ d) ಮೈಸೂರು.

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First Semester B.E./B.Tech. Degree Examination, June/July 2024
Mathematics – I for Civil Engineering Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Derive an expression for the radius of curvature for Cartesian curve $y = f(x)$.	6	L1	CO1
	b.	Show that the pedal equation of the curve $r^m \cos m\theta = a^m$ is $pr^{m-1} = a^m$.	7	L3	CO1
	c.	Show that the radius of curvature at any point of the Cardioid $r = a(1 - \cos\theta)$ varies as $\sqrt{r}$.	7	L3	CO1
OR					
Q.2	a.	With usual notations, prove that for the curve, $r = f(\theta)$, $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$	8	L3	CO1
	b.	For the curve $y = \frac{ax}{a+x}$, show that $\left(\frac{2\rho}{a} \right)^{\frac{2}{3}} = \left(\frac{x}{y} \right)^2 + \left(\frac{y}{x} \right)^2$.	7	L3	CO1
	c.	Using modern mathematical tool, write a programme to plot the sine and cosine curve.	5	L3	CO5
Module – 2					
Q.3	a.	Evaluate $\lim_{x \rightarrow 0} \left(\frac{1^x + 2^x + 3^x}{3} \right)^{\frac{1}{x}}$.	6	L1	CO2
	b.	If $u = e^{ax+by} f(ax-by)$, prove that $b \frac{\partial u}{\partial x} + a \frac{\partial u}{\partial y} = 2abu$	7	L2	CO2
	c.	If $u = x^2 + y^2 + z^2$, $v = xy + yz + zx$, $w = x + y + z$, find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$.	7	L2	CO2
OR					
Q.4	a.	If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$ find the value of $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z}$.	7	L1	CO2
	b.	Show that $f(x, y) = x^3 + y^3 - 3xy + 1$ is minimum at (1, 1).	7	L3	CO2
	c.	Using modern mathematical tool, write a programme to show that $u_{xx} + u_{yy} = 0$, given $u = e^x [x \cos y - y \sin y]$.	6	L3	CO5
Module – 3					
Q.5	a.	Solve : $(1-x^2) \frac{dy}{dx} - xy = 1$.	6	L3	CO3
	b.	Find the orthogonal trajectories of the family of confocal and coaxial parabolas $r = \frac{2a}{1 + \cos\theta}$.	7	L1	CO3
	c.	Solve $yp^2 + (x-y)p - x = 0$	7	L2	CO3

OR					
Q.6	a.	Solve $(xy^3 + y)dx + 2(x^2y^2 + x + y^4)dy = 0$.	6	L3	CO3
	b.	A metal ball is heated to a temperature of 100°C and at time $t = 0$ it is placed in heater which is maintained at 40°C . If temperature of the ball is reduced to 60°C in 4 min, find the time at which the temperature of the ball is 50°C .	7	L2	CO3
	c.	Find the general and singular solution of $(px - y)(x - py) = 2p$ by using the substitution $x^2 = u$ and $y^2 = v$.	7	L2	CO3
Module – 4					
Q.7	a.	Solve : $(D^2 - D + 2)y = \sin 2x$.	6	L3	CO4
	b.	Solve by the method of variation of parameters $y'' - 2y' + y = e^x \log x$.	7	L3	CO4
	c.	Solve : $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^4$.	7	L3	CO4
OR					
Q.8	a.	Solve : $(4D^4 - 8D^3 - 7D^2 + 11D + 6)y = 0$.	6	L3	CO4
	b.	Find the complete solution of, $(D^2 - 2D + 2)y = x + e^x$.	7	L3	CO4
	c.	Solve $(1 + x^2) \frac{d^2y}{dx^2} + (1 + x) \frac{dy}{dx} + y = \sin(2 \log(1 + x))$	7	L3	CO4
Module – 5					
Q.9	a.	Find the rank of the matrix, $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ -2 & -3 & 1 & 2 \\ -3 & -4 & 5 & 8 \\ 1 & 3 & 10 & 14 \end{bmatrix}$	6	L1	CO4
	b.	Investigate the values of λ and μ so that the equations $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ have, (i) No solution (ii) Unique solution (iii) Infinite solution	7	L2	CO4
	c.	Using the Rayleigh's power method, find the largest Eigen value and the corresponding Eigen vector of the matrix, $A = \begin{bmatrix} 1 & -3 & 2 \\ 4 & 4 & -1 \\ 6 & 8 & 5 \end{bmatrix}$ by taking $\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T$ as the initial Eigen vector perform four iterations.	7	L2	CO4
OR					
Q.10	a.	Apply Gauss-Seidal iterative method to solve the equations $10x + y + z = 12$, $x + 10y + z = 12$, $x + y + 10z = 12$ by taking the initial approximation to the solution as, $(0, 0, 0)$ (Carry out 4 iterations).	7	L2	CO4
	b.	Solve the system of equations by Gauss Elimination method. $x + y + z = 6$, $x - y + 2z = 5$, $3x + y + z = 8$	7	L3	CO4
	c.	Using modern mathematical tool write a programme to test the consistency of the equation $x + 2y - z = 1$, $2x + y + 4z = 2$, $3x + 3y + 4z = 1$	6	L3	CO5

CBCS SCHEME

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BMATC201

**Second Semester B.E/B.Tech. Degree Supplementary Examination,
June/July 2024**

Mathematics – II for Civil Engineering Stream

Time: 3 hrs.

Max. Marks:100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks , L: Bloom's level , C: Course outcomes.
 3. VTU Formula Hand Book is permitted.

Module – 1			M	L	C
1	a.	Evaluate : $\int_0^1 \int_0^{2-x} \int_0^y (x - 2y + z) dz dy dx$.	7	L3	CO1
	b.	Evaluate : $\int_0^1 \int_1^{2-x} xy dy dx$ by changing the order of integration.	7	L3	CO1
	c.	Show that : $\int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} \times \int_0^{\pi/2} \sin \theta d\theta = \pi$.	6	L2	CO1
OR					
2	a.	Evaluate : $\int_0^1 \int_x^{\sqrt{2-x^2}} \frac{x}{\sqrt{x^2 + y^2}} dy dx$ by changing into polars.	7	L3	CO1
	b.	Find the area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by double integration.	7	L3	CO1
	c.	Write a modern mathematical tool program to evaluate the double integral $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2) dy dx$.	6	L2	CO5
Module – 2					
3	a.	Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point P(2, -1, 2).	7	L3	CO2
	b.	Show that $\vec{f} \cdot \text{curl } \vec{f} = 0$ where $\vec{f} = (x + y + 1)\hat{i} + \hat{j} - (x + y)\hat{k}$.	7	L2	CO2
	c.	Find the constants 'a' and 'b' such that, $\vec{f} = (axy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (bxz^2 - y)\hat{k}$ is irrotational. Then find ϕ such that $\vec{f} = \nabla\phi$.	6	L3	CO2

OR

4	a.	If $\vec{F} = (3x^2 + 6y)\hat{i} - 14yz\hat{j} + 20xz^2\hat{k}$. Evaluate $\int_C \vec{F} \cdot d\vec{r}$ along the curve given by $x = t, y = t^2, z = t^3$ from $t = 0$ to $t = 1$.	7	L3	CO2
	b.	Evaluate : $\int_C (xy - x^2)dx + x^2ydy$ where 'C' is the closed curve formed by $y = 0, x = 1$ and $y = x$ by Green's theorem.	7	L3	CO2
	c.	Write a modern mathematical tool program to find the gradient of $\phi = xy^2 + 2yz - 7$.	6	L3	CO5

Module – 3

5	a.	Form the partial differential equation from $z = f(y + x) + g(y + 2x)$ by eliminating arbitrary functions.	7	L2	CO3
	b.	Solve $\frac{\partial^2 z}{\partial x^2} - 16z = 0$, given that $z = 0, \frac{\partial z}{\partial x} = 4 \sin y$ when $x = 0$.	7	L3	CO3
	c.	With usual notations derive the dimensional heat equation.	6	L2	CO3

OR

6	a.	Form the partial differential equation by eliminating arbitrary function 'ϕ' from $\phi(x + y + z, x^2 + y^2 - z^2) = 0$.	7	L3	CO4
	b.	Solve $\frac{\partial^2 z}{\partial x \partial y} = \sin x \cos y$, given $\frac{\partial z}{\partial y} = -2 \cos y$ when $x = 0$ and $z = 0$ when $y = n\pi$.	7	L3	CO4
	c.	Solve $x(y^2 - z^2)P + y(z^2 - x^2)\phi = z(x^2 - y^2)$.	6	L3	CO4

Module – 4

7	a.	Find the approximate value of the real root of the equation $x^3 - 3x = 4 = 0$ using Regula – Falsi method (carry out 3 iterations).	7	L3	CO4
	b.	Given $f(40) = 184, f(50) = 204, f(60) = 226, f(70) = 250, f(80) = 276, f(90) = 304$. Find $f(85)$ using suitable interpolation formula).	7	L3	CO4
	c.	Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's $1/3^{\text{rd}}$ rule taking four equal strips and hence deduce an approximate value of π .	6	L3	CO4

OR

8	a.	Use Newton – Raphson method to find a real root of $x \sin x + \cos x = 0$ near $x = \pi$ (carry out 3 iterations).	7	L3	CO4										
	b.	Applying Lagrange’s interpolation formula to find y when $x = 4$, given : <table><tr><td>x</td><td>0</td><td>2</td><td>3</td><td>6</td></tr><tr><td>y</td><td>707</td><td>819</td><td>866</td><td>966</td></tr></table>	x	0	2	3	6	y	707	819	866	966	7	L3	CO4
x	0	2	3	6											
y	707	819	866	966											
	c.	Evaluate $\int_4^{s2} \log_e^x dx$ taking six equal strips by applying Trapezoidal Rule.	6	L3	CO4										

Module – 5

9	a.	Use Taylor's series method to find $y(0.1)$ from $\frac{dy}{dx} = x^2 + y$ with $y(0) = 10$. (consider the terms upto fourth degree).	7	L3	CO4
	b.	Given $\frac{dy}{dx} = 1 + \frac{y}{x}$, $y = 2$ at $x = 1$. Find the approximate value of y at $x = 1.2$ by taking step size $h = 0.2$ applying modified Euler's method.	7	L3	CO4
	c.	Apply Milne's method to compute $y(0.4)$ correct to four decimal places given $\frac{dy}{dx} = x^2 + \frac{y}{2}$ and the data $y(1) = 2$, $y(1.1) = 2.2156$, $y(1.2) = 2.4659$, $y(1.3) = 2.7514$.	6	L3	CO4

OR

10	a.	Using modified Euler's method, find $y(0.2)$ by taking $h = 0.2$, given that $\frac{dy}{dx} = x + \sqrt{y}$ and $y = 1$ at $x = 0$ initially.	7	L3	CO4
	b.	Using Runge – Kutta method of fourth order, find $y(0.2)$ for the equation : $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 1$ taking $h = 0.2$.	7	L3	CO4
	c.	Write a modern mathematical tool program to solve $\frac{dy}{dx} = 3e^x + 2y$, $y(0) = 0$ by Runge – Kutta fourth order method.	6	L2	CO5

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BMATC201

Second Semester B.E./B.Tech. Degree Examination, June/July 2024 Mathematics – II for Civil Engineering Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz \, dz dy dx$.	07	L2	CO1
	b.	Evaluate $\int_0^1 \int_{\sqrt{y}}^1 dx dy$ by changing the order of integration.	07	L2	CO1
	c.	Derive the relation $\beta(m,n) = \frac{\overline{m} \, \overline{n}}{\overline{m+n}}$.	06	L3	CO1
OR					
Q.2	a.	Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing into polar coordinates.	07	L2	CO1
	b.	Evaluate $\iint xy(x+y) dy dx$ taken over the area between $y = x^2$ and $y = x$.	07	L2	CO1
	c.	Write a modern mathematical tool program to evaluate the double integral $\int_0^x \int_0^x (x^2 + y^2) dy dx$.	06	L3	CO5
Module – 2					
Q.3	a.	Find the unit vector normal to the surface $x^2y - 2xz + 2y^2z^4 = 10$ at $(2, 1, -1)$.	07	L2	CO2
	b.	Show that $\vec{F} = (y+z)\mathbf{i} + (z+x)\mathbf{j} + (x+y)\mathbf{k}$ is irrotational. Also find a scalar function ϕ such that $\vec{F} = \nabla\phi$.	07	L2	CO2
	c.	If $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$, find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$.	06	L3	CO2
OR					
Q.4	a.	Verify Green's theorem in a plane for $\int_C (3x^2 - 8y^2) dx + (4y - 6xy) dy$ where c is the boundary of the region enclosed by $y = \sqrt{x}$ and $y = x^2$.	07	L2	CO2
	b.	Use Stoke's theorem to evaluate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = (x^2 + y^2)\mathbf{i} - 2xy\mathbf{j}$ and c is bounded by the lines $x = \pm a$, $y = 0$ and $y = b$.	07	L2	CO2
	c.	Write a modern mathematical tool program to evaluate curl of $\vec{F} = xy^2\mathbf{i} + 2x^2yz\mathbf{j} - 3yz^2\mathbf{k}$.	06	L3	CO5
Module – 3					
Q.5	a.	Form the partial differential equation by eliminating the arbitrary function from the relation $z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$.	07	L2	CO3

	b.	Solve $\frac{\partial^2 z}{\partial x^2} = xy$, subject to the conditions that $\frac{\partial z}{\partial x} = \log(1 + y)$ when $x = 1$ and $z = 0$ when $x = 0$.	07	L2	CO3												
	c.	Solve $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$.	06	L3	CO3												
OR																	
Q.6	a.	Form the partial differential equation from $f(xy + z^2, x + y + z) = 0$.	07	L2	CO3												
	b.	Solve $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$.	07	L2	CO3												
	c.	With usual notations, derive one dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$.	06	L3	CO3												
Module – 4																	
Q.7	a.	Find a real root of the equation $x^3 - 2x - 5 = 0$ by Regula-Falsi method. Correct to three decimal places.	07	L2	CO4												
	b.	Find $y(1.4)$ for the given data: <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>10</td><td>26</td><td>58</td><td>112</td><td>194</td></tr></table>	x	1	2	3	4	5	y	10	26	58	112	194	07	L2	CO4
x	1	2	3	4	5												
y	10	26	58	112	194												
	c.	Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's $\frac{1}{3}^{\text{rd}}$ rule taking four equal strips.	06	L3	CO4												
OR																	
Q.8	a.	Use Newton-Raphson method to find a real root of $x \sin x + \cos x = 0$ near $x = \pi$. Carry out the iterations upto four decimal places of accuracy.	07	L2	CO4												
	b.	Determine $f(4)$ for the data given below by using Newton's divided difference formula <table><tr><td>x</td><td>0</td><td>2</td><td>3</td><td>6</td></tr><tr><td>f(x)</td><td>-4</td><td>2</td><td>14</td><td>158</td></tr></table>	x	0	2	3	6	f(x)	-4	2	14	158	07	L2	CO4		
x	0	2	3	6													
f(x)	-4	2	14	158													
	c.	Evaluate $\int_0^6 3x^2 dx$ dividing the interval $[0, 6]$ into six equal parts by applying Simpson's $\frac{3}{8}^{\text{th}}$ rule.	06	L3	CO4												
Module – 5																	
Q.9	a.	Find an approximate value of y when $x = 0.1$. If $\frac{dy}{dx} = x - y^2$ and $y = 1$ at $x = 0$. Using Taylor's series method.	07	L2	CO4												
	b.	Use modified Euler's method to find y at $x = 0.1$, given $\frac{dy}{dx} = 3x + \frac{y}{2}$ with $y(0) = 1$ taking $h = 0.1$. Perform three iterations.	07	L2	CO4												
	c.	Given $\frac{dy}{dx} = x^2 + y^2$, $y(0) = 1$, $y(0.1) = 1.1113$, $y(0.2) = 1.2507$, $y(0.3) = 1.426$. Compute $y(0.4)$, using Milne's predictor-corrector method.	06	L3	CO4												
OR																	
Q.10	a.	Using Runge-Kutta method of fourth order to solve $\frac{dy}{dx} = \frac{y-x}{y+x}$ with $y(0) = 1$ taking $h = 0.1$ to find $y(0.1)$.	07	L2	CO4												
	b.	Solve by using modified Euler's method $\frac{dy}{dx} = 1 + \frac{y}{x}$, $y = 2$ at $x = 1$. Find y at $x = 1.2$ by taking $h = 0.2$.	07	L2	CO4												
	c.	Write a modern mathematical tool to solve $\frac{dy}{dx} = e^{-x}$ with $y(0) = -1$ using Euler's method at $x = 0.2$ (0.2) 0.6.	06	L3	CO5												

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BMATE101

First Semester B.E./B.Tech. Degree Examination, June/July 2024
Mathematics - I for EEE Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	With usual notation, prove that $\rho = \frac{(r^2 + r_1^2)^{\frac{3}{2}}}{r^2 + 2r_1^2 - rr_2}$.	6	L1	CO1
	b.	Find the angle between the curves $r = a \log \theta$ and $r = \frac{a}{\log \theta}$.	7	L2	CO1
	c.	Find the Pedal equation of the curve $r^n = a^n \cos n\theta$.	7	L1	CO1
OR					
Q.2	a.	With usual notation, prove that $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$	8	L1	CO1
	b.	Show that the radius of curvature of $x = a \cos^3 \theta$ and $y = a \sin^3 \theta$.	7	L3	CO1
	c.	Using modern mathematical tool, write a programme / code to plot the curve $r = 2 \cos 2\theta $.	5	L3	CO5
Module – 2					
Q.3	a.	Evaluate : (i) $\lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x}$ (ii) $\lim_{x \rightarrow 1} (1-x^2)^{\frac{1}{\log(1-x)}}$.	6	L2	CO2
	b.	Calculate the $J \left(\frac{u, v, w}{x, y, z} \right)$, if $U = x + 2y + z$, $V = x + 2y + 3z$, $W = 2x + 3y + 5z$	7	L3	CO2
	c.	Find the extreme values of $\sin A + \sin B + \sin(A+B)$.	7	L3	CO2
OR					
Q.4	a.	Expand $\sqrt{1 - \sin 2x}$ by Maclaurin's series upto the term containing x^4 .	8	L2	CO2
	b.	If z is a function of x and y where $x = e^u + e^{-v}$ and $y = e^{-u} - e^v$ show that $z_u - z_v = xz_x - yz_y$.	7	L2	CO2
	c.	Using Modern mathematical tool. Write a programme/code to show that $u_{xx} + u_{yy} = 0$ given $u = e^x(x \cos y - y \sin y)$	5	L2	CO5
Module – 3					
Q.5	a.	Solve : $\frac{dy}{dx} + \frac{y}{x} = y^2 x$.	6	L2	CO3
	b.	Find the orthogonal trajectories of the family of curves $\frac{x^2}{a^2} + \frac{y^2}{b^2 + \alpha} = 1$, where α is a parameter.	7	L3	CO3
	c.	Solve : $xyp^2 - (x^2 + y^2)p + xy = 0$.	7	L1	CO3

OR					
Q.6	a.	Solve : $[2xy + y - \tan y]dx + [x^2 - x \tan^2 y + \sec^2 y] dy = 0$.	6	L2	CO3
	b.	A series circuit with resistance R, inductance L with electromotive force E, the current i and time t is given by $L \frac{di}{dt} + iR = E$. Find the current at any time t when initial current i = 0.	7	L3	CO3
	c.	Solve the equation $(px - y)(py + x) = 2p$ by reducing into Clairaut's form, taking the substitution $X = x^2$, $Y = y^2$.	7	L2	CO3
Module – 4					
Q.7	a.	Evaluate : $\int_0^{2\pi} \int_0^\pi \int_0^a r^4 (\sin \phi) dr d\phi d\theta$.	6	L2	CO4
	b.	Evaluate : $\int_0^1 \int_{\sqrt{y}}^{2-y} x^2 dx dy$.	7	L2	CO4
	c.	Define Beta and Gama function and show that $\beta(m, n) = 2 \int_0^{\frac{\pi}{2}} (\sin^{2m-1} \theta) (\cos^{2n-1} \theta) d\theta$.	7	L2	CO4
OR					
Q.8	a.	Evaluate by changing the order of integration $\int_0^a \int_y^a \frac{x}{x^2 + y^2} dy dx$.	6	L2	CO4
	b.	Find the volume bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y + z = 4$ and $z = 0$.	7	L2	CO4
	c.	Evaluate $\int_0^1 x^{\frac{3}{2}} (1-x)^{\frac{1}{2}} dx$, by expressing in terms of Gamma and Beta function.	7	L2	CO4
Module – 5					
Q.9	a.	Using Gauss-Jordan method, solve : $x + 3y - 2z = 7$; $x + 2y - 3z = 10$; $2x - y + z = 5$.	6	L3	CO5
	b.	Solve by Gauss-Seidal iteration method, $8x - y + z = 18$; $2x + 5y - 2z = 3$; $x + y - 3z = -16$; taking (0, 0, 0) as an initial approximate. (Carry out 4 iterations).	7	L3	CO5
	c.	Find the value of λ and μ the system of equations $x + y + z = 6$; $x + 2y + 3z = 10$; $x + 2y + \lambda z = \mu$ has (i) No solution (ii) Unique solution (iii) Infinite solution.	7	L3	CO5
OR					
Q.10	a.	Find the rank of matrix $\begin{bmatrix} 1 & 4 & 9 & 16 \\ 4 & 9 & 16 & 25 \\ 9 & 16 & 25 & 36 \\ 16 & 25 & 36 & 49 \end{bmatrix}$.	8	L2	CO5

	b. Test for consistency and solve : $x + 2y + 2z = 1$; $2x + y + z = 2$; $3x + 2y + 2z = 3$; $y + z = 0$	7	L3	CO5
	c. Using modern mathematical tool write a programme/code to find the largest eigen value of $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ by power method.	5	L3	CO5

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BMATE201

Second Semester B.E./B.Tech. Degree Supplementary Examination, June/July 2024

Mathematics – II for EEE Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Find the angle between the surfaces $xy^2y = 3x + z^2$ and $3x^2 - y^2 + 2z = 1$ at the point (1, -2, 1).		7	L2	CO1
	b.	If $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$, find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$.		7	L2	CO1
	c.	Show that the vector $\vec{F} = \frac{x\hat{i} + y\hat{j}}{x^2 + y^2}$ is both solenoidal and irrotational.		6	L3	CO1
OR						
Q.2	a.	Find the total work done by the force $\vec{F} = 3xy\hat{i} - 5z\hat{j} + 10x\hat{k}$ along the curve $x = t^2 + 1$; $y = 2t^2$, $z = t^3$ from $t = 1$ to $t = 2$.		7	L2	CO1
	b.	Using Green's theorem, evaluate $\int_c (xy + y^2)dx + x^2dy$ where 'c' is the closed curve of the region bounded by $y = x$ and $y = x^2$.		7	L3	CO1
	c.	Using modern mathematical tools, write the code to find the gradient of $\phi = x^2y + 2xz - 4$.		6	L2	CO5
Module – 2						
Q.3	a.	Define a Subspace. Show that the intersection of two subspaces of a vector V is also a subspace of V.		7	L2	CO2
	b.	Show that $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x, y, -z)$ is linear transformation.		7	L3	CO2
	c.	If $u = [2, -5, -1]^T$, $v = [-7, -4, 6]^T$, compute : i) $\langle u, v \rangle$ ii) $\ u\ ^2$ iii) $\ v\ ^2$ iv) $\ u + v\ ^2$.		6	L2	CO2
OR						
Q.4	a.	Define linearly independent and linearly dependent set of vectors. Test the vectors $v_1 = [3, 0, -6]^T$, $v_2 = [-4, 1, 7]^T$ and $v_3 = [-2, 1, 5]^T$ forms a basis.		7	L2	CO2
	b.	State Rank – Nullity Theorem. For the matrix $A = \begin{bmatrix} 1 & -4 & 9 & -7 \\ -1 & 2 & -4 & 1 \\ 5 & -6 & 10 & 7 \end{bmatrix}$, Find : i) Rank of A ii) Dim (Nul A) iii) Bases		7	L3	CO2

	c.	Using the modern mathematical tool, write the code to represent the reflection transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and to find the image of vector (10, 0) when it is reflected about y – axis.	6	L2	CO5												
Module – 3																	
Q.5	a.	Find the Laplace Transform of (i) $e^{-3t} \cos 2t$ ii) $\frac{\cos at - \cos bt}{t}$.	7	L2	CO3												
	b.	Find the Laplace Transform of the square wave function of period $2a$, defined by $f(t) = \begin{cases} k & 0 < t < a \\ -k & a < t < 2a \end{cases}$.	7	L2	CO3												
	c.	Explain $f(t) = \begin{cases} \cos t & 0 < t < \pi \\ \cos 2t & \pi < t < 2\pi \\ \cos 3t & t > 2\pi \end{cases}$ in term of the unit step function and hence find $L[f(t)]$.	6	L3	CO3												
OR																	
Q.6	a.	Find the inverse Laplace transformer of i) $\frac{2s-1}{s^2+4s+29}$ ii) $\frac{1}{(s-4)^2}$.	7	L2	CO3												
	b.	Using the convolution theorem, find the inverse Laplace transform of $\frac{1}{(s-1)(s^2+1)}$.	7	L3	CO3												
	c.	Solve by the Laplace transforms $y'' + k^2y = 0$, given that $y(0) = 2$, $y'(0) = 0$.	6	L2	CO3												
Module – 4																	
Q.7	a.	Find the real root of $x \log_{10} x = 1.2$ by Regula – Falsi method correct to 2 decimal places the root lies between (2, 3).	7	L2	CO4												
	b.	Find interpolating polynomial by Newton's divided difference formula for the data $f(1) = 4$, $f(3) = 32$, $f(4) = 55$ and $f(6) = 119$.	7	L2	CO4												
	c.	Evaluate using Simpson's $\frac{1}{3}$ rule $\int_0^6 \frac{e^x}{1+x} dx$ by taking six equal parts.	6	L2	CO4												
OR																	
Q.8	a.	Find the real root of the equation $\cos x = xe^x$, using Newton's – Raphson method, correct to 3 decimal places taking $x_0 = 0.5$.	7	L2	CO4												
	b.	Use Newton's backward interpolation formula to compute the value of y when $x = 6$, given that <table border="1" data-bbox="646 1854 933 1930"> <tr> <td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr> <tr> <td>y</td><td>1</td><td>-1</td><td>1</td><td>-1</td><td>1</td></tr> </table>	x	1	2	3	4	5	y	1	-1	1	-1	1	7	L3	CO4
x	1	2	3	4	5												
y	1	-1	1	-1	1												

	c.	Evaluate $\int_0^5 \frac{dx}{4x+5}$, by Trapezoidal rule, taking 6 ordinates.	6	L2	CO4
Module – 5					
Q.9	a.	Employ Taylors series method to find $y(0.2)$, given that $\frac{dy}{dx} = 2y + 3e^x$, $y(0) = 0$.	7	L3	CO4
	b.	Using Modified Euler's method, find $y(0.1)$ correct to 4 decimal places, given that $y' = x - y^2$, $y(0) = 1$, $h = 0.1$, perform 2 iterations.	7	L2	CO4
	c.	Employ Milne's predictor – corrector method given that $y' = x^2(1 + y)$ $y(1) = 1$, $y(1.1) = 1.233$, $y(1.2) = 1.548$, $y(1.3) = 1.979$ to find $y(1.4)$.	6	L3	CO4
OR					
Q.10	a.	Solve $y' = \log_{10}(x + y)$, by modified Euler's method at $x = 0.2$ and $x = 0.4$ with $h = 0.2$, perform 2 iterations at each stage.	7	L2	CO4
	b.	Use 4 th order Runge – Kutta method to solve $(x + y) y' = 1$ with $y(0.4) = 1$, at $x = 0.5$ correct to 4 decimal places.	7	L2	CO4
	c.	Using modern mathematical tools, write a code to find $y(0.1)$, given $y' = x - y$, $y(0) = 1$ by Taylors series.	6	L3	CO5

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Second Semester B.E./B.Tech. Degree Examination, June/July 2024 Mathematics – II for EEE Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Find the directional derivatives of $\phi = x^2yz + 4xz^2$, at $(1, 2, -1)$ along $2\mathbf{i} - \mathbf{j} - 2\mathbf{k}$.	7	L3	CO1
	b.	Find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$, where $\vec{F} = \text{grad}(x^3 + y^3 + z^3 - 3xyz)$.	7	L3	CO1
	c.	Show that the vector, $\vec{F} = \frac{x\hat{i} + y\hat{j}}{x^2 + y^2}$ is both solenoidal and irrotational.	6	L2	CO1
OR					
Q.2	a.	Find the work done in moving a particle in the Force field $F = 3x^2\hat{i} + (2xz - y)\hat{j} + z\hat{k}$ along the straight line from $(0, 0, 0)$ to $(2, 1, 3)$.	7	L3	CO1
	b.	Using Green's theorem, evaluate $\oint (xy + y^2)dx + x^2dy$ over the region bounded by the curves $y = x$ and $y = x^2$.	7	L3	CO1
	c.	Using modern mathematical tools, write a code to find the divergence and curl of the vector $x^2yz\hat{i} + y^2zx\hat{j} + z^2xy\hat{k}$.	6	L2	CO5
Module – 2					
Q.3	a.	Define a subspace. Show that the intersection of two subspaces of a vector space V is also a subspace of V .	7	L2	CO2
	b.	Define a basis for a vector space. Determine whether or not the vectors : $(2, 2, 1), (1, 3, 7), (1, 2, 2)$ form a basis of R^3 .	7	L2	CO2
	c.	Show that $T : R^2 \rightarrow R^3$ given by $T(x, y) = (x + y, x - y, y)$ is a linear transformation.	6	L2	CO2
OR					
Q.4	a.	Define linearly independent set of vectors and linearly dependent set of vectors. Show that the vectors $(1, 4, 9), (3, 1, 4), (9, 3, 12)$ are linearly dependent.	7	L2	CO2
	b.	Verify the Rank-Nullity theorem for $T : R^3 \rightarrow R^3$ defined by $T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$.	7	L2	CO2
	c.	Using the modern mathematical tool, write the code to represent the reflection transformation $T : R^2 \rightarrow R^2$ and to find the image of vector $(10, 0)$ when it is reflected about the y - axis.	6	L2	CO5
Module – 3					
Q.5	a.	Find the Laplace Transform of, (i) $e^{-3t}(2\cos 5t - 3\sin 5t)$ (ii) $\frac{\cos at - \cos bt}{t}$	7	L2	CO3

	b.	Find the Laplace transform of the triangular wave function, $f(t) = \begin{cases} t, & \text{if } 0 \leq t \leq a \\ 2a - t, & \text{if } a \leq t \leq 2a \end{cases}$	7	L2	CO3										
	c.	Express $f(t) = \begin{cases} t^2, & 1 < t \leq 2 \\ 4t, & t > 2 \end{cases}$ in terms of Heaviside unit step function and hence find $L(f(t))$.	6	L3	CO3										
OR															
Q.6	a.	Find $L^{-1} \left[\frac{2s^2 - 6s + 5}{s^3 - 6s^2 + 11s - 6} \right]$.	7	L2	CO3										
	b.	Find $L^{-1} \left[\frac{1}{s(s^2 + a^2)} \right]$ using convolution theorem.	7	L2	CO3										
	c.	Solve the differential equation by using Laplace Transform method. $y'' + 6y' + 9y = 12t^2 e^{-3t}$, $y(0) = y'(0) = 0$	6	L3	CO3										
Module – 4															
Q.7	a.	By Newton-Raphson method, find the root of $x \sin x + \cos x = 0$, near $x = \pi$. Carryout the iteration upto four decimal places of accuracy.	7	L2	CO4										
	b.	Using Lagrange's interpolation formula, find y at $x = 2$, using the points $(0, -12), (1, 0), (3, 6), (4, 12)$	7	L2	CO4										
	c.	Using Simpson's $\left(\frac{1}{3}\right)^{\text{rd}}$ rule, evaluate $\int_0^{0.6} e^{-x^2}$ by taking seven ordinates.	6	L3	CO4										
OR															
Q.8	a.	Find a real root of the equation $x^3 - 4x - 9 = 0$ correct to three decimal places by the method of False position in $(2, 3)$	7	L2	CO4										
	b.	Construct Newton's forward interpolation polynomial for the data : <table border="1"><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>f(x)</td><td>1</td><td>2</td><td>1</td><td>10</td></tr></table>	x	0	1	2	3	f(x)	1	2	1	10	7	L2	CO4
x	0	1	2	3											
f(x)	1	2	1	10											
	c.	Evaluate $\int_0^1 \frac{dx}{(1+x)^2}$ by using Simpson's $\left(\frac{3}{8}\right)^{\text{th}}$ rule, by taking 6 equal intervals.	6	L3	CO4										
Module – 5															
Q.9	a.	Use Taylor series method to find $y(0.2)$ from $\frac{dy}{dx} = 2y + 3e^x$, with $y(0) = 0$.	7	L3	CO5										
	b.	Using R-K method of order 4, solve $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$ with $y(0) = 1$ at $x = 0.2$.	7	L3	CO5										
	c.	Applying Milne's Predictor-Corrector method, find $y(0.4)$, from $\frac{dy}{dx} = 2e^x - y$, given that, $y(0) = 2$, $y(0.1) = 2.010$, $y(0.2) = 2.040$, $y(0.3) = 2.090$	6	L3	CO5										
OR															

Q.10	a.	Solve by using modified Euler's method, $y' = 1 + \frac{y}{x}$, $y(1) = 2$ at $x = 1.2$ and $x = 1.4$.	7	L3	CO5
	b.	Using the Runge-Kutta method of fourth order find $y(1.1)$, given $\frac{dy}{dx} = xy^{\frac{1}{3}}$, taking $h = 0.1$, $y(1) = 1$.	7	L3	CO5
	c.	Using modern mathematical tools, write a code to find $y(1.4)$, given $\frac{dy}{dx} = x^2 + \frac{y}{2}$, $y(1) = 2$, $y(1.1) = 2.2156$, $y(1.2) = 2.4649$, $y(1.3) = 2.7514$, by Milne's Predictor and Corrector method.	6	L3	CO5

First Semester B.E./B.Tech. Degree Examination, June/July 2024 Mathematics – I for Mechanical Engineering Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book is permitted.

3. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Find the angle of intersection of the curves $r = a(1 + \cos \theta)$, $r = b(1 - \cos \theta)$	06	L2	CO1
	b.	With the usual notations prove that for the given curve $r = f(\theta)$, $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$.	07	L2	CO1
	c.	Find the radius of curvature of the curve $x^4 + y^4 = 2$ at (1, 1)	07	L3	CO1
OR					
Q.2	a.	Find the pedal equation for the curve $a^m = r^m \cos m\theta$.	08	L2	CO1
	b.	Find the radius of curvature for the curves $x = a(\cos t + t \sin t)$ and $y = a(\sin t - t \cos t)$	07	L3	CO1
	c.	Using modern mathematical tool write a program/code to plot the curve $r = 2 \cos 2\theta $	05	L3	CO5
Module – 2					
Q.3	a.	Using Maclaurin's theorem prove that $\sqrt{1 + \sin x} = 1 + x - \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} - \dots$	06	L2	CO2
	b.	If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$ then find $6U_x + 4U_y + 3U_z = 0$	07	L2	CO2
	c.	Examine the function $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$ for extreme value.	07	L3	CO2
OR					
Q.4	a.	If $u = x + 3y^2 - z^3$, $v = x^2yz$, $w = 2x^2 - xy$, evaluate $\frac{\partial(u, v, w)}{\partial(x, y, z)}$	08	L3	CO2
	b.	If $u = x^3y^2 + x^2y^3$ where $x = at^2$, $y = 2at$ find $\frac{du}{dt}$	07	L3	CO2
	c.	Using modern mathematical tool write a program/code to evaluate $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x$	05	L3	CO5
Module – 3					
Q.5	a.	Solve $\frac{dy}{dx} + \frac{y \cos x + \sin y + y}{\sin x + x \cos y + x} = 0$	06	L2	CO3
	b.	Find the orthogonal trajectories of cardioid $r = a(1 + \cos \theta)$	07	L2	CO3
	c.	Solve $yp^2 + (x - y)p - x = 0$	07	L2	CO3

OR					
Q.6	a.	Solve $\frac{dy}{dx} + \frac{y}{x} = y^2 x$	06	L2	CO3
	b.	A body is originally at 80°C and cools down at 60°C in 20 minutes. If the temperature of the air is 40°C, find the temperature of the body after 40 minutes.	07	L3	CO3
	c.	Find the general and singular solution of $x^2 + xp - yp + 1 - y = 0$	07	L2	CO3
Module – 4					
Q.7	a.	Solve $(4D^4 - 8D^3 - 7D^2 + 11D + 6)y = 0$	06	L2	CO3
	b.	Solve $(D^2 - 6D + 9)y = 6e^{3x} - \log 2$	07	L3	CO3
	c.	Solve $(1+x)^2 \frac{d^2y}{dx^2} + (1+x) \frac{dy}{dx} + y = \sin 2[\log(1+x)]$	07	L2	CO3
OR					
Q.8	a.	Solve $(D^2 - 4D + 13)y = \cos 2x$	06	L2	CO3
	b.	Solve by variation of parameters $(D^2 + 1)y = \tan x$	07	L2	CO3
	c.	Solve $x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} + 4y = (1+x)^2$	07	L2	CO3
Module – 5					
Q.9	a.	Find the rank of the matrix $\begin{bmatrix} 21 & 22 & 23 & 24 \\ 22 & 23 & 24 & 25 \\ 23 & 24 & 25 & 26 \\ 24 & 25 & 26 & 27 \end{bmatrix}$	06	L2	CO4
	b.	For what value of λ and μ the system of equations $2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$, $2x + 3y + \lambda z = \mu$ has i) no solution ii) a unique solution iii) infinite number of solutions.	07	L2	CO4
	c.	Using Rayleigh's power method find the dominant eigen value and the corresponding eigen vector of $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$ by taking $[1, 0, 0]^T$ as the initial eigen vector. (Carry out 6 iterations).	07	L3	CO4
OR					
Q.10	a.	Solve the system of equation by Gauss Elimination method. $2x + y + 4z = 12$, $4x + 11y - z = 33$, $x + 2y + 5z = 20$	07	L3	CO4
	b.	Solve the system of equations using Gauss-Seidel method by taking (0,0,0) as an initial approximate root. $5x + 2y + z = 12$, $x + 4y + 2z = 15$, $x + 2y + 5z = 20$	08	L3	CO4
	c.	Using modern mathematical tool write a program/code to find the largest eigen value of $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ by power method.	05	L3	CO5

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**Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Mathematics – II for Mechanical Engineering Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Evaluate $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) dy dx dz$	7	L3	CO1
	b.	Evaluate $\int_0^3 \int_0^{\sqrt{4-y}} (x+y) dx dy$ by changing the order of integration.	7	L3	CO1
	c.	Show that $\int_0^\infty \sqrt{y} e^{-y^2} dy \times \int_0^\infty \frac{e-y^2}{\sqrt{y}} dy = \frac{\pi}{2\sqrt{2}}$	6	L2	CO1
OR					
Q.2	a.	Show that $\beta(m,n) = \frac{\Gamma_{(m)} \cdot \Gamma_{(n)}}{\Gamma_{(m+n)}}$	7	L2	CO1
	b.	Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} y\sqrt{x^2+y^2} dx dy$ by changing into polars.	7	L3	CO1
	c.	Write a modern mathematical tool program to find the volume of the tetrahedron bounded by the planes $x=0$, $y=0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.	6	L3	CO5
Module – 2					
Q.3	a.	Find the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at $(2, -1, 2)$.	7	L2	CO2
	b.	If $\vec{F} = (3x^2y - z)\hat{i} + (xz^3 + y)\hat{j} - 2x^3z^2\hat{k}$, find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$.	7	L2	CO2
	c.	Find the constants a , b and c such that the vector $\vec{F} = (x+y+az)\hat{i} + (bx+2y-z)\hat{j} + (x+cy+2z)\hat{k}$ is irrotational.	6	L2	CO2
OR					
Q.4	a.	Using Grun's theorem, evaluate $\int_C (2x^2 - y^2) dx + (x^2 + y^2) dy$, where C is the boundary of the region in the xy -plane enclosed by the x -axis and the upper-half of the circle $x^2 + y^2 = a^2$.	7	L3	CO2
1 of 3					

	b.	If $\vec{F} = 2xy\hat{i} + yz^2\hat{j} + xz\hat{k}$ and S is the rectangular parallelepiped bounded by $x = 0, y = 0, z = 0, x = 2, y = 1, z = 3$. Find the Flux across S.	7	L3	CO2												
	c.	Write the modern mathematical tool program to find the divergence of the vector field $\vec{F} = (3x^2 - 3yz)\hat{i} + (3y^2 - 3xz)\hat{j} + (3z^2 - 3xy)\hat{k}$	6	L3	CO5												
Module – 3																	
Q.5	a.	Form the partial differential equation by eliminating the arbitrary function ϕ from $\ell x + my + nz = \phi(x^2 + y^2 + z^2)$.	7	L2	CO3												
	b.	Solve $\frac{\partial^2 z}{\partial x \partial y} = \frac{x}{y}$, subject to the conditions that $\frac{\partial z}{\partial x} = \log_e x$ when $y = 1$ and $z = 0$ when $x = 1$.	7	L3	CO3												
	c.	Derive the dimensional heat equation in the standard form $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$.	6	L2	CO3												
OR																	
Q.6	a.	Form the partial differential equation by eliminating the arbitrary constants a and b from $2z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$.	7	L2	CO3												
	b.	Solve $\frac{\partial^2 z}{\partial x^2} = a^2 z$ given that when $x = 0$, $\frac{\partial z}{\partial x} = a \sin y$ and $\frac{\partial z}{\partial y} = 0$.	7	L3	CO3												
	c.	Solve $(y - z)p + (z - x)q = (x - y)$.	6	L2	CO3												
Module – 4																	
Q.7	a.	Find an approximate value of the root of the equation $\cos x = 3x - 1$ that lies between 0.5 and 1 correct to three decimal places using Regula false method.	7	L3	CO4												
	b.	The following table gives the distances (in miles) of the visible horizon for the given heights (in feet) above the earth's surface. <table border="1" data-bbox="518 1478 1061 1556"> <tr> <td>x</td><td>200</td><td>250</td><td>300</td><td>350</td><td>400</td></tr> <tr> <td>F(x)</td><td>15.04</td><td>16.81</td><td>18.42</td><td>19.9</td><td>21.27</td></tr> </table> Find y for $x = 218$.	x	200	250	300	350	400	F(x)	15.04	16.81	18.42	19.9	21.27	7	L3	CO4
x	200	250	300	350	400												
F(x)	15.04	16.81	18.42	19.9	21.27												
	c.	Evaluate $\int_0^\pi e^{\sin \theta} d\theta$ by using Simpson's $\left(\frac{1}{3}\right)^{rd}$ rule by taking 7 ordinates.	6	L3	CO4												
OR																	
Q.8	a.	Using the Newton-Raphson method, find the real root of the equation $e^x \sin x = 1$. (Here x is in radians).	7	L3	CO4												

	b.	Using Newton's divided difference formula, evaluate F(9) from the following table. <table><tr><td>x</td><td>5</td><td>7</td><td>11</td><td>13</td><td>17</td></tr><tr><td>F(x)</td><td>150</td><td>392</td><td>1452</td><td>2366</td><td>5202</td></tr></table>	x	5	7	11	13	17	F(x)	150	392	1452	2366	5202	7	L2	CO4
x	5	7	11	13	17												
F(x)	150	392	1452	2366	5202												
	c.	If y(0) = -12, y(1) = 0, y(3) = 6 and y(4) = 12, find the value of y at x = 2 using Lagranges method.	6	L3	CO4												

Module – 5

Q.9	a. Find by Taylor's series method the value of y at $x = 0.1$ to 4 decimal places from $\frac{dy}{dx} = x - y^2$, $y(0) = 1$.	7	L2	CO4
	b. Using the Runge-Kutta method of fourth order, find $y(1.1)$ given that $\frac{dy}{dx} = xy^{1/3}$ taking $h = 0.1$.	7	L3	CO4
	c. Given that $\frac{dy}{dx} = 2e^x - y$ and the data $y(0) = 2$, $y(0.1) = 2.010$, $y(0.2) = 2.040$, $y(0.3) = 2.090$. Compute y at $x = 0.4$ by applying Milne's method.	6	L3	CO4

OR

Q.10	a. Using the modified Euler's method, find $y(20.2)$ given that $\frac{dy}{dx} = \log_{10}(x/y)$ with $y(20) = 5$ taking $h = 0.2$.	7	L3	CO4
	b. Apply the Runge-Kutta method of fourth order, to find an approximate value of y at $x = 0.1$, given that $\frac{dy}{dx} = 3e^x + 2y$ with $y(0) = 0$ and $h = 0.1$	7	L3	CO4
	c. Using modern mathematical tools write a program to find y at $x = 1.4$, given $\frac{dy}{dx} = x^2 + \frac{y}{2}$, $y(1) = 2$, $y(1.1) = 2.2156$, $y(1.2) = 2.4649$, $y(1.3) = 2.7514$. Use corrector formula thrice using Milne's method.	6	L3	CO5

Second Semester B.E./B.Tech. Degree Examination, June/July 2024 Mathematics – II for Mechanical Engineering Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.
3. Mathematics hand book is permitted.*

Module – 1			M	L	C
Q.1	a.	Evaluate $\int_0^a \int_0^b \int_0^c (x^2 + y^2 + z^2) dx dy dz$.	6	L2	CO1
	b.	Change the order of integration in $\int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} dy dx$ and hence evaluate the same.	7	L2	CO1
	c.	Derive the relation between Beta and Gamma function.	7	L2	CO1
OR					
Q.2	a.	Evaluate $\int_0^1 \int_0^{\sqrt{1-y^2}} (x^2 + y^2) dx dy$ by changing to polar coordinates.	7	L2	CO1
	b.	Using double integration, find the volume of the tetrahedron bounded by the planes $x = 0, y = 0, z = 0$ and $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$.	7	L2	CO1
	c.	Write a modern mathematical program to evaluate the integral $\int_0^3 \int_0^{3-x} \int_0^{3-x-y} xyz dz dy dx$.	6	L3	CO5
Module – 2					
Q.3	a.	If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then show that $\nabla r^n = nr^{n-2}\vec{r}$.	7	L2	CO2
	b.	Find the directional derivative of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ along $2\hat{i} - \hat{j} - 2\hat{k}$.	7	L2	CO2
	c.	If $\vec{F} = (x + y + az)\hat{i} + (bx + 2y - z)\hat{j} + (x + cy + 2z)\hat{k}$, find a, b, c such that $\text{curl } \vec{F} = \vec{0}$.	6	L2	CO2
OR					
Q.4	a.	Find the work done in moving a particle in the force field $\vec{F} = 3x^2\hat{i} + (2xz - y)\hat{j} + z\hat{k}$ along the straight line from $(0, 0, 0)$ to $(2, 1, 3)$.	7	L3	CO2

	b.	Using Green's theorem, evaluate $\int_C (xy + y^2)dx + x^2dy$, where C is the closed curve of the region bounded by $y = x$ and $y = x^2$.	7	L2	CO2												
	c.	Write a modern mathematical tool program to find the divergence of $\vec{F} = x^2yz\hat{i} + y^2zx\hat{j} + z^2xy\hat{k}$.	6	L3	CO2												
Module – 3																	
Q.5	a.	Form the partial differential equation from the relation $z = f(y + 2x) + g(y - 3x)$.	6	L1	CO3												
	b.	Solve $\frac{\partial^2 z}{\partial x^2} = xy$ subject to the conditions $\frac{\partial z}{\partial x} = \log(1 + y)$ when $x = 1$ and $z = 0$ when $x = 0$.	7	L2	CO3												
	c.	Derive one dimensional heat equation.	7	L2	CO3												
OR																	
Q.6	a.	Form the partial differential equation from the relation $f(xy + z^2, x + y + z) = 0$	6	L2	CO3												
	b.	Solve $\frac{\partial^2 z}{\partial x^2} + z = 0$ given that when $x = 0$, $z = e^y$ and $\frac{\partial z}{\partial x} = 1$.	7	L2	CO3												
	c.	Solve $(mz - ny)p + (nx - lz)q = ly - mx$.	7	L2	CO3												
Module – 4																	
Q.7	a.	Find the real root of the equation $\cos x - xe^x = 0$ in $(0.5, 0.6)$ using the Regula – Falsi method correct to four decimal places, carryout there iterations.	7	L2	CO4												
	b.	The population of a town is given by the table <table border="1"> <tr> <td>Year</td><td>1951</td><td>1961</td><td>1971</td><td>1981</td><td>1991</td></tr> <tr> <td>Population in thousands</td><td>19.96</td><td>39.65</td><td>58.81</td><td>77.21</td><td>94.61</td></tr> </table> Using Newton's forward interpolation formula, calculate the population in the year 1955.	Year	1951	1961	1971	1981	1991	Population in thousands	19.96	39.65	58.81	77.21	94.61	7	L2	CO4
Year	1951	1961	1971	1981	1991												
Population in thousands	19.96	39.65	58.81	77.21	94.61												
	c.	Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using Simpson's $1/3^{rd}$ rule. [Take 6 equal parts].	6	L3	CO4												
OR																	
Q.8	a.	Find a real root of the equation $x^3 + 5x - 11 = 0$ near to $x = 1$ using Newton – Raphson method. Carryout three iterations.	7	L3	CO4												

	b.	Using Newton's divided difference formula, evaluate f(4) from the following table <table><tr><td>x</td><td>0</td><td>2</td><td>3</td><td>6</td></tr><tr><td>f(x)</td><td>-4</td><td>2</td><td>14</td><td>158</td></tr></table>	x	0	2	3	6	f(x)	-4	2	14	158	7	L3	CO4
x	0	2	3	6											
f(x)	-4	2	14	158											
	c.	Evaluate $\int_0^1 \frac{dx}{1+x}$ taking seven ordinates by applying Simpson's (3/8) th rule.	6	L3	CO4										
Module – 5															
Q.9	a.	Using Taylor's series method, find y(0.1) considering upto fourth degree term if y(x) satisfies the equation $\frac{dy}{dx} = x - y^2, y(0) = 1$.	6	L2	CO4										
	b.	Using Runge-Kutta method of fourth order, find y(0.2) for the equation $\frac{dy}{dx} = \frac{y-x}{y+x}, y(0) = 1$ taking h = 0.2.	7	L3	CO4										
	c.	Given that $\frac{dy}{dx} = x - y^2$ and the data y(0) = 0, y(0.2) = 0.02, y(0.4) = 0.0795, y(0.6) = 0.1762. Compute y at x = 0.8 applying Milne's method.	7	L3	CO4										
OR															
Q.10	a.	Using modified Euler's method, compute y(1.1) given that $\frac{dy}{dx} + \frac{y}{x} = \frac{1}{x^2}$ and y = 1 at x = 1 by taking h = 0.1.	7	L3	CO5										
	b.	Apply Runge-Kutta fourth order method, to find an approximate value of y when x = 0.2, given that $\frac{dy}{dx} = x + y$ and y = 1 when x = 0.	7	L3	CO5										
	c.	Using modern mathematical tools with a program to find y when x = 1.4, given $\frac{dy}{dx} = x^2 + (y/2), y(1) = 2, y(1.1) = 2.2156, y(1.2) = 2.4649, y(1.3) = 2.7514$ using predictor corrector method.	6	L3	CO5										

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First Semester B.E./B.Tech. Degree Examination, June/July 2024
Mathematics for CSE Stream – I

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	With usual notation prove that $\tan \phi = r \frac{d\theta}{dr}$.	6	L2	CO1
	b.	Find the angle between the curves $r = \frac{a}{1 + \cos \theta}$ and $r = \frac{b}{1 - \cos \theta}$.	7	L2	CO1
	c.	Show that the radius of curvature at any point θ on the cycloid $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$ is $4 \cos \left(\frac{\theta}{2} \right)$.	7	L3	CO1
OR					
Q.2	a.	Find the pedal equation of the curve $r(1 - \cos \theta) = 2a$.	7	L2	CO1
	b.	Find the radius of curvature for the curve $y^2 = \frac{a^2(a - x)}{x}$ at the point $(a, 0)$.	8	L3	CO1
	c.	Using modern mathematical tool write a program / code to plot the sine and cosine curve.	5	L3	CO5
Module – 2					
Q.3	a.	Expand $\log(\sec x)$ upto the term containing x^4 using Maclausin's series.	6	L2	CO2
	b.	If $u = \log(\tan x + \tan y + \tan z)$, show that $\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$.	7	L2	CO2
	c.	Find the extreme values of the function $f(x, y) = x^2 + y^2 + 6x - 12$.	7	L3	CO2
OR					
Q.4	a.	Evaluate i) $\lim_{x \rightarrow 0} \frac{(a^x + b^x)^{1/x}}{2}$ ii) $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^2}$.	7	L2	CO2
	b.	If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$.	8	L2	CO2
	c.	Using modern mathematical tool write a program/code to evaluate $\lim_{x \rightarrow \infty} (1 + 1/x)^x$.	5	L3	CO2

1 of 3

Module – 3

Q.5	a.	Solve : $\frac{dy}{dx} + \frac{y}{x} = y^2x$.	6	L2	CO3
	b.	Find the orthogonal trajectories of $r = a(1 + \cos\theta)$, where a is a parameter.	7	L3	CO3
	c.	Find the general solution of the equation $(px-y)(py+x) = 2p$ by reducing into Clairaut's form by taking the substitution $X = x^2$, $Y = y^2$.	7	L2	CO3

OR

Q.6	a.	Solve $(y \log y) dx + (x - \log y) dy = 0$.	6	L2	CO3
	b.	Prove that the system of parabolas $y^2 = 4a(x + a)$ is self-orthogonal.	7	L3	CO3
	c.	Solve : $xyp^2 - (x^2 + y^2)p + xy = 0$.	7	L2	CO3

Module – 4

Q.7	a.	i) Find the last digit of 7^{2013} ii) Find the last digit of 13^{37}	6	L2	CO4
	b.	i) Find the remainder when $175 \times 113 \times 53$ is divided by 11. ii) Find the remainder when 2^{23} is divided by 47.	7	L2	CO4
	c.	Encrypt the message STOP using RSA with key (2537, 13) using the prime numbers 43 and 59.	7	L3	CO4

OR

Q.8	a.	Solve $2x + 6y \equiv 1 \pmod{7}$ $4x + 3y \equiv 2 \pmod{7}$.	6	L2	CO4
	b.	Using Fermat's little theorem, show that $8^{30} - 1$ is divisible by 31.	7	L2	CO4
	c.	Show that $4(29)! + 5!$ is divisible by 31.	7	L3	CO4

Module – 5

Q.9	a.	Find the Rank of the matrix $\begin{bmatrix} 2 & -1 & -3 & -1 \\ 1 & 2 & 3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$	6	L2	CO5
	b.	Solve the system of Equations by Gauss-Jordon method $x + y + z = 9$ $2x + y - z = 0$ $2x + 5y + 7z = 52$.	7	L3	CO5
	c.	Using power method, find the largest eigen value and the corresponding eigen vector of the matrix $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}$. Carry out six iterations.	7	L3	CO5

OR

Q.10	a.	Solve the following system of equations by Gauss-Siedel method. $27x + 6y - z = 85$, $6x + 15y + 2z = 72$, $x + y + 54z = 110$. Carry out three iterations.	7	L3	CO5
	b.	Investigate for what values of λ , μ the equations $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ have i) No solution ii) Unique solution iii) Infinite number of solutions.	8	L3	CO5
	c.	Using modern mathematical tool, write a program/code to test the consistency of the equation $x + 2y - z = 1$ $2x + y + 4z = 2$ $3x + 3y + 4z = 1$	5	L3	CO5

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BMATS201

**Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Mathematics – II for CSE Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book is permitted.

3. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1				M	L	C
Q.1	a.	Evaluate $\int_{-1}^{+1} \int_0^z \int_{x-z}^{x+z} (x+y+z) dy dx dz$.		7	L2	CO1
	b.	Evaluate $\int_0^a \int_{x/a}^{\sqrt{x/a}} (x^2 + y^2) dy dx$ by changing the order of integration.		7	L3	CO1
	c.	Show that $\beta(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$.		6	L2	CO1
OR						
Q.2	a.	Evaluate $\int_0^1 \int_0^{\sqrt{1-y^2}} (x^2 + y^2) dy dx$ by changing into polar co-ordinates.		7	L3	CO1
	b.	Find the area between the parabolas $x^2 = y$ and $y^2 = x$ using double integration.		7	L3	CO1
	c.	Using mathematical number's, write a code to find the area of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by double integration.		6	L3	CO5
Module – 2						
Q.3	a.	Find the directional derivative of $\phi = 4xz^3 - 3x^2y^2z$ at $(2, -1, 2)$ in the direction of the vector $2\hat{i} - 3\hat{j} + 6\hat{k}$.		7	L2	CO2
	b.	If $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$ find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$.		7	L2	CO2
	c.	Prove that spherical co-ordinate system is orthogonal.		6	L3	CO2
OR						
Q.4	a.	Find the angle between the normals to the surface $xy = z^2$ at the points $(4, 1, 2)$ and $(3, 3, -3)$.		7	L2	CO2
	b.	If $\vec{F} = (x + y + az)\hat{i} + (bx + 2y - z)\hat{j} + (x + cy + 2z)\hat{k}$, find a, b, c such that $\text{curl } \vec{F} = 0$.		7	L2	CO2
	c.	Using mathematical tool write a code to find the curl of $\vec{F} = x^3\hat{i} + y^3\hat{j} + z^3\hat{k}$.		6	L3	CO5
Module – 3						
Q.5	a.	Prove that the set $w = \left\{ \frac{(x, y, z)}{x - 3y + 4z = 0} \right\}$ is a subspace of $V_3(R)$.		7	L2	CO3

	b.	Express the matrix $M = \begin{bmatrix} 4 & 7 \\ 7 & 9 \end{bmatrix}$ as a linear combination of the matrices, $P = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $Q = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $R = \begin{bmatrix} 1 & 1 \\ 4 & 5 \end{bmatrix}$	7	L2	CO3														
	c.	Find the basis and dimension of subspace spanned by the vectors, $\{(1,-2,3), (1,-3,4)(-1,1,-2)\}$ of $V_3(R)$.	6	L3	CO3														
OR																			
Q.6	a.	Find the matrix of linear transformation, $T: R^2 \rightarrow R^2$ defined by, $T(x, y) = (2x + 3y, 4x - 5y)$ with respect to the basis, $B_1 = \{(1, 2), (2, 5)\}$ of R^2 .	7	L2	CO3														
	b.	The transformation $G: R^3 \rightarrow R^3$ is defined as $G(x, y, z) = (x + 2y - z, y + z, x + 2y - 2z)$. Find the basis and dimension of $\text{Im}(G)$.	7	L2	CO3														
	c.	If $f(t) = t + 2$, $g(t) = 3t - 2$, $h(t) = t^2 - 2t - 3$ and $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$, find (i) $\langle f, g \rangle$ (ii) $\langle f, h \rangle$ (iii) $\ f\ $ and $\ g\ $	6	L3	CO3														
Module – 4																			
Q.7	a.	Find the real root of the equation $x^3 - 2x - 5 = 0$, correct to three decimal places using Regula-Falsi method. Carry out three iteration.	7	L2	CO4														
	b.	Using Newton's forward interpolation formula find y at $x = 5$ from the following table: <table><tr><td>x:</td><td>0</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td></tr><tr><td>y:</td><td>0</td><td>4</td><td>56</td><td>204</td><td>496</td><td>980</td></tr></table>	x:	0	2	4	6	8	10	y:	0	4	56	204	496	980	7	L2	CO4
x:	0	2	4	6	8	10													
y:	0	4	56	204	496	980													
	c.	Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Simpson's $\frac{3}{8}$ rule taking six equal intervals.	6	L3	CO4														
OR																			
Q.8	a.	Find the real root of the equation, $xe^x - 2 = 0$, correct to three decimal places using Newton-Raphson method. Carry out three iterations.	7	L2	CO4														
	b.	Using Lagrange's interpolation formula find $f(4)$ given, <table><tr><td>x :</td><td>0</td><td>2</td><td>3</td><td>6</td></tr><tr><td>f(x) :</td><td>-4</td><td>2</td><td>14</td><td>158</td></tr></table>	x :	0	2	3	6	f(x) :	-4	2	14	158	7	L2	CO4				
x :	0	2	3	6															
f(x) :	-4	2	14	158															
	c.	Evaluate $\int_0^1 \frac{x}{1+x^2} dx$ by trapezoidal rule considering six equal intervals.	6	L3	CO4														
Module – 5																			
Q.9	a.	Employ Taylor's series method to obtain approximate value of y at $x = 0.1$ for the differential equation, $\frac{dy}{dx} = 2y + 3e^x$, $y(0) = 0$	7	L2	CO4														
	b.	Apply Runge-Kutta method of 4 th order to find an approximate value of y at $x = 0.2$, given that $\frac{dy}{dx} = \frac{y-x}{y+x}$, $y(0) = 1$.	7	L2	CO4														
	c.	Apply Milne's method to find $y(1.4)$ given $\frac{dy}{dx} = x^2 + \frac{y}{2}$ and the following data : $y(1) = 2$, $y(1.1) = 2.2156$, $y(1.2) = 2.4649$, $y(1.3) = 2.7514$.	6	L3	CO5														

OR					
Q.10	a.	Using modified Euler's method, find $y(0.1)$ given $\frac{dy}{dx} = x - y^2$, $y(0) = 1$. Carry out 3 iterations.	7	L2	CO4
	b.	Using Runge-Kutta method of fourth order solve $\frac{dy}{dx} = \frac{1}{x+y}$, $y(0.4) = 1$ at $x = 0.5$.	7	L2	CO4
	c.	Using mathematical tool, write a code to solve the differential equation $\frac{dy}{dx} = x^2 + y^2$ with $y(0) = 0$, using Taylor's series method at $x = 0.1$.	6	L3	CO5

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Second Semester B.E./B.Tech. Degree Examination, June/July 2024

Mathematics – II for CSE Stream

Time: 3 hrs.

Max. Marks: 100

- Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.*
 2. M : Marks , L: Bloom's level , C: Course outcomes.
 3. VTU Hand book is permitted.

Module – 1				M	L	C
Q.1	a.	Evaluate $\int_0^a \int_0^x \int_0^{x+y} e^{(x+y+z)} dz dy dx$.		7	L2	CO1
	b.	By changing the order of integration evaluate $\int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} xy dy dx$.		7	L3	CO1
	c.	With usual notation, prove that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.		6	L2	CO1
OR						
Q.2	a.	Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} y^2 \sqrt{x^2+y^2} dx dy$ by changing into polar coordinates.		7	L3	CO1
	b.	Find the area bounded between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ by double integration.		7	L2	CO1
	c.	Using Mathematical tool, write the code to find the volume bounded by the sphere $x^2 + y^2 + z^2 = a^2$ by double integration.		6	L3	CO5
Module – 2						
Q.3	a.	Find the directional derivative of $\phi = xy^3 + yz^3$ at the point (2, -1, 1) in the direction of the vector $\hat{i} + 2\hat{j} + 2\hat{k}$.		7	L2	CO2
	b.	Verify whether the vector $\vec{F} = \frac{x\hat{i} + y\hat{j}}{x^2 + y^2}$ is both solenoidal and irrotational.		7	L2	CO2
	c.	Prove that the cylindrical coordinate system is orthogonal.		6	L2	CO2
OR						
Q.4	a.	If $\vec{F} = \nabla(x^3 + y^3 + z^3 - 3xyz)$ find $\text{div } \vec{F}$ and $\text{curl } \vec{F}$.		7	L2	CO2
	b.	Find the angle between the normal's to the surface $x^2yz = 1$ at the points (-1, 1, 1) and (1, -1, -1).		7	L3	CO2
	c.	Using mathematical tool write the code to find divergence and curl of the vector $\vec{F} = (4xy - z^3)\hat{i} + 2x^2\hat{j} - 3xz^2\hat{k}$.		6	L3	CO5

Module – 3

Q.5	a.	Let W be a subset of $V_3(R)$ consisting of vectors of the form (a, a^2, b) where the second component is the square of the first. Is W a subspace of $V_3(R)$.	7	L2	CO3
	b.	Let P_n be the vector space of real polynomial functions of degree $\leq n$. Verify that the transformation $T : P_2 \rightarrow P_1$ defined by $T(ax^2 + bx + c) = (a + b)x + c$ is linear.	7	L2	CO3
	c.	Find the Kernel and range of the linear transformation $T : R^3 \rightarrow R^2$ defined by $T(x, y, z) = (x + y, z)$.	6	L2	CO3

OR

Q.6	a.	Determine whether or not each of the following $x_1 = (2, 2, 1)$, $x_2 = (1, 3, 7)$, $x_3 = (1, 2, 3)$ forms a basis in R^3 .	7	L2	CO3
	b.	Verify Rank-nullity theorem for the transformation $T : R^3 \rightarrow R^3$ defined by $T(x, y, z) = (x + 2y - z, y + z, x + y - 2z)$.	7	L2	CO3
	c.	The inner product of the polynomials $f(t) = t + 2$, $g(t) = 3t - 2$ in $p(t)$ is given by $\langle f, g \rangle = \int_0^1 f(t)g(t)dt$. Find i) $\langle f, g \rangle$ ii) $\ f\ $ iii) $\ g\ $	6	L2	CO3

Module – 4

Q.7	a.	Find an approximate root of the equation $\cos x = 3x - 1$ correct to four decimal places using Regula Falsi method between 0.5 and 0.7.	7	L2	CO4												
	b.	The area 'A' of a circle of diameter 'd' is given by the following table: <table border="1" data-bbox="502 1176 1029 1254"> <tr> <td>d:</td><td>80</td><td>85</td><td>90</td><td>95</td><td>100</td></tr> <tr> <td>A:</td><td>5026</td><td>5674</td><td>6362</td><td>7088</td><td>7854</td></tr> </table> Using appropriate Newton's interpolation formula for equispaced values of x , find area of the circle corresponding to the diameter 105.	d:	80	85	90	95	100	A:	5026	5674	6362	7088	7854	7	L2	CO4
d:	80	85	90	95	100												
A:	5026	5674	6362	7088	7854												
	c.	Evaluate $I = \int_0^5 \frac{1}{4x+5} dx$ by Simpson's $1/3^{rd}$ rule by considering 10 sub intervals. Hence find an approximate value of $\log 5$.	6	L3	CO4												

OR

Q.8	a.	Find the real root of $x \log_{10} x = 1.2$ correct to four decimals that lies near 2.5 using Newton Raphson method.	7	L2	CO4												
	b.	Fit a polynomial for the following data using Newton's divided difference formula: <table border="1" data-bbox="571 1780 965 1859"> <tr> <td>x:</td><td>-4</td><td>-1</td><td>0</td><td>2</td><td>5</td></tr> <tr> <td>y:</td><td>1245</td><td>33</td><td>5</td><td>9</td><td>1335</td></tr> </table>	x:	-4	-1	0	2	5	y:	1245	33	5	9	1335	7	L2	CO4
x:	-4	-1	0	2	5												
y:	1245	33	5	9	1335												
	c.	Use trapezoidal rule to find $\int_0^{0.6} e^{-x^2} dx$ by taking seven ordinates.	6	L3	CO4												

Module – 5

Q.9	a.	Employ Taylor's series method to obtain approximate solution at $x = 0.1$ and $x = 0.2$ for the initial value problem $\frac{dy}{dx} = 2y + 3e^x$, $y(0) = 0$.	7	L2	CO4
	b.	Apply Runge-Kutta method of fourth order to find an approximate solution at $x = 0.1$ given $\frac{dy}{dx} = 3x + y/2$, $y(0) = 1$.	7	L2	CO4
	c.	Apply Milne's predictor – corrector method to solve the equation $(y^2 + 1)dy - x^2dx = 0$ at $x = 1$ given $y(0) = 1$, $y(0.25) = 1.0026$, $y(0.5) = 1.0206$, $y(0.75) = 1.0679$.	6	L2	CO4
OR					
Q.10	a.	Apply modified Euler's method to find solution at $x = 0.1$ by taking $h = 0.1$ given $y' = x^2 + y^2$, $y(0) = 0$.	7	L2	CO4
	b.	Find an approximate solution of $\frac{dy}{dx} = \frac{y^2 - x^2}{y^2 + x^2}$, $y(0) = 1$ at $x = 0.2$ using Runge-Kutta method of order four.	7	L2	CO4
	c.	Write the mathematical tool code to solve $\frac{dy}{dx} = x^2 + y$, $y(0) = 10$ using Taylor's series method at $x = 0.1(0.1)0.3$. Consider the terms upto fourth degree.	6	L3	CO5

CBCS SCHEME

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BPHYC102/202

**First/Second Semester B.E./B.Tech. Degree Supplementary Examination,
June/July 2024**

Applied Physics for Civil Engineering Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book is permitted.

3. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Define simple harmonic motion. Give example. Write the differential equation for SHM and explain the terms.	05	L2	CO1
	b.	With a neat diagram, explain the construction and working of Reddy tube and hence mention any four applications of shock waves.	10	L2	CO1
	c.	A free particle is executing SHM in a straight line. The maximum velocity it attains during any oscillation is 62.8 m/s. Find the frequency of oscillation if its amplitude is 0.5 m.	05	L3	CO3
OR					
Q.2	a.	What are forced oscillations? Derive the expression for amplitude of vibration of a body undergoing forced vibrations.	10	L2	CO1
	b.	Define Mach number and hence give the classification of waves on the basis of Mach number.	05	L2	CO1
	c.	A vibrating system of natural frequency 500 CPS is forced to vibrate with a periodic force/unit mass of amplitude 100×10^{-5} N/kg in the presence of a damping/unit mass of 0.01×10^{-3} rad/s. Calculate the maximum amplitude of vibration of the system.	05	L3	CO5
Module – 2					
Q.3	a.	Define elongation strain coefficient, compression strain coefficient and Poisson's ratio. Mention the equation for the same.	06	L2	CO1
	b.	Derive the relation between Young's modulus rigidity modulus and Poisson's ratio.	09	L2	CO1
	c.	Calculate the extension produced in a wire of length 2m and radius 0.013×10^{-2} m due to a force of 14.7 N applied along its length. Given $Y = 2.1 \times 10^{11}$ Nm ⁻² .	05	L3	CO5
OR					
Q.4	a.	What are beams? Discuss I – shaped beam in brief and hence mention its advantages, disadvantages and uses.	09	L2	CO1
	b.	What is fracture? Distinguish between brittle fracture and ductile fracture.	06	L2	CO1
	c.	Calculate the force required to produce an extension of 1 mm in steel wire of length 2m and diameter 1 mm. Given $Y = 2 \times 10^{11}$ N/m ² .	05	L3	CO5
Module – 3					
Q.5	a.	Define reverberation time. Based on the assumptions made by Sabine deduce the expression for reverberation time.	09	L2	CO2
	b.	Discuss the remedial measures taken to improve the acoustic quality in the auditorium.	06	L2	CO2
	c.	A class room is having dimensions $20 \times 15 \times 5$ m ³ . The reverberation time is 3.5 s. Calculate total absorption of surface and average absorption coefficient.	05	L3	CO2

OR

Q.6	a.	What is photometry? Explain different photometric quantities.	07	L2	CO2
	b.	Define absorption coefficient and absorption power. Explain in detail how the absorption coefficient of a material is measured using Sabine's formula.	08	L2	CO2
	c.	A lecture hall has volume 600 m^3 . Its floor area is 120 m^2 , walls area is 220 m^2 and ceiling area is 120 m^2 . The walls, floor and ceiling are covered by materials of absorption coefficient 0.03, 0.06 and 0.80 respectively. Calculate reverberation time.	05	L3	CO2

Module – 4

Q.7	a.	Expand the term LASER and discuss the interaction of radiation with matter.	07	L2	CO3
	b.	With neat diagrams, explain the construction and working of semiconductor LASER diagram. Mention the applications of semiconductor laser.	08	L2	CO3
	c.	A pulsed laser emits photons of wavelength 780 nm with average power/pulse 20 mW . Calculate the number of photons contained if each pulse duration is $10 \times 10^{-9} \text{ s}$.	05	L3	CO3

OR

Q.8	a.	Define angle of acceptance and numerical aperture (NA). Derive the expression for NA.	09	L2	CO3
	b.	Explain the construction and working of fiber optic temperature sensor.	06	L2	CO3
	c.	An optical fiber has a core material with RI 1.55 and its cladding material has a RI of 1.50. Calculate its numerical aperture and angle of acceptance.	05	L3	CO3

Module – 5

Q.9	a.	Discuss the classification of Earthquakes.	09	L2	CO4
	b.	Discuss the engineering structures to withstand earthquakes.	06	L2	CO4
	c.	Calculate the intensity of the earthquake of magnitude 6.5. Assume the base intensity is I_0 .	05	L3	CO4

OR

Q.10	a.	Describe various causes for land sliding.	08	L2	CO4
	b.	Write a note on forest fires and fire protection.	07	L2	CO4
	c.	The intensity of one earthquake is 100 times the intensity of the other. If the magnitude of the first earthquake is 8.9, estimate the magnitude of the other.	05	L3	CO4

BPHYC102/202					
Module – 3					
Q.5	a.	Define photometry and explain photometric quantities.	10	L2	CO2
	b.	Elucidate the impact of noise in multi-storied buildings.	5	L2	CO2
	c.	For an empty assembly hall of size $20 \times 15 \times 10$ cubic meter with absorption coefficient 0.106. Calculate reverberation time.	5	L3	CO2
OR					
Q.6	a.	Define reverberation and reverberation time and hence derive sabines formula.	10	L2	CO2
	b.	Mention the conditions for good acoustics.	5	L2	CO2
	c.	Define the five spectral quantities.	5	L3	CO2
Module – 4					
Q.7	a.	Discuss the interaction of radiation with matter and hence explain laser action.	8	L2	CO3
	b.	Explain propagation of light through optical fiber and hence derive an expression for numerical aperture and angle of acceptance.	7	L2	CO3
	c.	Calculate the numerical aperture and acceptance angle for an optical fiber of RI of core 1.5 and RI of cladding 1.48 placed in water of RI 1.33.	5	L3	CO5
OR					
Q.8	a.	Enumerate the requisites of a laser system and describe the construction and working of semiconductor laser with a neat sketch and energy level diagram.	9	L2	CO3
	b.	Define attenuation in fiber with the expression for attenuation coefficient and describe the various fiber losses.	6	L2	CO3
	c.	Calculate the number of photons emitted per second for a laser with power output 10mW, given the wave length of fiber 690 nanometer.	5	L3	CO5
Module – 5					
Q.9	a.	Discuss the classification of earthquakes.	9	L2	CO4
	b.	Enumerate the causes and adverse effect of Tsunami waves.	6	L2	CO4
	c.	Calculate the intensity of earthquake of magnitude 6.5 assuming the base intensity as I_0 .	5	L3	CO4
OR					
Q.10	a.	Discuss the landslides and describe the causes for landslides.	8	L2	CO4
	b.	Discuss the engineering structures to withstand earthquakes and Tsunami waves.	7	L2	CO4
	c.	The intensity of one earthquake is 100 times the intensity of the other. If the magnitude of the first earthquake is 8.9, estimate the magnitude of the other.	5	L3	CO4

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BPHYE102/202

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024
Applied Physics for EEE Stream

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks , L: Bloom's level , C: Course outcomes.
 3. VTU Formula Handbook is permitted.*

Module – 1			M	L	C
Q.1	a.	State de Broglie hypothesis and set up one – dimensional time independent Schrodinger Wave equation.	9	L2	CO1
	b.	Define a Wavepacket and explain the terms Phase velocity and Group velocity and mention their expressions.	6	L2	CO1
	c.	In measurement of position and momentum that involved an uncertainty of 0.003%, the speed of an electron was found to be 800ms^{-1} . Calculate the corresponding uncertainty that arises in determining its position.	5	L3	CO1
OR					
Q.2	a.	Using the time independent Schrodinger wave equation, obtain the expression for the normalized wave function for a particle in one dimensional potential well of infinite height.	9	L2	CO1
	b.	Using uncertainty principle, show that an electron cannot exist within the nucleus of an atom.	6	L2	CO1
	c.	An electron is bound in an one dimensional potential well of width $1\text{\AA}$, but of infinite wall height. Find its energy values in the ground and in the first two excited states.	5	L3	CO1
Module – 2					
Q.3	a.	Define Fermi energy and Fermi factor. Discuss the variation of Fermi factor with temperature and energy.	9	L2	CO2
	b.	Define Dielectric polarization and describe the different polarization mechanisms.	7	L2	CO2
	c.	Calculate the probability of an electron occupying an energy level 0.02eV above Fermi level at 200K in a material.	4	L3	CO2
OR					
Q.4	a.	Derive Clausius – Mossotti equation.	6	L2	CO2
	b.	Explain BCS theory of Superconductivity. Write a short note on Maglev vehicles.	9	L2	CO2
	c.	An elemental solid dielectric material has polarizability $7 \times 10^{-40} \text{Fm}^2$. Assuming the internal field to be Lorentz field, calculate the dielectric constant for the material if the material has $3 \times 10^{28} \text{atoms/m}^3$.	5	L3	CO2

Module – 3					
Q.5	a.	Define Induced absorption , Spontaneous emission and Stimulated emission. Obtain an expression for energy density of radiation under equilibrium condition in terms of Einstein's co-efficient.	10	L2	CO1
	b.	Explain different types attenuations in optical fibers.	6	L2	CO1
	c.	A medium in thermal equilibrium at temperature 300K has two energy levels with a wavelength separation of 1 μ m. Find the ratio of population densities of the upper and lower levels.	4	L3	CO1
OR					
Q.6	a.	Describe the construction and working of carbon dioxide laser.	8	L2	CO1
	b.	With neat diagram, derive an expression for numerical aperture of an optical fiber and arrive at the condition for propagation.	7	L2	CO1
	c.	An optical fiber has a core material with refractive index 1.55 and its cladding material has a refractive index of 1.50. The light is launched into it in air. Calculate its numerical aperture, the acceptance angle and also the fractional index change.	5	L3	CO1
Module – 4					
Q.7	a.	Describe the vector operator ∇ and explain the concepts of gradient, divergence and curl.	7	L2	CO3
	b.	Explain the Gauss's law in electrostatics and magnetism. Express the same in their differential forms.	8	L2	CO3
	c.	Given $\vec{A} = (3x^2 + y + az) \hat{a}_x + (bx - 5y^3 - 2z) \hat{a}_y + (2x + cy + 3z^2) \hat{a}_z$. For what values of a, b, and c the vector $\vec{A}$ is irrotational?	5	L3	CO3
OR					
Q.8	a.	Derive Gauss's divergence theorem and also mention the Stoke's theorem.	7	L2	CO3
	b.	Derive the electromagnetic wave equation using Maxwell's equation in free space.	8	L2	CO3
	c.	Determine the constant C such that, the vector $\vec{A} = (x + ay) \hat{a}_x + (y + bz) \hat{a}_y + (x + cz) \hat{a}_z$ is Solenoidal.	5	L3	CO3
Module – 5					
Q.9	a.	Show that the Fermi level lies in the middle of the energy gap for an intrinsic semiconductor.	6	L2	CO4
	b.	What is Hall effect? Obtain the expression for Hall voltage in terms of Hall co-efficient.	9	L2	CO4

	c.	In a diffraction grating experiment, the laser light undergoes third order diffraction with diffraction angle of 11.7° . The grating constant is 10^{-5}m and the distance between the grating and laser source is 1m, find the wavelength of laser light?	5	L3	CO5
OR					
Q.10	a.	Obtain the expression for electrical conductivity ion extrinsic and intrinsic semiconductors.	7	L2	CO4
	b.	Describe with energy band diagram, the construction and working of a semiconductor diode laser.	8	L2	CO4
	c.	Determine the resonance frequency of an LCR series circuit with inductance = 0.5 henry , Capacitance = 0.45 , Microfarad and resistance = 300Ω .	5	L3	CO5

COs and Pos Mapping

COs	POs											
	1	2	3	4	5	6	7	8	9	10	11	12
CO1	3	2	-	-	-	-	-	-	-	-	-	2
CO2	3	2	-	-	-	-	-	-	-	-	-	2
CO3	3	2	-	-	-	-	-	-	-	-	-	2
CO4	3	2	-	-	1	-	-	-	-	-	-	2
CO5	3	2	1	-	2	-	-	3	3	-	-	2

Note : Level – 3 : Highly Mapped , Level – 2 : Moderately Mapped ,
Level – 1 : Low Mapped.

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BPHYM102/202

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024
Applied Physics for ME Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

3. VTU Hand book is permitted.

Module – 1			M	L	C
Q.1	a.	Define Spring Constant. Obtain an expression for equivalent force constant for two springs in series and parallel combination. What is the expression for time period of oscillation in above cases?	8	L2	CO1
	b.	What is Mach Number? Distinguish subsonic ultrasonic, supersonic and hypersonic flow.	8	L2	CO1
	c.	In Reddy tube experiment, it was found that the time taken to travel between the two sensors is 195 microseconds. If the distance between the two sensors is 100mm. Find Mach Number. Given velocity of sound under the same conditions is 340m/s.	4	L3	CO1
OR					
Q.2	a.	Obtain a differential equation for a body undergoes forced oscillation and mention the expression for amplified and phase of oscillation.	8	L2	CO1
	b.	What are shock waves? Describe the construction and working of Reddy shock tube with neat diagram.	7	L2	CO1
	c.	A spring undergoes extension of 5cm for a load of 50gm. Find its frequency of oscillation. If it set for vertical oscillation with a load of 200gm, attached to its bottom. Ignore the mass of the spring.	5	L3	CO5
Module – 2					
Q.3	a.	Explain the types of strain. Explain the nature of elasticity with the help of stress- strain diagram.	8	L2	CO1
	b.	Define Poisson's ratio, Young's modulus, Bulk modulus. Derive the relation for 'Y', 'η' and 'σ'.	8	L2	CO1
	c.	Calculate the force required to produce an extension of 3mm in steel wire of length 3m and diameter 2mm ($y = 20 \times 10^{10} \text{N/m}^2$).	4	L3	CO1
OR					
Q.4	a.	What is bending moment? Derive the expression for the bending moment in terms of moment of Inertia. Hence, arrive the expression for a bending moment of circular and rectangular cross sections.	8	L2	CO1
	b.	Discuss the different types of beams and explain 'I' section Girder and their Engineering applications.	8	L2	CO1
	c.	A metal wire of length 1.5m is loaded and an elongation of 2mm is produced. If the diameter of wire is 1mm. find the change in diameter of wire when elongated (where $\sigma = 0.24$).	4	L3	CO1
Module – 3					
Q.5	a.	What is Seeback effect? Explain variation of Thermo-emf with temperature and obtain the relation between inversion temperature and neutral temperature.	8	L2	CO2

	b.	What are thermo couples? Explain laws of thermo – electricity with neat sketch.	8	L2	CO2
	c.	EMF of a thermo couple is 1200 microvolts when working between 0°C and 100°C. Its neutral temperature is 300°C. Find the value of ‘a’ and ‘b’ for it.	4	L3	CO2
OR					
Q.6	a.	What are thermo-eletric materials? Explain Low, mid and high temperature thermo electric materials and mention their applications.	8	L	CO
	b.	Describe the construction and working of Thermo electric generator and mention the applications of it.	8	L	CO
	c.	The emf in microvolts of a thermocouple one junction of which is at 0°C is given by $e = 1600T - 4T^2$, where T°C is the temperature of hot junction. Find neutral temperature and Piltier co-efficient.	4	L	CO
Module – 4					
Q.7	a.	Explain the construction and working of porous plug experiment with neat diagram.	8	L2	CO3
	b.	Describe the Lindey’s air liquifier.	8	L2	CO3
	c.	Calculate the inversion temperature of gas. Given $a = 0.244 \text{ atm L}^2/\text{mol}^2$, $b = 0.027 \text{ L/mol}$ and $R = 0.0821 \text{ Latm/K/mol}$.	4	L3	CO3
OR					
Q.8	a.	Explain Joule-Thomson effect. Derive $\Delta T = \left[\frac{P_1 - P_2}{C_p} \right] \left(\frac{2a}{RT} - b \right)$	8	L2	CO3
	b.	What is Cryogenics? Explain the applications of cryogenics in aerospace and food processing.	8	L2	CO3
	c.	A platinum resistance thermometer has resistance of 2Ω and 0°C and 3Ω at 100°C. What will be the temperature when resistance indicates 5Ω.	4	L3	CO3
Module – 5					
Q.9	a.	Explain the construction and working of X-ray diffractometer and discuss the Scherrer’s method for determining Crystallite size.	8	L2	CO4
	b.	Describe construction and working of SEM and mention its applications.	8	L2	CO4
	c.	An X-ray undergoes First order Bragg’s diffraction by the crystal with ‘d’ spacing 3.6 Å at a glancing angle 12° calculate wavelength of X-rays.	4	L3	CO4
OR					
Q.10	a.	What are Nonomaterials and Nano composites? Discuss their classification based on their dimensional constraints.	8	L2	CO4
	b.	Describe the principle, construction and working of X-ray photo Electron spectroscopy and mention its advantages and applications.	8	L2	CO4
	c.	Determine the Crystallite size using Scherrer’s relation. Given $\lambda = 1.5 \text{ Å}$, $2\theta = 35^\circ$ and $k = 0.94$ peak full width half maxima = 0.5°.	4	L3	CO4

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BPHYS102/202

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024
Applied Physics for CSE Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

3. VTU Hand book is permitted.

Module – 1			M	L	C
Q.1	a.	Explain the construction and working of semiconductor LASER with a neat sketch and energy level diagram.	9	L2	CO1
	b.	Discuss different types of optical fibers based on modes of propagation and RI profile.	6	L2	CO1
	c.	An optical fiber has refractive index of core and cladding of 1.55 and 1.50, respectively. Calculate its numerical aperture and angle of acceptance if it is kept in air.	5	L3	CO5
OR					
Q.2	a.	Obtain the expression for energy density of radiation in term of Einstein's A and B coefficients.	8	L2	CO1
	b.	Define numerical aperture and derive an expression for numerical aperture of an optical fiber.	7	L2	CO1
	c.	In a diffraction grating experiment the Laser light undergoes first order diffraction at an angle of 19.3° . Find the wavelength of Laser light. Given the grating constant $d = 1.98 \times 10^{-6} \text{m}$.	5	L3	CO5
Module – 2					
Q.3	a.	Setup one dimensional time independent Schrodinger wave equation.	8	L2	CO
	b.	State Heisenberg's uncertainty principle and apply the same to prove the non-existence of free electron inside the nucleus.	7	L2	CO
	c.	An electron is bound in an infinite potential well of width 0.18nm. Find its energy values in the first two allowed energy states.	5	L3	CO2
OR					
Q.4	a.	Obtain an expression for Eigen function and Eigen energy values for a particle in an infinite potential well of width 'a'.	9	L2	CO2
	b.	What is wave function? Mention the properties of wave function and give its significance.	6	L2	CO2
	c.	Calculate the kinetic energy of a neutron in eV. Given: de -Broglie wave length is $1 \text{ \AA}$ and mass of neutron, $m_n = 1.674 \times 10^{-27} \text{Kg}$.	5	L3	CO2
Module – 3					
Q.5	a.	Distinguish between classical computing and Quantum computing.	6	L2	CO2
	b.	Explain the CNOT gate and its operation on four different input states.	6	L2	CO2
	c.	Apply Pauli matrices on the state $ 0\rangle$ and $ 1\rangle$.	8	L3	CO2
OR					
Q.6	a.	Explain the working of T-gate mentioning its matrix representation and truth table.	7	L2	CO2
	b.	Explain Orthogonality and Orthonormality with an example of each.	8	L2	CO2

	c.	A linear operator 'X' operates such that $X 0\rangle = 0\rangle$ and $X 1\rangle = i 1\rangle$. Find the matrix representation of 'X'.	5	L3	CO2
Module – 4					
Q.7	a.	Mention the failures of classical free electron theory and explain the assumptions of Quantum free electron theory of metals.	7	L2	CO3
	b.	Explain Meissner's effect and the variation of critical field with temperature.	8	L2	CO3
	c.	A lead wire has a critical field of 6.5×10^3 A/m at 0 Kelvin. The critical temperature is 7.18K. At what temperature the critical field will drop to 4.5×10^3 A/m.	5	L3	CO3
OR					
Q.8	a.	Define Fermi factor and explain the variation of Fermi factor with temperature and energy.	8	L2	CO3
	b.	Differentiate Type – I and Type – II superconductors.	8	L2	CO3
	c.	Calculate the probability of occupation of an energy level 0.02eV above level at temperature 27°C.	4	L3	CO3
Module – 5					
Q.9	a.	Explain the importance of (i) size and scale and (ii) weight and strength, in animation.	7	L2	CO4
	b.	Mention the general pattern of Monte – Carlo method and hence explain the procedure to find the value of ' π '.	8	L2	CO4
	c.	In the case of animating a jump, the jump height is 2.5m and jump magnification is 5. Calculate the push height and push acceleration. Given gravitational acceleration is 10m/s.	5	L3	CO5
OR					
Q.10	a.	Describe jumping and parts of jump.	9	L2	CO4
	b.	Distinguish between descriptive and inferential statistics.	6	L2	CO4
	c.	On a particular place, volcanic eruption occurs once in every 100years on an average. Calculate the probability of volcanic eruption in a 100 years interval for $K = 0, 1$ and 2 , assuming the Poisson's model appropriate.	5	L3	CO5

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First Semester B.E./B.Tech Degree Examination, June/July 2024
Introduction to Python Programming

Time: 3 hrs.

Max. Marks: 100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.**

		Module – 1	M	L	C
Q.1	a.	Define comparison operator and list its type. Give the difference between == and = operator.	4	L1	CO1
	b.	Explain flow control statement in detail with if, else, while loop and for loop.	10	L2	CO1
	c.	Build a function to calculate factorial of a number. Develop a program to compute binomial coefficient.	6	L3	CO1
OR					
Q.2	a.	Define exception in python programming and give the basic form of an exception handling block with an example.	8	L1	CO1
	b.	Explain how to define function and to make a function call by passing an argument with an example.	4	L2	CO1
	c.	Develop a python program to read the name and year of birth of a person and to display whether the person is a senior citizen or not.	8	L3	CO1
Module – 2					
Q.3	a.	Explain in detail about append() and index() function with respect to list in python.	6	L2	CO2
	b.	Develop suitable python program with nested list to explain copy() and deepcopy() methods.	6	L3	CO2
	c.	Tuples are immutable. Explain with an example.	8	L2	CO2
OR					
Q.4	a.	Explain the below methods in list with suitable code : i) remove() ii) sort() iii) reverse().	6	L2	CO2
	b.	Outline python dictionaries with some of their methods.	8	L2	CO2
	c.	Explain Nested dictionaries with an example.	6	L2	CO2
1 of 2					

Module – 3

Q.5	a.	Develop a code to print to most frequently appearing words in a text file.	10	L3	CO3
	b.	Explain below python string handling function with example : i) split() ii) rjust() iii) partition() iv) join() v) startwith().	10	L2	CO3

OR

Q.6	a.	Explain the method to restore the data to variable from the hard drive.	10	L2	CO3
	b.	Develop a program to sort the contents of a text file and write the sorted content into a separate text file.	10	L3	CO3

Module – 4

Q.7	a.	Explain various shell utilities function.	10	L2	CO3
	b.	Develop a program to read and to extract all the files and folder into a ZIP file by using relevant methods.	10	L3	CO3

OR

Q.8	a.	Explain permanent delete and safe delete with a suitable python programming.	10	L2	CO3
	b.	Define Assertion. Explain the use of Assertion in a Traffic light simulation with a python program.	10	L2	CO3

Module – 5

Q.9	a.	Define pure function and modifier. Explain the role of pure function and modifier in application development.	10	L2	CO3
	b.	Explain the methods <code>_int_</code> and <code>_str_</code> with example.	10	L2	CO3

OR

Q.10	a.	Define operator overloading. Explain with suitable python program.	10	L2	CO4
	b.	Define polymorphism and give a suitable python program.	10	L1	CO4

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CBCS SCHEME

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BPLCK205B / BPLCKB205

Second Semester B.E./B.Tech. Degree Examination, June/July 2024

Introduction to Python Programming

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1				M	L	C
Q.1	a.	Explain with example print(), input() and len().		6	L2	CO1
	b.	Explain elif, for, while statement in python with example.		6	L2	CO1
	c.	Develop a program to generate Fibonacci sequence of Length(N).Read N from the console.		8	L3	CO1
OR						
Q.2	a.	What are functions? Explain python function with parameters and return statement.		6	L2	CO1
	b.	How to handle exception in python with example.		6	L2	CO1
	c.	Explain Local and Global scope with variables for each.		8	L2	CO1
Module – 2						
Q.3	a.	Explain the use of in and not in operator in list with examples.		6	L2	CO2
	b.	Explain negative Indexing, slicing, index(), append(), remove(), pop(), insert() and sort() with suitable example.		8	L2	CO2
	c.	Write about Mutable and Immutable data type in list.		6	L2	CO2
OR						
Q.4	a.	Explain the following list methods with examples. i) index () ii) append() iii) insert() iv) sort() v) reverse() vi) List concatenation and Replication.		10	L2	CO2
	b.	Develop a program to read the student details like Name, USN and Marks in three subjects. Display the student details, total marks and percentage with suitable messages.		10	L3	CO2
Module – 3						
Q.5	a.	Illustrate with example opening of a file with open() function, reading the contents of the file with read() and writing to files with write ().		10	L2	CO3
	b.	Explain how to save variable with the shelve module.		10	L2	CO3

OR					
Q.6	a.	Explain the following string methods with examples. i) isalpha() ii) isalnum() iii) isdecimal() iv) isspace() v) istitle().	10	L2	CO3
	b.	Explain about in and not in operators in string.	5	L2	CO3
	c.	Explain about pyperclip module.	5	L2	CO3
Module – 4					
Q.7	a.	What are Assertions? Write the contents of an assert statement. Explain them with examples.	10	L2	CO3
	b.	Develop a program with a function named DivExp which takes Two parameters a, b and returns a value c(c = a/b), write suitable assertion for a > 0 in function DivExp and raise an exception for when b = 0. Develop a suitable program which reads two values from the console and calls a function DivExp.	10	L3	CO3
OR					
Q.8	a.	Explain about files and folders can be copied using shutil module.	10	L2	CO3
	b.	Explain about Debug control window.	10	L2	CO3
Module – 5					
Q.9	a.	Explain about class and objects.	10	L2	CO4
	b.	Explain about pure function and modifier.	10	L2	CO4
OR					
Q.10	a.	Explain the concept of prototyping Vs planning.	10	L2	CO4
	b.	Explain _init_ and _str_ methods with examples.	10	L2	CO4

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CBCS SCHEME

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BPLCK205C/BPLCKC205

Second Semester B.E/B.Tech. Degree Examination, June/July 2024

Basics of Java Programming

Time: 3 hrs.

Max. Marks:100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.**

Module – 1			M	L	C
1	a.	Explain object oriented principles.	6	L2	CO1
	b.	Explain different lexical issues in Java.	8	L2	CO1
	c.	Give different syntax of array allocation in Java.	6	L2	CO1
OR					
2	a.	Explain different promotion rules in Java.	8	L2	CO1
	b.	Write a Java program for multiplication of two arrays.	8	L2	CO1
	c.	What is type casting? Illustrate with an example.	4	L2	CO1
Module – 2					
3	a.	Explain the following operations with an example : i) << ii) >> iii) ? iv) &.	8	L2	CO2
	b.	Explain Four different types of if statements in Java.	8	L2	CO2
	c.	Explain the nested switch statement in Java.	4	L2	CO2
OR					
4	a.	Demonstrate working of break with labels in Java.	6	L2	CO2
	b.	Write a program to sort the elements using for loop.	6	L2	CO2
	c.	Explain any five Boolean logical operators with an example.	8	L2	CO2
Module – 3					
5	a.	What is Class? How to create objects for the class?	6	L2	CO3
	b.	What are constructors? Explain two types of constructors.	6	L2	CO3
	c.	Explain static variable and static methods in Java.	8	L2	CO3
OR					
6	a.	Explain the following key words : i) this ii) super iii) final iv) static.	8	L3	CO3
	b.	Write a Java program demonstrating method overloading.	6	L3	CO3
	c.	Explain memory allocation and use of garbage collector in Java.	6	L3	CO3
Module – 4					
7	a.	Explain different types of inheritance in Java.	6	L2	CO3
	b.	What is abstract class? Explain with an example.	6	L2	CO3
	c.	Explain the working of final class and final method in Java.	8	L2	CO3
OR					
8	a.	Write example program for method overriding.	6	L2	CO3
	b.	What is super? Explain the use of super with suitable example.	6	L2	CO3
	c.	Explain dynamic dispatch using abstract class in Java using example program.	8	L2	CO3
1 of 2					

Module – 5

9	a.	What is Package? How to create and import package in Java. Explain with an example.	8	L2	CO4
	b.	Explain the working of interface in Java.	6	L2	CO4
	c.	What is an exception? Give an example.	6	L2	CO4
OR					
10	a.	What is nested try? Explain with an example program.	6	L2	CO4
	b.	Explain try and catch in Java with an example.	6	L2	CO4
	c.	Demonstrate throw, throws and finally in Java with an example.	8	L2	CO4

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[illegible]

Question Paper Version : A

First/Second Semester B.E./B.Tech./B.Design Degree Examination, June/July 2024
Professional Writing Skills in English

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Choose the part of the sentence that has an error :

1. If I had known (a) this yesterday (b) I will have helped him (c) no error (d)
2. Having received your letter (a) this morning, we was writing (b) to thank you for the same (c) no error (d)
3. A large number of peoples (a) have gathered (b) to greek the leader (c) no error (d)
4. Driving for five hundred kilometers in a day (a) are a tiring proposition (b) no error (c)

Directions : Fill in the blanks with suitable phrasal verbs :

- 5.** His arrogance _____ his ruin.
a) brought in b) brought up
c) brought about d) none of these
- 6.** Henry _____ the magazine quickly.
a) Looked through b) Looked up
c) Looked in d) None of these

Directions : Choose a suitable verb to complete the sentence :

- [illegible]

8. Either the teacher or the student _____ to be blamed for his performance in the final exam.
 a) are b) is
 c) were d) none of these

Directions : Fill in the blanks with suitable tenses :

9. Suddenly she gave a loud scream and _____ to the ground.
 a) had fallen b) has fallen
 c) fell d) none of these
10. The room _____ but the police failed to find anything suspicious.
 a) Searched b) was searched
 c) had searched d) none of these

Directions : Do as directed :

11. Okay, see you _____ the concert.
 a) at b) in
 c) for d) none of these
12. The children were _____ at having been informed about the trip.
 a) thrilling b) thrills
 c) thrilled d) none of these
13. _____ you ever been to Kashmir?
 a) Have b) Did
 c) Had d) None of these
14. Don't narrate _____ stories, they scare me.
 a) ghastly b) ghostly
 c) both 'a' and 'b' d) none of these
15. That _____ be true. He wouldn't do something like that.
 a) wouldn't b) shouldn't
 c) can't d) none of these

Directions : Rearrange the sentence :

16. defined as a place (a) where man is passive (b) and the rest of the nature is active (c) a sanctuary may be (d)
 a) a, b, c, d b) d, c, a, b
 c) b, c, d, a d) d, a, b, c
17. are simply at a loss (a) of 500 and 1000 rupee notes (b) all the corrupt politicians and their cronies (c) after the demonetization (d)
 a) c, a, d, b b) b, a, c, d
 c) d, c, b, a d) b, c, a, d

Directions : Change the voice of the following sentences :

18. You need to clean your shoes properly.
 a) Your shoes are needed to clean properly.
 b) You are needed to clean your shoes properly
 c) Your shoes need to be cleaned properly
 d) Your shoes are needed by you to clean properly
19. James watt discovered the energy of steam.
 a) The energy of steam discovered James watt
 b) The energy of steam was discovered by James watt.
 c) James watt discovering the energy of steam
 d) James watt had been discovered energy by the steam

Directions : Convert the following sentences from Direct to Indirect speech :

20. "I am sorry", he said.
 a) He apologized that he was sorry
 b) He cried that he was to be sorry
 c) He demanded that he was sorry
 d) All of these

Directions : Do as directed :

21. A sentence that introduces the topic or the main idea to the readers is called.
 a) Topic sentence
 b) First sentence
 c) Both 'a' and 'b'
 d) None of these
22. A paragraph which is written after analyzing a situation is called,
 a) Descriptive paragraph
 b) Analytical paragraph
 c) Illustrative paragraph
 d) narrative paragraph
23. The important parts of an essay are :
 a) Introduction
 b) Body
 c) Conclusion
 d) All of these
24. _____ is a gist of any passage written in as few words as possible.
 a) Essay writing
 b) Precis writing
 c) Analytical writing
 d) None of these
25. Which among the following is not a feature of reports?
 a) Focuses on facts and data
 b) Is written for a specific purpose
 c) Includes irrelevant information
 d) Is structured in an organized way
26. Reports which are submitted at regular intervals is called,
 a) Routine report
 b) Periodic report
 c) Both 'a' and 'b'
 d) None of these
27. A technical report establishes a,
 a) illogical conclusion
 b) logical conclusion
 c) personal prejudice
 d) misplaced learning

28. _____ is drafted in response to an advertisement or demand.
a) Solicited proposal
b) Unsolicited proposal
c) Both 'a' and 'b'
d) None of these
29. Childhood is a time when there are _____ responsibilities to make life difficult. If a child _____ good parents, he is fed, looked _____ and loved.
a) many, had, up
b) few, has, after
c) little, have, at
d) all of these
30. Listening is a vital skill which helps in enhancing our learning.
a) True
b) False
31. Which among the following are barriers to listening?
a) Forged attention
b) Poor interpersonal relations
c) Premature evaluation
d) All of these
32. Which among the following should not be implemented for effective listening?
a) Having an open mind
b) Not being prejudiced
c) Employing critical thinking
d) Asking irrelevant questions
33. A business letter must be,
a) written in proper format
b) short and concise
c) polite in tone
d) All of these
34. The date on a business letter should appear after the salutation.
a) True
b) False
35. This format of the letter has the heading, dateline, complementary closure and signature right aligned.
a) Block format
b) Informal format
c) Modified block format
d) None of these
36. Which among the following is not to be mentioned in a resume?
a) Educational qualification
b) Work experience
c) Strengths
d) Weaknesses
37. The _____ format of resume lists your work history with dates, with your most recent employer and job title listed first.
a) Historical
b) Functional
c) Chronological
d) All of these
38. BCC in an email refers to :
a) British council careers
b) Blind carbon copy
c) Booked carbon copy
d) None of these
39. The cover letter is written,
a) To introduce oneself as the suitable candidate for the job.
b) To give biographical details of the candidate.
c) To try for the job
d) To let the employer know of our writing skills

40. _____ communication is a direct, written or oral communication that occurs between two or more persons
 a) Interpersonal b) Extra-personal c) Intrapersonal d) None of these
41. In a group discussion one must communicate with,
 a) Hostility b) Arrogance
 c) Long sentences d) Knowledge
42. Which among the following should not be followed while appearing for an interview?
 a) Arriving late to the venue b) Knowing your resume
 c) Being formally dressed d) Knowledge of the company
43. When giving a presentation in front of an audience you should do all of the following except :
 a) Speak loud and clear b) Provide handouts if needed
 c) Dress professionally d) Lack of eye contact with the audience
44. A group discussion checks and monitors,
 a) Leadership skills b) Listening ability
 c) Confidence d) All of these
45. Communication helps to make accurate decisions and influence organizational performance positively.
 a) True b) False

Directions : Fill in the Blanks :

46. What actually scared us _____ the fact that there was no one around to help us.
 a) were b) was
 c) with d) all of these
47. He was the man _____ they thought was dead.
 a) of b) who c) whom d) all of these
48. The flowers smell _____.
 a) Sweet b) Sweetly c) Sweeter d) Sweetest
49. The Guptas are travelling _____ plane.
 a) at b) in c) on d) by
50. Which of the following sentence does not contain misplaced modifier?
 a) Tired after a long day at work, Rita napped with her cat.
 b) Happy that school was over, the afternoon was quite relaxing
 c) We glued together the vase we broke quietly
 d) My uncle had to see a doctor with indigestion

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01-08-2024 01:32:43pm

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Question Paper Version : B

First/Second Semester B.E./B.Tech./B.Design Degree Examination, June/July 2024

Professional Writing Skills in English

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Directions : Fill in the Blanks :

- What actually scared us _____ the fact that there was no one around to help us.
a) were
b) was
c) with
d) all of these
- He was the man _____ they thought was dead.
a) of
b) who
c) whom
d) all of these
- The flowers smell _____.
a) Sweet
b) Sweetly
c) Sweeter
d) Sweetest
- The Guptas are travelling _____ plane.
a) at
b) in
c) on
d) by
- Which of the following sentence does not contain misplaced modifier?
a) Tired after a long day at work, Rita napped with her cat.
b) Happy that school was over, the afternoon was quite relaxing
c) We glued together the vase we broke quietly
d) My uncle had to see a doctor with indigestion
- Which among the following are barriers to listening?
a) Forged attention
b) Poor interpersonal relations
c) Premature evaluation
d) All of these
- Which among the following should not be implemented for effective listening?
a) Having an open mind
b) Not being prejudiced
c) Employing critical thinking
d) Asking irrelevant questions

8. A business letter must be,
 - a) written in proper format
 - b) short and concise
 - c) polite in tone
 - d) All of these
9. The date on a business letter should appear after the salutation.
 - a) True
 - b) False
10. This format of the letter has the heading, dateline, complementary closure and signature right aligned.
 - a) Block format
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 - d) None of these
11. Which among the following is not to be mentioned in a resume?
 - a) Educational qualification
 - b) Work experience
 - c) Strengths
 - d) Weaknesses
12. The _____ format of resume lists your work history with dates, with your most recent employer and job title listed first.
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 - d) All of these
13. BCC in an email refers to :
 - a) British council careers
 - b) Blind carbon copy
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14. The cover letter is written,
 - a) To introduce oneself as the suitable candidate for the job.
 - b) To give biographical details of the candidate.
 - c) To try for the job
 - d) To let the employer know of our writing skills
15. _____ communication is a direct, written or oral communication that occurs between two or more persons
 - a) Interpersonal
 - b) Extra-personal
 - c) Intrapersonal
 - d) None of these
16. In a group discussion one must communicate with,
 - a) Hostility
 - b) Arrogance
 - c) Long sentences
 - d) Knowledge
17. Which among the following should not be followed while appearing for an interview?
 - a) Arriving late to the venue
 - b) Knowing your resume
 - c) Being formally dressed
 - d) Knowledge of the company
18. When giving a presentation in front of an audience you should do all of the following except :
 - a) Speak loud and clear
 - b) Provide handouts if needed
 - c) Dress professionally
 - d) Lack of eye contact with the audience
19. A group discussion checks and monitors,
 - a) Leadership skills
 - b) Listening ability
 - c) Confidence
 - d) All of these

21. "I am sorry", he said.

a) He apologized that he was sorry b) He cried that he was to be sorry

c) He demanded that he was sorry d) All of these

22. A sentence that introduces the topic or the main idea to the readers is called.

 - a) Topic sentence
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 - d) None of these
23. A paragraph which is written after analyzing a situation is called,

 - a) Descriptive paragraph
 - b) Analytical paragraph
 - c) Illustrative paragraph
 - d) narrative paragraph
24. The important parts of an essay are :

 - a) Introduction
 - b) Body
 - c) Conclusion
 - d) All of these
25. _____ is a gist of any passage written in as few words as possible.

 - a) Essay writing
 - b) Precis writing
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 - d) None of these
26. Which among the following is not a feature of reports?

 - a) Focuses on facts and data
 - b) Is written for a specific purpose
 - c) Includes irrelevant information
 - d) Is structured in an organized way
27. Reports which are submitted at regular intervals is called,

 - a) Routine report
 - b) Periodic report
 - c) Both 'a' and 'b'
 - d) None of these
28. A technical report establishes a,

 - a) illogical conclusion
 - b) logical conclusion
 - c) personal prejudice
 - d) misplaced learning
29. _____ is drafted in response to an advertisement or demand.

 - a) Solicited proposal
 - b) Unsolicited proposal
 - c) Both 'a' and 'b'
 - d) None of these
30. Childhood is a time when there are _____ responsibilities to make life difficult
child _____ good parents, he is fed, looked _____ and loved.

 - a) many, had, up
 - b) few, has, after
 - c) little, have, at
 - d) all of these
31. Listening is a vital skill which helps in enhancing our learning.

 - a) True
 - b) False

Directions : Change the voice of the following sentences :

32. You need to clean your shoes properly.
 a) Your shoes are needed to clean properly.
 b) You are needed to clean your shoes properly
 c) Your shoes need to be cleaned properly
 d) Your shoes are needed by you to clean properly
33. James watt discovered the energy of steam.
 a) The energy of steam discovered James watt
 b) The energy of steam was discovered by James watt.
 c) James watt discovering the energy of steam
 d) James watt had been discovered energy by the steam

Directions : Rearrange the sentence :

34. defined as a place (a) where man is passive (b) and the rest of the nature is active (c) a sanctuary may be (d)
 a) a, b, c, d
 b) d, c, a, b
 c) b, c, d, a
 d) d, a, b, c
35. are simply at a loss (a) of 500 and 1000 rupee notes (b) all the corrupt politicians and their cronies (c) after the demonetization (d)
 a) c, a, d, b
 b) b, a, c, d
 c) d, c, b, a
 d) b, c, a, d

Directions : Do as directed :

36. Okay, see you _____ the concert.
 a) at
 b) in
 c) for
 d) none of these
37. The children were _____ at having been informed about the trip.
 a) thrilling
 b) thrills
 c) thrilled
 d) none of these
38. _____ you ever been to Kashmir?
 a) Have
 b) Did
 c) Had
 d) None of these
39. Don't narrate _____ stories, they scare me.
 a) ghastly
 b) ghostly
 c) both 'a' and 'b'
 d) none of these
40. That _____ be true. He wouldn't do something like that.
 a) wouldn't
 b) shouldn't
 c) can't
 d) none of these

Directions : Fill in the blanks with suitable tenses :

41. Suddenly she gave a loud scream and _____ to the ground.
 a) had fallen
 b) has fallen
 c) fell
 d) none of these

- 42.** The room _____ but the police failed to find anything suspicious.
a) Searched b) was searched
c) had searched d) none of these

Directions : Choose a suitable verb to complete the sentence :

43. One of our employees _____ retiring today.
a) are b) is
c) were d) none of these
44. Either the teacher or the student _____ to be blamed for his performance in the final exam.
a) are b) is
c) were d) none of these

Directions : Fill in the blanks with suitable phrasal verbs :

- 45.** His arrogance _____ his ruin.
a) brought in b) brought up
c) brought about d) none of these
- 46.** Henry _____ the magazine quickly.
a) Looked through b) Looked up
c) Looked in d) None of these

Choose the part of the sentence that has an error :

47. If I had known (a) this yesterday (b) I will have helped him (c) no error (d)
48. Having received your letter (a) this morning, we was writing (b) to thank you for the same (c) no error (d)
49. A large number of peoples (a) have gathered (b) to greet the leader (c) no error (d)
50. Driving for five hundred kilometers in a day (a) are a tiring proposition (b) no error (c)

* * * * *

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Question Paper Version : C

First/Second Semester B.E./B.Tech./B.Design Degree Examination, June/July 2024
Professional Writing Skills in English

Time: 1 hr.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Directions : Choose a suitable verb to complete the sentence :

1. One of our employees _____ retiring today.
a) are b) is
c) were d) none of these
2. Either the teacher or the student _____ to be blamed for his performance in the final exam.
a) are b) is
c) were d) none of these

Directions : Rearrange the sentence :

3. defined as a place (a) where man is passive (b) and the rest of the nature is active (c) a sanctuary may be (d)
a) a, b, c, d
b) d, c, a, b
c) b, c, d, a
d) d, a, b, c
4. are simply at a loss (a) of 500 and 1000 rupee notes (b) all the corrupt politicians and their cronies (c) after the demonetization (d)
a) c, a, d, b
b) b, a, c, d
c) d, c, b, a
d) b, c, a, d
5. Which among the following are barriers to listening?
a) Forged attention
b) Poor interpersonal relations
c) Premature evaluation
d) All of these
6. Which among the following should not be implemented for effective listening?
a) Having an open mind
b) Not being prejudiced
c) Employing critical thinking
d) Asking irrelevant questions

7. A business letter must be,
 - a) written in proper format
 - b) short and concise
 - c) polite in tone
 - d) All of these
8. The date on a business letter should appear after the salutation.
 - a) True
 - b) False
9. This format of the letter has the heading, dateline, complementary closure and signature right aligned.
 - a) Block format
 - b) Informal format
 - c) Modified block format
 - d) None of these
10. Which among the following is not to be mentioned in a resume?
 - a) Educational qualification
 - b) Work experience
 - c) Strengths
 - d) Weaknesses
11. The _____ format of resume lists your work history with dates, with your most recent employer and job title listed first.
 - a) Historical
 - b) Functional
 - c) Chronological
 - d) All of these
12. BCC in an email refers to :
 - a) British council careers
 - b) Blind carbon copy
 - c) Booked carbon copy
 - d) None of these
13. The cover letter is written,
 - a) To introduce oneself as the suitable candidate for the job.
 - b) To give biographical details of the candidate.
 - c) To try for the job
 - d) To let the employer know of our writing skills
14. _____ communication is a direct, written or oral communication that occurs between two or more persons
 - a) Interpersonal
 - b) Extra-personal
 - c) Intrapersonal
 - d) None of these
15. In a group discussion one must communicate with,
 - a) Hostility
 - b) Arrogance
 - c) Long sentences
 - d) Knowledge
16. Which among the following should not be followed while appearing for an interview?
 - a) Arriving late to the venue
 - b) Knowing your resume
 - c) Being formally dressed
 - d) Knowledge of the company
17. When giving a presentation in front of an audience you should do all of the following except :
 - a) Speak loud and clear
 - b) Provide handouts if needed
 - c) Dress professionally
 - d) Lack of eye contact with the audience
18. A group discussion checks and monitors,
 - a) Leadership skills
 - b) Listening ability
 - c) Confidence
 - d) All of these

19. Communication helps to make accurate decisions and influence organizational performance positively.
a) True b) False

Choose the part of the sentence that has an error :

20. If I had known (a) this yesterday (b) I will have helped him (c) no error (d)
21. Having received your letter (a) this morning, we was writing (b) to thank you for the same (c) no error (d)
22. A large number of peoples (a) have gathered (b) to greek the leader (c) no error (d)
23. Driving for five hundred kilometers in a day (a) are a tiring proposition (b) no error (c)

Directions : Fill in the blanks with suitable tenses :

24. Suddenly she gave a loud scream and _____ to the ground.
a) had fallen b) has fallen
c) fell d) none of these
25. The room _____ but the police failed to find anything suspicious.
a) Searched b) was searched
c) had searched d) none of these

Directions : Change the voice of the following sentences :

26. You need to clean your shoes properly.
a) Your shoes are needed to clean properly.
b) You are needed to clean your shoes properly
c) Your shoes need to be cleaned properly
d) Your shoes are needed by you to clean properly
27. James watt discovered the energy of steam.
a) The energy of steam discovered James watt
b) The energy of steam was discovered by James watt.
c) James watt discovering the energy of steam
d) James watt had been discovered energy by the steam

Directions : Fill in the Blanks :

28. What actually scared us _____ the fact that there was no one around to help us.
a) were b) was
c) with d) all of these
29. He was the man _____ they thought was dead.
a) of b) who c) whom d) all of these
30. The flowers smell _____.
a) Sweet b) Sweetly c) Sweeter d) Sweetest

31. The Guptas are travelling _____ plane.
a) at b) in c) on d) by
32. Which of the following sentence does not contain misplaced modifier?
a) Tired after a long day at work, Rita napped with her cat.
b) Happy that school was over, the afternoon was quite relaxing
c) We glued together the vase we broke quietly
d) My uncle had to see a doctor with indigestion

Directions : Fill in the blanks with suitable phrasal verbs :

33. His arrogance _____ his ruin.
a) brought in b) brought up
c) brought about d) none of these
34. Henry _____ the magazine quickly.
a) Looked through b) Looked up
c) Looked in d) None of these

Directions : Do as directed :

35. Okay, see you _____ the concert.
a) at b) in
c) for d) none of these
36. The children were _____ at having been informed about the trip.
a) thrilling b) thrills
c) thrilled d) none of these
37. _____ you ever been to Kashmir?
a) Have b) Did
c) Had d) None of these
38. Don't narrate _____ stories, they scare me.
a) ghastly b) ghostly
c) both 'a' and 'b' d) none of these
39. That _____ be true. He wouldn't do something like that.
a) wouldn't b) shouldn't
c) can't d) none of these

Directions : Convert the following sentences from Direct to Indirect speech :

40. "I am sorry", he said.
a) He apologized that he was sorry b) He cried that he was to be sorry
c) He demanded that he was sorry d) All of these

Directions : Do as directed :

41. A sentence that introduces the topic or the main idea to the readers is called.
a) Topic sentence b) First sentence
c) Both 'a' and 'b' d) None of these

42. A paragraph which is written after analyzing a situation is called,
 a) Descriptive paragraph b) Analytical paragraph
 c) Illustrative paragraph d) narrative paragraph
43. The important parts of an essay are :
 a) Introduction b) Body
 c) Conclusion d) All of these
44. _____ is a gist of any passage written in as few words as possible.
 a) Essay writing b) Precis writing
 c) Analytical writing d) None of these
45. Which among the following is not a feature of reports?
 a) Focuses on facts and data b) Is written for a specific purpose
 c) Includes irrelevant information d) Is structured in an organized way
46. Reports which are submitted at regular intervals is called,
 a) Routine report b) Periodic report
 c) Both 'a' and 'b' d) None of these
47. A technical report establishes a,
 a) illogical conclusion b) logical conclusion
 c) personal prejudice d) misplaced learning
48. _____ is drafted in response to an advertisement or demand.
 a) Solicited proposal b) Unsolicited proposal
 c) Both 'a' and 'b' d) None of these
49. Childhood is a time when there are _____ responsibilities to make life difficult. If a child _____ good parents, he is fed, looked _____ and loved.
 a) many, had, up b) few, has, after
 c) little, have, at d) all of these
50. Listening is a vital skill which helps in enhancing our learning.
 a) True b) False

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Question Paper Version : D

First/Second Semester B.E./B.Tech./B.Design Degree Examination, June/July 2024
Professional Writing Skills in English

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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a) at
b) in
c) for
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2. The children were _____ at having been informed about the trip.
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CBCS SCHEME

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Question Paper Version : A

**First/Second Semester B.E./B.Tech./B.Design Degree Examination,
June/July 2024**

Scientific Foundations of Health

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
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-
1. According to WHO, what is health?
 - a) Health is a state of complete physical, mental and social well being and not the absence of disease.
 - b) Health is not a state of complete physical, mental and not the absence of disease.
 - c) Complete physical, mental and social well being and not the presence of disease
 - d) None of these
 2. Advantages of good health
 - a) Reduces confidence
 - b) Improves brain health and confidence
 - c) Increased stress
 - d) Reduces performance
 3. What is balanced nutrition?
 - a) A healthy nutrition diet provides the body with non essential nutrition.
 - b) A healthy nutrition diet provides the body with essential nutrition: fluid, macronutrients such as protein, vitamins, adequate fibre and food energy.
 - c) Nutrition is adequate fibre and food energy
 - d) None of these
 4. What are the examples of positive attitude?
 - a) Teasing others
 - b) Looking at the glass half full
 - c) Looking at the glass half empty
 - d) None of these
 5. The only disability in life is a _____
 - a) Bad attitude
 - b) Positive attitude
 - c) Commenting on life
 - d) Making fun on others

6. What factors influencing Good Health?
 - a) State of our environment and genetics
 - b) Income and education
 - c) Relationship with friends and family
 - d) All of these
7. What is the key to having a positive attitude?
 - a) Always do your best
 - b) Do only what you think you can
 - c) Think you cant do anything
 - d) Believe in yourself
8. Individual personality is influenced by
 - a) Experiences
 - b) Environment
 - c) Inherited characteristics
 - d) All of these
9. Well being also involved reducing the risk of an injury or health issue by
 - a) Minimizing hazards in the work place
 - b) Using contraceptive when having sex
 - c) Avoiding the use of tobacco, alcohol and illegal drug
 - d) All of these
10. How does personality affect health?
 - a) Failure in handle stress
 - b) Failure in activity level
 - c) Avoiding bad habits
 - d) None of these
11. The source of antioxidant and minerals
 - a) Vegetables and fruit
 - b) Coke and pizza
 - c) Mutton and meat
 - d) Baked foods
12. Which of the following is whole grain food?
 - a) Brown rice and wild rice
 - b) Oat's meal
 - c) Ragi ball and Barley malt
 - d) All of these
13. Excessive intake of food leads to ____
 - a) Obesity
 - b) Fit body
 - c) Over smartness
 - d) Lowers body calorie
14. Which of the following components are major nutrients in our food?
 - a) Carbohydrates
 - b) Lipids and proteins
 - c) Vitamins and Minerals
 - d) All of these
15. Egg is a rich source of ____
 - a) Proteins
 - b) Vitamins
 - c) Minerals
 - d) None of these
16. Potatoes, beans, pulses and oats are rich in ____
 - a) Proteins
 - b) Vitamins
 - c) Minerals
 - d) Carbohydrates
17. Which of the following food items provides dietary fibre?
 - a) Pulses
 - b) Whole grain
 - c) fruits and vegetables
 - d) Pizza
18. Which of the following food components are rich in fat?
 - a) Rice and Maize
 - b) Milk, egg and beans
 - c) Butter, cheese and oil
 - d) None of these

19. Which of the following vitamin helps in blood clotting
a) Vitamin – A
b) Vitamin – C
c) Vitamin – D
d) Vitamin – K
20. Guava, Lemon, Orange and Tomato are rich in
a) Vitamin – A
b) Vitamin – B
c) Vitamin – C
d) Vitamin – D
21. How many minutes should an average person walk a day?
a) 30 minutes
b) 40 minutes
c) 20 minutes
d) 60 minutes
22. What is the recommended daily water intake?
a) 0.5 litre
b) 1 litre
c) 2 litres
d) 4 litres
23. Oral communication ensures _____ and _____
a) Fluency and speed
b) Adequate response and immediate response
c) Speedy interaction and speed response
d) Fast and attention
24. The primary goal of communication is to
a) To create barriers
b) To create noise
c) To effect a change
d) None of these
25. Body language play a important role in _____
a) Communication
b) Judgement
c) Both A and B
d) None of these
26. Ways to improve the communication skill:
a) Active listening skill
b) Passive listening skill
c) Both A and B
d) None of these
27. Which of the following is not a communication skill?
a) Listening
b) Feedback
c) Conversation
d) Playing game
28. Which of the following is barrier of communication?
a) No interest in conversation
b) Ability to listen
c) Transparency and Trust
d) None of these
29. What is not a basic instincts of Human life?
a) Self perseverance
b) Sexual instinct
c) Social instinct
d) Making wealth and property all time.
30. Which of the following social engineering changes the Health behaviours?
a) Banning the use of certain drugs
b) Providing purified water to people
c) Legalizations can be passed to make environment healthier
d) All of these

31. Factors leading to addiction
a) Environment
b) Genes
c) Both a and b
d) None of these
32. Which of the following is not a type of addiction?
a) Drug
b) Alcohol
c) Behavioural
d) Yoga practicing
33. Symptoms of substance use disorder
a) Impaired control
b) Social problem
c) Having intense usage for the drug
d) All of these
34. Recognizing the addictions
a) Problems at work
b) Physical health issue
c) Changes in behavior
d) All of these
35. Which of the following is a protective factor of addictive people?
a) Good self control
b) Parental monitoring and support
c) Positive relationships
d) All of these
36. Which is an example of health hazard?
a) Cardiovascular disease
b) Cancer
c) Stroke
d) All of these
37. What are the different sources of infection?
a) Patients
b) Health care workers
c) Visitors and Household workers
d) All of these
38. Which one is a type of infective agent?
a) Bacteria
b) Virus
c) Both a and b
d) None of these
39. How to reduce risk for good health?
a) Be a non-smoker
b) Be physically active every day
c) Achieve a healthy weight
d) All of these
40. Which is not the risk factor for addictive people?
a) Aggressive behavior in childhood
b) Lack of parental supervision
c) Good social skills
d) Poor social skills
41. Which is an example for acute health change condition?
a) Heart disease
b) Cancer
c) Diabetes
d) Cold and cough
42. Health is _____
a) Money
b) Wealth
c) Gold
d) Diamond
43. Which of the following are not major health issues?
a) Malnutrition
b) Obesity
c) Anaemia
d) Fever
44. Preventive measures against disease for good health
a) Avoid touching your face
b) Stay home if you are sick
c) Avoid contaminated food and water
d) All of these

45. Who proposed the biopsychosocial model?
a) Freud b) Sontag c) Engel d) None of these
46. Which of the following charters defined health promotion as “the process of enabling people to increase control over and to improve their health”?
a) Charter of United Nations (1945) b) Tokyo Charter (1946)
c) Ottawa Charter (1986) d) None of these
47. HIV is a retrovirus that infects and colonizes cells in the ____
a) Immune system
b) Immune system and central nervous system
c) Endocrine system
d) Immune system and Endocrine system
48. In 2015, the global prevalence of HIV infection was ____
a) 0.2% b) 0.4% c) 0.6% d) 0.8%
49. In H.J. Eysenck’s twentieth – century theory of personality, how did he reinterpret the classical phlegmatic temperament?
a) Stable extrovert b) Stable introvert
c) Unstable extrovert d) Unstable introvert
50. In which year WHO did the first express “the right to health” as a fundamental human right?
a) 1946 b) 1952 c) 1987 d) 2000

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CBCS SCHEME

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Question Paper Version : B

**First/Second Semester B.E./B.Tech./B.Design Degree Examination,
June/July 2024**

Scientific Foundations of Health

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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a) Good self control b) Parental monitoring and support
c) Positive relationships d) All of these
16. Which is an example of health hazard?
a) Cardiovascular disease b) Cancer
c) Stroke d) All of these
17. What are the different sources of infection?
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c) Visitors and Household workers d) All of these
18. Which one is a type of infective agent?
a) Bacteria b) Virus c) Both a and b d) None of these
19. How to reduce risk for good health?
a) Be a non-smoker b) Be physically active every day
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25. Egg is a rich source of _____
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26. Potatoes, beans, pulses and oats are rich in _____
 a) Proteins b) Vitamins c) Minerals d) Carbohydrates
27. Which of the following food items provides dietary fibre?
 a) Pulses b) Whole grain
 c) fruits and vegetables d) Pizza
28. Which of the following food components are rich in fat?
 a) Rice and Maize b) Milk, egg and beans
 c) Butter, cheese and oil d) None of these
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38. Individual personality is influenced by
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c) Inherited characteristics
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d) None of these
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a) 30 minutes
b) 40 minutes
c) 20 minutes
d) 60 minutes
42. What is the recommended daily water intake?
a) 0.5 litre
b) 1 litre
c) 2 litres
d) 4 litres

43. Oral communication ensures _____ and _____
a) Fluency and speed
b) Adequate response and immediate response
c) Speedy interaction and speed response
d) Fast and attention
44. The primary goal of communication is to
a) To create barriers
b) To create noise
c) To effect a change
d) None of these
45. Body language play a important role in _____
a) Communication
b) Judgement
c) Both A and B
d) None of these
46. Ways to improve the communication skill:
a) Active listening skill
b) Passive listening skill
c) Both A and B
d) None of these
47. Which of the following is not a communication skill?
a) Listening
b) Feedback
c) Conversation
d) Playing game
48. Which of the following is barrier of communication?
a) No interest in conversation
b) Ability to listen
c) Transparency and Trust
d) None of these
49. What is not a basic instincts of Human life?
a) Self perseverance
b) Sexual instinct
c) Social instinct
d) Making wealth and property all time.
50. Which of the following social engineering changes the Health behaviours?
a) Banning the use of certain drugs
b) Providing purified water to people
c) Legalizations can be passed to make environment healthier
d) All of these

* * * * *

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Question Paper Version : C

**First/Second Semester B.E./B.Tech./B.Design Degree Examination,
June/July 2024**

Scientific Foundations of Health

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
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-
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 - a) Environment
 - b) Genes
 - c) Both a and b
 - d) None of these
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 - b) Alcohol
 - c) Behavioural
 - d) Yoga practicing
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 - b) Social problem
 - c) Having intense usage for the drug
 - d) All of these
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 - b) Physical health issue
 - c) Changes in behavior
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c) Avoiding the use of tobacco, alcohol and illegal drug
d) All of these
20. How does personality affect health?
a) Failure in handle stress
b) Failure in activity level
c) Avoiding bad habits
d) None of these
21. Which is an example for acute health change condition?
a) Heart disease
b) Cancer
c) Diabetes
d) Cold and cough
22. Health is _____
a) Money
b) Wealth
c) Gold
d) Diamond
23. Which of the following are not major health issues?
a) Malnutrition
b) Obesity
c) Anaemia
d) Fever
24. Preventive measures against disease for good health
a) Avoid touching your face
b) Stay home if you are sick
c) Avoid contaminated food and water
d) All of these
25. Who proposed the biopsychosocial model?
a) Freud
b) Sontag
c) Engel
d) None of these
26. Which of the following charters defined health promotion as “the process of enabling people to increase control over and to improve their health”?
a) Charter of United Nations (1945)
b) Tokyo Charter (1946)
c) Ottawa Charter (1986)
d) None of these
27. HIV is a retrovirus that infects and colonizes cells in the _____
a) Immune system
b) Immune system and central nervous system
c) Endocrine system
d) Immune system and Enderine system
28. In 2015, the global prevalence of HIV infection was _____
a) 0.2%
b) 0.4%
c) 0.6%
d) 0.8%
29. In H.J. Eysenck’s twentieth – century theory of personality, how did he reinterpret the classical phlegmatic temperament?
a) Stable extrovert
b) Stable introvert
c) Unstable extrovert
d) Unstable introvert
30. In which year WHO did the first express “the right to health” as a fundamental human right?
a) 1946
b) 1952
c) 1987
d) 2000

43. Excessive intake of food leads to _____
a) Obesity
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47. Which of the following food items provides dietary fibre?
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Question Paper Version : D

**First/Second Semester B.E./B.Tech./B.Design Degree Examination,
June/July 2024**

Scientific Foundations of Health

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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Question Paper Version : A

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024 Universal Human Values Course

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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-
1. The purpose of value –Education is to
 - a) Foster universal core values
 - b) Make syllabus easy
 - c) Develop values in individual
 - d) Both A and C
 2. Self exploration uses two mechanisms i) Natural Acceptance ii) -----?
 - a) Experiential validation
 - b) Reason
 - c) Logical Thinking
 - d) Theoretical concept
 3. Once we know what is valuable to us, these values becomes the basis, the anchor for ____
 - a) Knowledge
 - b) Actions
 - c) Society
 - d) None of these
 4. To fulfill Human Aspirations, what are necessary
 - a) Both values and skills
 - b) Values
 - c) Skills
 - d) None of these
 5. Which the following are the encompassing principles underlying the successful implementation of value education?
 A) Conviction B) Connection C) Critical thinking D) Commitment
 choose the most appropriate answer from the options given below :
 - a) A, C and D only
 - b) B, C and D only
 - c) A, B and D only
 - d) None of these
 6. Value and skills should go hand in hand
 - a) True
 - b) False
 - c) Cannot tell
 - d) None of these

7. Are the content of self – exploration
 - a) Program
 - b) Desire
 - c) Both a and b
 - d) None
8. Human life is lived at four levels individual, Family, Society and _____
 - a) Nature
 - b) Nurture
 - c) World
 - d) Universe
9. Any course content on value education needs to be
 - a) Universal
 - b) Rational
 - c) Natural
 - d) All of these
10. Value education enables us to
 - a) To understand our needs
 - b) Visualize our goals correctly
 - c) Indicate the direction for their fulfillment
 - d) All of the above
11. Harmony should be maintained in
 - a) Between body and life
 - b) Between self and society
 - c) Between life and environment
 - d) All of these
12. I being the
 - a) does, seer and Enjoyer
 - b) doer
 - c) seer
 - d) enjoyer
13. Which of the following is NOT response of the self?
 - a) Knowing
 - b) Assuming
 - c) Recognizing
 - d) Preconditioning
14. Activities of self (I) are
 - a) Happiness
 - b) Prosperity
 - c) Desire, thought and expectation
 - d) None
15. The requirement of body is right utilization and nurturing
 - a) Desire
 - b) Protection
 - c) Thought
 - d) Expectation
16. The _____ is an instrument of _____
 - a) I, Body
 - b) Body, I
 - c) Both a and b
 - d) None
17. The activity of desire, thought and expecting together is called as
 - a) Body
 - b) Health
 - c) Imagination
 - d) Future
18. Imaging is _____ with time
 - a) Continuous
 - b) Discontinuous
 - c) Random
 - d) Different
19. Where there is harmony among the parts of the body it is known as
 - a) Swasthya
 - b) Sanyam
 - c) Prosperity
 - d) None
20. Knowing means having the
 - a) Assumption
 - b) Right understanding
 - c) Right feeling
 - d) None
21. Harmony should be maintained in
 - a) Between body and life
 - b) Between self and society
 - c) Between life and environment
 - d) All of these

22. The foundational value in relationship is
 - a) Respect
 - b) Love
 - c) Trust
 - d) Glory
23. Ensuring right understanding and feeling in the others is called
 - a) Care
 - b) Affection
 - c) Gratitude
 - d) Guidance
24. Harmony in the family is the building block for harmony in the
 - a) Society
 - b) Individual
 - c) Friend
 - d) Relative
25. The total numbers of feelings in human relationship
 - a) 5
 - b) 10
 - c) 9
 - d) 8
26. Comprehensive human goal is right understanding prosperity, trust (fearlessness) and
 - a) Co-existence
 - b) Happiness
 - c) Abhay
 - d) None
27. There is justice in relationship when there is
 - a) Mutual fulfillment
 - b) Self regulation
 - c) Freedom
 - d) None
28. The extension of family is
 - a) Self
 - b) Body
 - c) Society
 - d) Nature
29. The feeling of relatedness to all human beings is called
 - a) Love
 - b) Affection
 - c) Gratitude
 - d) Respect
30. Acceptance of excellence in others is called
 - a) Reverence
 - b) Glory
 - c) Gratitude
 - d) Guidance
31. All the units of nature can be classified into _____ orders
 - a) Two
 - b) Three
 - c) Four
 - d) Six
32. Which of the following does not form an order in nature?
 - a) BIO
 - b) Animal
 - c) Consciousness
 - d) Human
33. Which of the following statements is true,
 - a) Material units have only two kinds of activities recognizing and fulfilling
 - b) Material units have three kinds of activities assuming, recognizing and fulfilling
 - c) Material units have only four kinds of activities knowing, assuming, recognizing and fulfilling
 - d) None of the statement
34. Which of the following statement is not true?
 - a) There is inter connectedness in nature
 - b) There is recyclability and self regulation in nature
 - c) There is struggle for survival in nature
 - d) There is mutual fulfillment in nature
35. According to quantity, which of the following is true for the orders in nature
 - a) Bio order >> Physical order >> Animal order >> Human order
 - b) Animal order >> Bio order >> Physical order >> Human order
 - c) Physical order >> Bio order >> Animal order >> Human order
 - d) None of the above
36. What are the fundamental components of ecosystems?
 - a) Plants and Animals
 - b) Air and water
 - c) Rocks and minerals
 - d) All of these

37. The third order of nature is
 a) Material order b) Animal order c) Plant order d) Human order
38. The activities in human body are
 a) Composition b) Decomposition c) Respiration d) All of these
39. The systems in nature are
 a) Cyclic b) Mutually fulfilling
 c) Both a and b d) None of these
40. The natural characteristics/Svabhava of a human being are
 a) Perseverance b) Bravery c) Generosity d) All of these
41. The only effective way to ensure professional ethics is by developing
 a) Knowledge b) Ethical conduct
 c) Ethical competence d) Professional activities
42. How does unethical practices in various professions can be resolved
 a) Through skills b) Through knowledge
 c) Through practical d) Via right understanding
43. What provides clear guidance and policy frame work conducive to the development of an un-fragmented human society and a universal human order
 a) Humanistic education b) Humanistic constitution
 c) Profession d) Ethical Human conduct
44. The right understanding gained through self exploration also enables us to identify the definitiveness of human conduct. What is this called?
 a) Ethical Human conduct b) Values
 c) Policy d) Utility values
45. Primary step to move towards the holistic alternative is to develop the right understanding among humans and the commitment to
 a) Do practical b) Remain calm c) Live accordingly d) Teach others
46. The right understanding helps us identify the comprehensive human goal in terms of
 a) Samadhan b) Samridhi c) Sah-astitva d) All of these
47. The humanistic education will facilitate the process of self exploration which will lead to continuous
 a) Education b) Self evolution c) Development d) People friendly
48. The values of human being can be enumerated as
 a) Nine b) Thirty c) Eighteen d) Twenty four
49. Which of the following is not a characteristic of professionalism?
 a) Kindness b) Competency c) Morality d) Complacency
50. There are six characteristics of a professional style which is not a professional style?
 a) Ethical b) Emotional c) Responsible d) Intellectual.

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Question Paper Version : B

Fourth Semester B.E. Degree Examination, June/July 2024 Universal Human Values Course

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. The only effective way to ensure professional ethics is by developing

a) Knowledge	b) Ethical conduct
c) Ethical competence	d) Professional activities
 2. How does unethical practices in various professions can be resolved

a) Through skills	b) Through knowledge
c) Through practical	d) Via right understanding
 3. What provides clear guidance and policy frame work conducive to the development of an unfragmented human society and a universal human order

a) Humanistic education	b) Humanistic constitution
c) Profession	d) Ethical Human conduct
 4. The right understanding gained through self exploration also enables us to identify the definitiveness of human conduct. What is this called?

a) Ethical Human conduct	b) Values
c) Policy	d) Utility values
 5. Primary step to move towards the holistic alternative is to develop the right understanding among humans and the commitment to

a) Do practical	b) Remain calm
c) Live accordingly	d) Teach others
 6. The right understanding helps us identify the comprehensive human goal in terms of

a) Samadhan	b) Samridhi	c) Sah-astitva	d) All of these
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7. The humanistic education will facilitate the process of self exploration which will lead to continuous
 - a) Education
 - b) Self evolution
 - c) Development
 - d) People friendly
8. The values of human being can be enumerated as
 - a) Nine
 - b) Thirty
 - c) Eighteen
 - d) Twenty four
9. Which of the following is not a characteristic of professionalism?
 - a) Kindness
 - b) Competency
 - c) Morality
 - d) Complacency
10. There are six characteristics of a professional style which is not a professional style?
 - a) Ethical
 - b) Emotional
 - c) Responsible
 - d) Intellectual.
11. All the units of nature can be classified into _____ orders
 - a) Two
 - b) Three
 - c) Four
 - d) Six
12. Which of the following does not form an order in nature?
 - a) BIO
 - b) Animal
 - c) Consciousness
 - d) Human
13. Which of the following statements is true,
 - a) Material units have only two kinds of activities recognizing and fulfilling
 - b) Material units have three kinds of activities assuming, recognizing and fulfilling
 - c) Material units have only four kinds of activities knowing, assuming, recognizing and fulfilling
 - d) None of the statement
14. Which of the following statement is not true?
 - a) There is inter connectedness in nature
 - b) There is recyclability and self regulation in nature
 - c) There is struggle for survival in nature
 - d) There is mutual fulfillment in nature
15. According to quantity, which of the following is true for the orders in nature
 - a) Bio order >> Physical order >> Animal order >> Human order
 - b) Animal order >> Bio order >> Physical order >> Human order
 - c) Physical order >> Bio order >> Animal order >> Human order
 - d) None of the above
16. What are the fundamental components of ecosystems?
 - a) Plants and Animals
 - b) Air and water
 - c) Rocks and minerals
 - d) All of these
17. The third order of nature is
 - a) Material order
 - b) Animal order
 - c) Plant order
 - d) Human order
18. The activities in human body are
 - a) Composition
 - b) Decomposition
 - c) Respiration
 - d) All of these
19. The systems in nature are
 - a) Cyclic
 - b) Mutually fulfilling
 - c) Both a and b
 - d) None of these
20. The natural characteristics/Svabhava of a human being are
 - a) Perseverance
 - b) Bravery
 - c) Generosity
 - d) All of these

21. Harmony should be maintained in
 - a) Between body and life
 - b) Between self and society
 - c) Between life and environment
 - d) All of these
22. I being the
 - a) does, seer and Enjoyer
 - b) doer
 - c) seer
 - d) enjoyer
23. Which of the following is NOT response of the self?
 - a) Knowing
 - b) Assuming
 - c) Recognizing
 - d) Preconditioning
24. Activities of self (I) are
 - a) Happiness
 - b) Prosperity
 - c) Desire, thought and expectation
 - d) None
25. The requirement of body is right utilization and nurturing
 - a) Desire
 - b) Protection
 - c) Thought
 - d) Expectation
26. The _____ is an instrument of _____
 - a) I, Body
 - b) Body, I
 - c) Both a and b
 - d) None
27. The activity of desire, thought and expecting together is called as
 - a) Body
 - b) Health
 - c) Imagination
 - d) Future
28. Imaging is _____ with time
 - a) Continuous
 - b) Discontinuous
 - c) Random
 - d) Different
29. Where there is harmony among the parts of the body it is known as
 - a) Swasthya
 - b) Sanyam
 - c) Prosperity
 - d) None
30. Knowing means having the
 - a) Assumption
 - b) Right understanding
 - c) Right feeling
 - d) None
31. The purpose of value –Education is to
 - a) Foster universal core values
 - b) Make syllabus easy
 - c) Develop values in individual
 - d) Both A and C
32. Self exploration uses two mechanisms i) Natural Acceptance ii) -----?
 - a) Experiential validation
 - b) Reason
 - c) Logical Thinking
 - d) Theoretical concept
33. Once we know what is valuable to us, these values becomes the basis, the anchor for _____
 - a) Knowledge
 - b) Actions
 - c) Society
 - d) None of these
34. To fulfill Human Aspirations, what are necessary
 - a) Both values and skills
 - b) Values
 - c) Skills
 - d) None of these

35. Which the following are the encompassing principles underlying the successful implementation of value education?
A) Conviction B) Connection C) Critical thinking D) Commitment
choose the most appropriate answer from the options given below :
a) A, C and D only b) B, C and D only c) A, B and D only d) None of these
36. Value and skills should go hand in hand
a) True b) False c) Cannot tell d) None of these
37. Are the content of self – exploration
a) Program b) Desire c) Both a and b d) None
38. Human life is lived at four levels individual, Family, Society and _____
a) Nature b) Nurture c) World d) Universe
39. Any course content on value education needs to be
a) Universal b) Rational c) Natural d) All of these
40. Value education enables us to
a) To understand our needs b) Visualize our goals correctly
c) Indicate the direction for their fulfillment d) All of these
41. Harmony should be maintained in
a) Between body and life b) Between self and society
c) Between life and environment d) All of these
42. The foundational value in relationship is
a) Respect b) Love c) Trust d) Glory
43. Ensuring right understanding and feeling in the others is called
a) Care b) Affection c) Gratitude d) Guidance
44. Harmony in the family is the building block for harmony in the
a) Society b) Individual c) Friend d) Relative
45. The total numbers of feelings in human relationship
a) 5 b) 10 c) 9 d) 8
46. Comprehensive human goal is right understanding prosperity, trust (fearlessness) and
a) Co-existence b) Happiness c) Abhay d) None
47. There is justice in relationship when there is
a) Mutual fulfillment b) Self regulation c) Freedom d) None
48. The extension of family is
a) Self b) Body c) Society d) Nature
49. The feeling of relatedness to all human beings is called
a) Love b) Affection c) Gratitude d) Respect
50. Acceptance of excellence in others is called
a) Reverence b) Glory c) Gratitude d) Guidance

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Question Paper Version : C

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024
Universal Human Values Course

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
- 1 All the units of nature can be classified into _____ orders
a) Two b) Three c) Four d) Six
 - 2 Which of the following does not form an order in nature?
a) BIO b) Animal c) Consciousness d) Human
 - 3 Which of the following statements is true,
a) Material units have only two kinds of activities recognizing and fulfilling
b) Material units have three kinds of activities assuming, recognizing and fulfilling
c) Material units have only four kinds of activities knowing, assuming, recognizing and fulfilling
d) None of the statement
 - 4 Which of the following statement is not true?
a) There is inter connectedness in nature
b) There is recyclability and self regulation in nature
c) There is struggle for survival in nature
d) There is mutual fulfillment in nature
 - 5 According to quantity, which of the following is true for the orders in nature
a) Bio order >> Physical order >> Animal order >> Human order
b) Animal order >> Bio order >> Physical order >> Human order
c) Physical order >> Bio order >> Animal order >> Human order
d) None of the above

- 21 The only effective way to ensure professional ethics is by developing
 a) Knowledge b) Ethical conduct
 c) Ethical competence d) Professional activities
- 22 How does unethical practices in various professions can be resolved
 a) Through skills b) Through knowledge
 c) Through practical d) Via right understanding
- 23 What provides clear guidance and policy frame work conducive to the development of an un-fragmented human society and a universal human order
 a) Humanistic education b) Humanistic constitution
 c) Profession d) Ethical Human conduct
- 24 The right understanding gained through self exploration also enables us to identify the definitiveness of human conduct. What is this called?
 a) Ethical Human conduct b) Values
 c) Policy d) Utility values
- 25 Primary step to move towards the holistic alternative is to develop the right understanding among humans and the commitment to
 a) Do practical b) Remain calm c) Live accordingly d) Teach others
- 26 The right understanding helps us identify the comprehensive human goal in terms of
 a) Samadhan b) Samridhi c) Sah-astitva d) All of these
- 27 The humanistic education will facilitate the process of self exploration which will lead to continuous
 a) Education b) Self evolution c) Development d) People friendly
- 28 The values of human being can be enumerated as
 a) Nine b) Thirty c) Eighteen d) Twenty four
- 29 Which of the following is not a characteristic of professionalism?
 a) Kindness b) Competency c) Morality d) Complacency
- 30 There are six characteristics of a professional style which is not a professional style?
 a) Ethical b) Emotional c) Responsible d) Intellectual.
- 31 Harmony should be maintained in
 a) Between body and life
 b) Between self and society
 c) Between life and environment
 d) All of the above
- 32 The foundational value in relationship is
 a) Respect b) Love c) Trust d) Glory
- 33 Ensuring right understanding and feeling in the others is called
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- 34 Harmony in the family is the building block for harmony in the
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- 35 The total numbers of feelings in human relationship
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- 36 Comprehensive human goal is right understanding prosperity, trust (fearlessness) and
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a) Mutual fulfillment b) Self regulation c) Freedom d) None
- 38 The extension of family is
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- 39 The feeling of relatedness to all human beings is called
a) Love b) Affection c) Gratitude d) Respect
- 40 Acceptance of excellence in others is called
a) Reverence b) Glory c) Gratitude d) Guidance
- 41 Harmony should be maintained in
a) Between body and life
b) Between self and society
c) Between life and environment
d) All of these
- 42 I being the
a) does, seer and Enjoyer b) doer
c) seer d) enjoy
- 43 Which of the following is NOT response of the self?
a) Knowing b) Assuming
c) Recognizing d) Preconditioning
- 44 Activities of self (I) are
a) Happiness b) Prosperity
c) Desire, thought and expectation d) None
- 45 The requirement of body is right utilization and nurturing
a) Desire b) Protection c) Thought d) Expectation
- 46 The _____ is an instrument of _____
a) I, Body b) Body, I c) Both a and b d) None
- 47 The activity of desire, thought and expecting together is called as
a) Body b) Health c) Imagination d) Future
- 48 Imaging is _____ with time
a) Continuous b) Discontinuous c) Random d) Different
- 49 Where there is harmony among the parts of the body it is known as
a) Swasthya b) Sanyam c) Prosperity d) None
- 50 Knowing means having the
a) Assumption
b) Right understanding
c) Right feeling
d) None

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Question Paper Version : D

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024
Universal Human Values Course

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

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b) Make syllabus easy
c) Develop values in individual
d) Both A and C
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a) Experiential validation b) Reason
c) Logical Thinking d) Theoretical concept
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a) Knowledge b) Actions c) Society d) None of these
- 44 To fulfill Human Aspirations, what are necessary
a) Both values and skills b) Values
c) Skills d) None of these
- 45 Which the following are the encompassing principles underlying the successful implementation of value education?
A) Conviction B) Connection C) Critical thinking D) Commitment
choose the most appropriate answer from the options given below :
a) A, C and D only b) B, C and D only c) A, B and D only d) None of these
- 46 Value and skills should go hand in hand
a) True b) False c) Cannot tell d) None of these
- 47 Are the content of self – exploration
a) Program b) Desire c) Both a and b d) None
- 48 Human life is lived at four levels individual, Family, Society and _____
a) Nature b) Nurture c) World d) Universe
- 49 Any course content on value education needs to be
a) Universal b) Rational c) Natural d) All of these
- 50 Value education enables us to
a) To understand our needs
b) Visualize our goals correctly
c) Indicate the direction for their fulfillment
d) All of the above

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CBCS 2022 – SCHEME

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BCEDK103/203

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024

COMPUTER AIDED ENGINEERING DRAWING

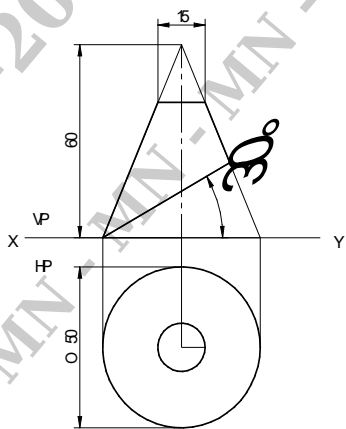
Time: 3 Hours

(COMMON TO ALL BRANCHES)

Max.Marks:100

Note: 1. Answer all four full question.

2. Grid sheets may be provided for making preparatory sketches.

Module – 1		
Q. No.		Marks
1 a	A point P is 25 mm above HP and 20 mm in front of VP. Another point Q is on HP and 30 mm behind VP. The distance between their projectors measured parallel to the line of intersection of VP and HP is 50mm. Find the distance between the top views of points P and Q.	8
1 b	The top view of a line 75 mm long measures 50 mm, the end P is 30 mm in front of VP and 15mm above HP. The ends Q is 15 mm in front of VP and above HP. Draw the projections of the line and find its true inclinations with HP and VP.	12
Module – 2		
2	A square prism 35 mm sides of base and 60 mm axis length rests on HP on one of its corners of the base such that the two base edges containing the corner on which it rests make equal inclinations with HP. Draw the projections of the prism when the axis of the prism is inclined to HP at 40° and appears to be inclined to VP at 45° .	30
Module – 3		
3	A hemisphere of 40 mm diameter is supported co-axially on the vertex of a cone of base diameter 60 mm and axis length 50 mm. The flat circular face of the hemisphere is facing upside. Draw the isometric projection of the combination of solids.	25
Module - 4		
4	Draw the development of the lateral surface of the cone, whose front view is as shown in following figure Q(4)  Figure Q(4)	25

Examiner 1:

Name:

Signature:

Examiner 2:

Name:

Signature:

CBCS 2022 – SCHEME

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BCEDK103/203**First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024****COMPUTER AIDED ENGINEERING DRAWING****Time: 3 Hours****(COMMON TO ALL BRANCHES)****Max.Marks:100****Note:** 1. Answer all four full question.

2. Grid sheets may be provided for making preparatory sketches.

Module – 1		
Q. No.		Marks
1 a	A point A is 40 mm in front of VP and is situated in the fourth quadrant. Its shortest distance from the Intersection of XY and X_1Y_1 is 45 mm. Draw its projections. Also find its distance from HP.	8
1 b	The front view of a 90 mm long line which is inclined at 45° to the XY line, measures 65 mm. End A is 15 mm above the XY line and is in VP. Draw the projections of the line and find its inclinations with HP and VP.	12
Module – 2		
2	A pentagonal pyramid 25 mm sides of base and 60 mm axis length rests on HP on one of the edges of the base which is inclined to VP at 30° . Draw the projections of the pyramid when the axis is inclined to HP at 40° .	30
Module – 3		
3	A cone of base diameter 50 mm and height 40 mm is placed centrally on the top face of a square slab side-80 mm and height 20 mm. Draw the isometric projection of the combination.	25
Module - 4		
4	Draw the development of the lateral surface of a funnel consisting of a cylinder and a frustum of a cone. The diameter of the cylinder is 20 mm and the top face diameter of the funnel is 80 mm. the height of the frustum and cylinders are equal to 60 mm and 40 mm respectively.	25

Examiner 1:

Name:

Signature:

Examiner 2:

Name:

Signature:

CBCS 2022 – SCHEME**BCEDK103/203**

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First / Second Semester B.E. Degree Examination, June/July 2024**COMPUTER AIDED ENGINEERING DRAWING****Time: 3 Hours****(COMMON TO ALL BRANCHES)****Max.Marks:100****Note:** 1. Answer all four full question

2. Grid sheets may be provided for making preparatory sketches

Module – 1		
Q. No.		Marks
1	A circular lamina of 50 mm diameter rests on HP such that one of its diameters is inclined at 30^0 to VP and 45^0 to HP. Draw its top and front views in this position.	20
Module – 2		
2	A hexagonal prism 25 mm sides of base and 50 mm axis length rests on HP on one of its base edges. Draw the projections of the prism when the axis is inclined to HP at 45^0 and appears to be inclined to VP 40^0 .	30
Module – 3		
3	The frustum of a square pyramid of base sides 40 mm and top face side 20 mm and height 60 mm rest on the center of the top of a square block of side 60 mm and height 20 mm. The base edges of the pyramid are parallel to the top edges of the square block. Draw the isometric projection of the combination of the solids.	25
Module - 4		
4	A square prism of base side 40 mm and axis length 65 mm is resting on HP on its base with all the vertical faces being equally inclined to VP. It is cut by an inclined plane 60^0 to HP and perpendicular to VP and is passing through the point on the axis at a distance 15 mm from the top face. Draw the development of the lower portion of the prism.	25

Examiner 1:

Name:

Signature:

Examiner 2:

Name:

Signature:

CBCS 2022 – SCHEME**BCEDK103/203**USN

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First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024**COMPUTER AIDED ENGINEERING DRAWING****Time: 3 Hours (COMMON TO ALL BRANCHES) Max.Marks:100**

- Note:** 1. Answer all four full question
2. Grid sheets may be provided for making preparatory sketches

Module - 1		
Q. No.		Marks
1	A $30^\circ - 60^\circ$ set square of 60 mm longest side is so kept such that the longest side is in HP, making an angle of 30° with VP. The set square itself is inclined at 45° to HP. Draw the projections of the setsquare.	20
Module - 2		
2	A Square pyramid 35 mm sides of base and 65 mm axis length rests on HP on one of its edges of the base which is inclined to VP at 30° . Draw the projections of the pyramid when the axis is inclined to HP at 45° .	30
Module - 3		
3	A hemisphere of 40 mm diameter is supported co-axially on the vertex of a cone of base diameter 60 mm and axis length 50 mm. The flat circular face of hemisphere is facing upside. Draw the isometric projection of the combination of solids.	25
Module - 4		
4	A square prism of base sides 30 mm and axis length 60 mm is resting on HP with all the vertical faces equally inclined to VP. It is cut by an inclined plane 60° to HP and perpendicular to VP and is passing through a point on the axis at a distance of 50 mm from the base. Obtain the development for the truncated portion of the solid.	25

Examiner 1:
Name:
Signature:

Examiner 2:
Name:
Signature: